

# Optimal Parameter-Free Second-order Acceleration with Inexact Hessians under Generalized Smoothness

Artem Agafonov<sup>1</sup>Aslan Ashabokov<sup>2</sup>

Alexander D'yakonov

Martin Takáč<sup>1</sup>Alexander Gasnikov<sup>2</sup>

Dmitry Kamzolov

<sup>1</sup>MBZUAI    <sup>2</sup>MIRAI

## Abstract

We study second-order convex optimization under generalized Hessian smoothness, where the local Hessian Lipschitz constant may grow linearly with the gradient norm. For inexact Hessians, we develop adaptive algorithms, including an accelerated Monteiro-Svaiter-type method whose complexity matches the lower bound for second-order methods with  $\delta$ -inexact Hessians in the classical Lipschitz-Hessian setting. Our methods are parameter-free in the sense that they require no problem-specific parameters as input, including the Hessian inexactness level  $\delta$  and smoothness constants. Instead, their backtracking adapts a single regularization parameter that jointly captures Hessian inexactness and local Hessian smoothness. The resulting rates split into two regimes. Far from the solution, the gradient-dependent smoothness term yields geometric convergence. Near the solution, the rate is governed by the ordinary Hessian smoothness and the inexactness level  $\delta$ , recovering the classical Lipschitz-Hessian guarantees.

## 1 Introduction

We study second-order methods for the problem  $\min_{x \in \mathbb{R}^d} f(x)$ , where  $f \in \mathcal{C}^2$  is a convex function. We assume that a solution  $x^* \in \mathbb{R}^d$  exists and we denote  $f^* := f(x^*)$ . Given an initial point  $x_0$ , we define  $R := \|x_0 - x^*\|$  as a distance to the solution. The classical theory assumes exact Hessians that are Lipschitz continuous (Nesterov & Polyak, 2006; Nesterov, 2018):

$$\|\nabla^2 f(y) - \nabla^2 f(x)\| \leq L\|y - x\|, \quad x, y \in \mathbb{R}^d. \quad (\text{H})$$

In this setting, optimal methods find an  $\varepsilon$ -solution in  $\mathcal{O}((LR^3/\varepsilon)^{2/7})$  iterations (Kovalev & Gasnikov, 2022; Carmon et al., 2022), matching the lower bound (Arjevani et al., 2019). We consider two extensions of this theory.

*Generalized smoothness.* In first-order optimization,  $(L_0, L_1)$ -smoothness lets the local Lipschitz-gradient constant grow with the gradient norm (Zhang et al., 2020; Crawshaw et al., 2022). We study its second-order analogue (Xie et al., 2024; Jerad, 2026): for some  $M_0 \geq 0$ ,  $M_1 \geq 0$ , and  $\rho > 0$ ,

$$\|\nabla^2 f(y) - \nabla^2 f(x)\| \leq (M_0 + M_1 \|\nabla f(x)\|) \|y - x\|, \quad x, y \in \mathbb{R}^d : \|y - x\| \leq \rho. \quad (\text{LH})$$

It reduces to (H) for  $M_1 = 0$  and  $\rho = +\infty$ , and it covers problems such as Poisson regression and  $\ell_p$  regression with  $p \geq 4$ , for which (H) fails (see examples in Section 2.1).

*Inexact Hessians.* Exact Hessians are often too expensive. In practice, methods use approximations

obtained by subsampling (Kohler & Lucchi, 2017; Xu et al., 2020), quasi-Newton updates (Kamzolov et al., 2025b), or lazy updates (Doikov et al., 2023). We assume that the approximation  $H(x)$  is symmetric and satisfies

$$\|H(x) - \nabla^2 f(x)\| \leq \delta. \quad (\text{IH})$$

**Parameter-free methods.**<sup>1</sup> Both extensions make assumptions more realistic, but they add constants that a method has to know: (LH) involves  $M_0$ ,  $M_1$ , and  $\rho$ , and (IH) adds  $\delta$ . These constants are rarely known. Moreover, under (LH) they are not unique. Poisson regression satisfies (LH) for every  $\rho > 0$ , with  $M_0$  and  $M_1$  growing in  $\rho$  (Example 2.5). A method that takes them as input must fix  $\rho$  in advance, whereas the guarantees of a parameter-free method hold for the best  $\rho$ .

To the best of our knowledge, even under (H), no parameter-free method attains the optimal complexity with inexact Hessians. The method of Carmon et al. (2022) is optimal and adaptive, but its analysis requires exact Hessians. The optimal method of Chen et al. (2026) requires  $L$ ,  $\delta$ , and  $R$  as input. Under generalized smoothness, existing convex guarantees are non-accelerated (Semenov et al., 2026; Jerad, 2026). In view of these results, we ask two questions:

*Can a parameter-free method attain the optimal complexity with inexact Hessians?  
Can such a method be extended to generalized Hessian smoothness?*

**Our main result.** We answer both questions affirmatively with a single method (Algorithm 3). Using the Monteiro–Svaiter condition (Monteiro & Svaiter, 2013) as an acceptance test, we do not estimate the Hessian inexactness and the local Lipschitz constant of the Hessian separately, but rather adapt them as a single regularization parameter. Under (H) and (IH), the proposed method needs  $\mathcal{O}\left((LR^3/\varepsilon)^{2/7} + \sqrt{\delta R^2/\varepsilon}\right)$  Hessian evaluations, up to an additive logarithmic term independent of  $\varepsilon$ . This matches the lower bound (Agafonov et al., 2024b). Under (LH), the same method converges linearly far from the solution and then at the same optimal rate with  $L = M_0$ . Table 1 compares our main methods with closest competitors and the lower bound.

## 1.1 Related Work

**Beyond classical smoothness.** In first-order optimization, empirical observations on neural networks have motivated  $(L_0, L_1)$ -smoothness assumption (Zhang et al., 2020; Crawshaw et al., 2022), which is studied for both convex (Koloskova et al., 2023; Li et al., 2023; Tyurin, 2025; Gorbunov et al., 2025) and nonconvex (Koloskova et al., 2023; Chen et al., 2023; Vankov et al., 2025) objectives. In particular, convex accelerated under  $(L_0, L_1)$ -smoothness has recently received considerable attention (Gorbunov et al., 2025; Vankov et al., 2025; Tyurin, 2026), including parameter and line-search free methods (Borodich & Kovalev, 2026). Second-order methods under generalized smoothness have been developed in nonconvex (Xie et al., 2024; Fang et al., 2026) and convex (Semenov et al., 2026; Jerad, 2026) settings, some under the global variant of (LH)

$$\|\nabla^2 f(y) - \nabla^2 f(x)\| \leq (M_0 + M_1 \|\nabla f(x)\|) \|y - x\|, \quad x, y \in \mathbb{R}^d. \quad (\text{GH})$$

**Inexact Hessians.** Under (H), many methods with inexact Hessians augment cubic regularization with a quadratic term that compensates for the Hessian error (Ghadimi et al., 2017; Agafonov et al., 2024a;b). These methods take both  $L$  and  $\delta$  as input. Kamzolov et al. (2025b) adapt the quadratic coefficient but still require a cubic coefficient  $\mathcal{M} \geq 2L$ . Antonakopoulos et al. (2022)

---

<sup>1</sup>By parameter-free, we mean requiring no problem constants as input and converging for any initialization.

Table 1: Comparison of our methods with the fastest known second-order convex methods and the lower bound with inexact Hessians

| Method                                                    | Assump.                   | Param.<br>Free | Hessian complexity                                                                                                                                               |
|-----------------------------------------------------------|---------------------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lower bound<br>(Agafonov et al., 2024b)                   | (H)<br>(IH)               | —              | $\Omega((\frac{LR^3}{\varepsilon})^{2/7} + \sqrt{\frac{\delta R^2}{\varepsilon}})$                                                                               |
| Optimal MS Acceleration<br>(Carmon et al., 2022)          | (H)                       | ✓              | $\mathcal{O}((\frac{LR^3}{\varepsilon})^{2/7})$                                                                                                                  |
| Optimal inexact Newton<br>(Chen et al., 2026)             | (H)<br>(IH)               | ✗              | $\mathcal{O}((\frac{LR^3}{\varepsilon})^{2/7} + \sqrt{\frac{\delta R^2}{\varepsilon}})$                                                                          |
| Gradient-Regularized Newton<br>(Semenov et al., 2026)     | (GH)<br>(IH) <sup>a</sup> | ✓              | $\tilde{\mathcal{O}}(\sqrt{\frac{M_0 D^3}{\varepsilon}} + \sqrt{M_1 D} + \frac{\delta D^2}{\varepsilon})^b$                                                      |
| AR2-LS (Jerad, 2026)                                      | (LH)                      | ✓              | $\tilde{\mathcal{O}}(\varepsilon^{-1/2})$                                                                                                                        |
| Adaptive regularized Newton<br>(This work, Thm. 3.1)      | (LH)<br>(IH)              | ✓              | $\mathcal{O}(\sqrt{\frac{M_0 D^3}{\varepsilon}} + \frac{\delta D^2}{\varepsilon}) + \tilde{\mathcal{O}}(1 + \sqrt{M_1 D} + \frac{D}{\rho})$                      |
| Inexact adaptive A-NPE<br>(This work, Thm. E.1)           | (LH)<br>(IH)              | ✓              | $\mathcal{O}((\frac{M_0 R^3}{\varepsilon})^{1/3} + \sqrt{\frac{\delta R^2}{\varepsilon}}) + \tilde{\mathcal{O}}(1 + \sqrt{M_1 R} + \frac{R}{\rho})$              |
| Inexact adaptive<br>damped A-NPE<br>(This work, Thm. 4.1) | (LH)<br>(IH)              | ✓              | $\mathcal{O}((\frac{M_0 R^3}{\varepsilon})^{2/7} + \sqrt{\frac{\delta R^2}{\varepsilon}}) + \tilde{\mathcal{O}}(1 + M_1^{1/3} R^{2/3} + (\frac{R}{\rho})^{2/3})$ |

$D = \text{diam}\{f \leq f(x_0)\} < \infty$ ,  $G = \sup_{f \leq f(x_0)} \|\nabla f\|$ .  $\tilde{\mathcal{O}}$  hides logarithmic factors. Parameter-free methods require no problem parameters. <sup>a</sup>  $H_k \succeq 0$  is also assumed. <sup>b</sup> Combines Corollary 2 and Appendix F.6 of Semenov et al. (2026).

propose Extra-Newton, a parameter-free but suboptimal method. Semenov et al. (2026) propose an adaptive but non-accelerated method for generalized Hessian smoothness with inexact Hessians. Recently, Chen et al. (2026) proposed an optimal method matching the lower bound (Agafonov et al., 2024b), but their method is non-adaptive.

**Adaptive second-order methods.** With exact Hessians and under (H), adaptive cubic regularization estimates the cubic coefficient by backtracking (Nesterov & Polyak, 2006; Cartis et al., 2011; Grapiglia & Nesterov, 2020). Gradient-regularized Newton methods replace the cubic subproblem with a linear system (Mishchenko, 2023; Doikov & Nesterov, 2023), and their adaptive variants tune the regularization coefficient in the same way (Doikov et al., 2024). Carmon et al. (2022) obtain an optimal adaptive method by searching over the regularization of a Newton step using Monteiro–Svaiter acceleration (Monteiro & Svaiter, 2013).

## 1.2 Contributions

- **Far/near rate separation under (LH).** We analyze an adaptive regularized Newton method and three accelerated variants. Far from the solution, the methods converge linearly at a rate that depends on  $M_1$  and  $\rho$ . Near the solution, they recover the standard rates under (H) (see Table 1).
- **Inexact Hessians and adaptivity.** The three methods listed in Table 1 allow inexact Hessians and remain parameter-free. A single regularization parameter in the Newton step accounts for

both the Hessian error and the local smoothness, and methods adapt it by backtracking on the Monteiro-Svaiter condition.

- **Optimal adaptive acceleration.** Algorithm 3 combines this backtracking with the damped acceleration scheme of Carmon et al. (2022). Under (LH) and (IH), it converges linearly during the first  $\tilde{\mathcal{O}}(1 + M_1^{1/3}R^{2/3} + (R/\rho)^{2/3})$  iterations and then needs  $\mathcal{O}((M_0R^3/\varepsilon)^{2/7} + \sqrt{\delta R^2/\varepsilon})$  more iterations. Under (H), the first phase reduces to an additive logarithmic term, and the bound matches the lower bound (Agafonov et al., 2024b). The number of gradient evaluations has the same order up to additive terms at most logarithmic in  $1/\varepsilon$ .
- **Revisited analysis of optimal acceleration.** Our convergence proof is new, and we discuss its differences from the analyses of Carmon et al. (2022) and Chen et al. (2026) in Section 4.3.
- **Recovering local quadratic rate.** For strongly convex functions, our adaptive regularized Newton converges globally linearly and does not accumulate Hessian errors. If the errors vanish (e.g., with lazy Hessian updates), it converges superlinearly. Under a stronger, relative accuracy condition on the Hessian approximation, it also converges locally quadratically (Section 3.2).

## 2 Preliminaries

Throughout the paper,  $\|\cdot\|$  denotes the Euclidean norm for vectors and the induced operator norm for matrices. For both conditions (GH) and (LH), we define  $L(x) := M_0 + M_1\|\nabla f(x)\|$ . When using  $H(x)$  satisfying (IH), we assume w.l.o.g. that it is symmetric, since replacing it by symmetric matrix  $(H(x) + H^T(x))/2$  does not increase  $\delta$ . We set  $\log_+ t := \max\{0, \log t\}$  and  $\log_{b,+} t := \max\{0, \log_b t\}$  for  $t > 0$  and  $b > 1$ . Under (GH) we take  $\rho = +\infty$ , and use  $a/(+\infty) = 0$  for finite  $a \geq 0$  and  $a/0 = +\infty$  for  $a > 0$ . In particular,  $1/\sqrt{M_1} = +\infty$  if  $M_1 = 0$ .

### 2.1 Generalized Hessian Smoothness

**Relations between assumptions.** Functions satisfying (H), such as logistic regression, also satisfy (GH) and (LH). Examples below show that the inclusions (H)  $\subset$  (GH)  $\subset$  (LH) are strict. In particular, setting  $x = x^*$  in (GH) gives  $\|\nabla^2 f(y)\| \leq \|\nabla^2 f(x^*)\| + M_0\|y - x^*\|$ . Thus, the Hessian norm can grow at most linearly with the distance from  $x^*$ , as under (H). However, (GH) is more general than (H), since the coefficient controlling the Hessian variation at  $x$  may grow with  $\|\nabla f(x)\|$ . The local condition (LH) does not imply this global linear bound and therefore covers a broader class of functions.

**Example 2.1.** The function  $f(x) = x^2 + \int_0^x (x-t) \sin(t^2)dt$  is 1-strongly convex and satisfies (GH) with  $M_0 = M_1 = 2$ , but it does not satisfy (H).

**Example 2.2.** The 1-strongly convex function  $f(x) = \cosh x - 1$  satisfies (LH) for every  $\rho > 0$  with  $M_0 = \sinh \rho$  and  $M_1 = e^\rho$ , but it does not satisfy (GH).

**Example 2.3.** The convex function  $f(x) = e^x - x - 1$  satisfies (LH) for every  $\rho > 0$  with  $M_0 = M_1 = e^\rho$ , but it does not satisfy (GH).

**Example 2.4.** For  $\mu \geq 0$ , consider  $f(x) = \frac{\mu}{2}\|x\|^2 + \frac{1}{4}\|x\|^4$  for  $x \in \mathbb{R}^d$ . The function is convex for  $\mu = 0$  and  $\mu$ -strongly convex for  $\mu > 0$ . For every  $\rho > 0$ , it satisfies (LH) with  $M_0 = 3\rho + 4$  and  $M_1 = 2$ , but it does not satisfy (GH).

**Example 2.5** (Poisson regression). Let  $a_i \in \mathbb{R}^d$  with  $[a_i]_1 = 1$  and  $b_i \in \mathbb{N}$ ,  $i = 1, \dots, n$ . The Poisson regression loss  $f(x) = \sum_{i=1}^n (e^{\langle a_i, x \rangle} - b_i \langle a_i, x \rangle)$  is convex and satisfies (LH) for every  $\rho > 0$  with  $M_0 = \|b\|_1 c^3 e^{c\rho}$  and  $M_1 = c^3 e^{c\rho}$ , where  $c = \max_i \|a_i\|$ , but it does not satisfy (GH).

**Example 2.6** ( $\ell_p$  regression). Let  $p > 3$ ,  $A \in \mathbb{R}^{n \times d}$ ,  $A \neq 0$ , and  $b \in \mathbb{R}^n$ . The  $\ell_p$  regression loss  $f(x) = \frac{1}{p} \|Ax - b\|_p^p$  is convex and satisfies (LH) for every  $\rho > 0$  with

$$M_0 = (p-1)(p-2)\|A\|^3 (1 + 2\|b\|_p + \|A\|\rho)^{p-3}, \quad M_1 = 3\sqrt{n}M_0/\sigma_+(A),$$

where  $\sigma_+(A)$  is the smallest positive singular value of  $A$ , but it does not satisfy (GH).

For  $f \in C^3(\mathbb{R}^d)$ , condition (LH) implies the pointwise bound (Xie et al., 2024)

$$\|\nabla^3 f(x)\|_{\text{op}} := \sup_{\|u\|=1} \|\nabla^3 f(x)[u]\| \leq M_0 + M_1 \|\nabla f(x)\|, \quad x \in \mathbb{R}^d. \quad (\text{TD})$$

**Proposition 2.7.** Suppose that, for some  $L_0 > 0$  and  $L_1 \geq 0$ ,  $\|\nabla^2 f(x)\| \leq L_0 + L_1 \|\nabla f(x)\|$ . If (TD) holds with constants  $\widehat{M}_0$  and  $\widehat{M}_1$ , then, for every  $\rho > 0$ , (LH) holds with

$$M_0(\rho) = \widehat{M}_0 + \widehat{M}_1 L_0 \rho e^{L_1 \rho}, \quad M_1(\rho) = \widehat{M}_1 e^{L_1 \rho}.$$

## 2.2 Globally Convergent Second-order Methods under Inexact Hessians

With an inexact Hessian, cubic regularization commonly uses the model

$$\omega_x^{\mathcal{M}, \widehat{\delta}}(s) := f(x) + \langle \nabla f(x), s \rangle + \frac{1}{2} \langle H(x)s, s \rangle + \frac{\widehat{\delta}}{2} \|s\|^2 + \frac{\mathcal{M}}{6} \|s\|^3, \quad (1)$$

with  $s = y - x$ . The quadratic term compensates for Hessian error and the cubic term controls the Taylor remainder. Under (H),  $\widehat{\delta} \geq \delta$  and  $\mathcal{M} \geq L$  ensure model convexity and majorization (Agafonov et al., 2024a, Theorem 1). In Nesterov-type acceleration these coefficients also enter the estimating sequence (Nesterov, 2008; Ghadimi et al., 2017).

For a parameter-free method, a useful observation is that the first-order optimality condition of the cubic model depends on these two regularizers only through their sum:

$$(H(x) + \tau I)s = -\nabla f(x), \quad \tau = \widehat{\delta} + \frac{\mathcal{M}}{2} \|s\|.$$

This suggests treating  $\tau$  as a single regularization parameter, rather than adapting  $\widehat{\delta}$  and  $M$  separately. The difficulty here is that  $\tau$  depends on unknown step length  $\|s\|$ . It is a circular dependence, which arises in the accelerated Newton proximal extragradient (A-NPE) method (Monteiro & Svaiter, 2013) and its optimal variants (Kovalev & Gasnikov, 2022; Carmon et al., 2022; Chen et al., 2026). In particular, it can be resolved by searching over the regularization parameter while solving only quadratic subproblems. This perspective motivates our approach: instead of separately adapting two regularization parameters, we search directly for a regularized Newton step that satisfies a verifiable Monteiro-Svaiter condition (Monteiro & Svaiter, 2013).

**Definition 2.8.** A Monteiro-Svaiter (MS) oracle at  $x$  returns  $y = x + s$  and  $\tau > 0$  such that

$$\|\nabla f(x + s) + \tau s\| \leq \tau \|s\|/2.$$

Under (H), inexact cubic step  $y = x + \arg\min_s \omega_x^{\mathcal{M}, \widehat{\delta}}(s)$  with  $\widehat{\delta} \geq 2\delta$  and  $\mathcal{M} \geq 2L$  is an instance of MS oracle with  $\tau = \widehat{\delta} + \mathcal{M}\|s\|/2$  (Chen et al., 2026, Lemma 3.1). Under exact Hessian the same Lipschitz assumption, the gradient-regularized Newton step (Mishchenko, 2023; Doikov & Nesterov, 2023) is also an MS oracle for  $\tau = \sqrt{M\|\nabla f(x)\|} > 0$  and  $M \geq L$ .

### 3 Basic Method

---

**Algorithm 1** MSBACKTRACK( $x, \tau$ )

---

```

1: Input:  $x \in \mathbb{R}^d$  and  $\tau > 0$ .
2: loop
3:   if PD( $x, \tau$ ) then
4:     Solve  $(H(x) + \tau I)s = -\nabla f(x)$ .
5:     if MS( $x, \tau, s$ ) then
6:       return  $(x + s, \tau)$ .
7:   Set  $\tau := 2\tau$ .
```

---



---

**Algorithm 2** Adaptive regularized Newton

---

```

1: Input:  $x_0 \in \mathbb{R}^d$  and  $\eta_0 > 0$ .
2: for  $k \geq 0$  do
3:    $(x_{k+1}, \tau_k) := \text{MSBACKTRACK}(x_k, \eta_k)$ .
4:   Set  $\eta_{k+1} := \frac{\tau_k}{2} \min \left\{ 1, \frac{\|\nabla f(x_{k+1})\|}{\|\nabla f(x_k)\|} \right\}$ .
```

---

We begin by proposing a basic non-accelerated adaptive method under MS oracle. First, we define two tests for our backtracking procedure

$$\text{PD}(x, \tau) : H(x) + \tau I \succ 0, \quad \text{MS}(x, \tau, s) : \|\nabla f(x + s) + \tau s\| \leq \frac{\tau}{2} \|s\|. \quad (2)$$

The tests do not require  $\delta$ ,  $L(x)$ , or  $\rho$ . The PD test is checked first, so the linear system is solved only when  $H(x) + \tau I$  is positive definite. Algorithm 1 defines MSBACKTRACK, which ensures that tests (2) hold. The resulting Adaptive regularized Newton methods is listed as Algorithm 2.

At each outer iteration, the method evaluates one  $H(x)$  and keeps it fixed throughout backtracking. For example, an SVD-based implementation can reuse the same factorization across all trial values of  $\tau$ . The gradient  $\nabla f(x_k)$  is reused throughout backtracking, and  $\nabla f(x_{k+1})$  is reused in the update of  $\eta_{k+1}$  and at the next iteration. The halving in line 4 prevents overregularization, which is important in our convergence analysis (see proof sketch below). Consequently,  $\tau$  is allowed to both increase and decrease across outer iterations. The gradient-dependent scaling factor is not needed for convex convergence, but allows us to recover local quadratic convergence under exact Hessians.

#### 3.1 Convex analysis

Let  $D = \max_{x \in \mathcal{L}(x_0)} \|x - x_0\|$  be the radius of the level set  $\mathcal{L}(x_0) = \{x \in \mathbb{R}^d : f(x) \leq f(x_0)\}$ . Since Algorithm 2 is monotone (Lemma C.1), all iterates remain in  $\mathcal{L}(x_0)$ . Define

$$B := \max \left\{ 1, 2D/\rho, 6\sqrt{M_1}D \right\}, \quad \varepsilon_{\text{tr}} := \max \left\{ M_0 D^3 / B^2, \delta D^2 / B \right\}, \quad (3)$$

where  $\varepsilon_{\text{tr}}$  marks the switch from the linear rate to the rate of inexact cubic Newton.

**Theorem 3.1.** *Suppose that  $f$  is convex, (IH) holds, and either (GH) or (LH) holds. Let  $\ell_0 := \max \left\{ 1, \frac{2D\|\nabla f(x_0)\|}{\varepsilon_{\text{tr}}}, \frac{\eta_0 D^2}{\sqrt{6B\varepsilon_{\text{tr}}}} \right\}$ . For every  $\varepsilon > 0$ , Algorithm 2 generates an iterate  $x_N$  satisfying  $f(x_N) - f^* \leq \varepsilon$  within the following number of iterations*

$$N = \mathcal{O} \left( 1 + \log \ell_0 + B \log_+ \frac{f(x_0) - f^*}{\max\{\varepsilon, \varepsilon_{\text{tr}}\}} + \sqrt{\frac{M_0 D^3}{\varepsilon}} + \frac{\delta D^2}{\varepsilon} \right). \quad (4)$$

Theorem 3.1 bounds the number of outer iterations  $N$ , which equals the number of inexact Hessian evaluations. In Appendix C.5, we show that the total number of gradient evaluations is  $\mathcal{O}(N + \log_+ \frac{B\|\nabla f(x_0)\|}{D\eta_0})$ , matching  $N$  up to a constant factor and an additive logarithmic term.

The complexity (4) separates the dependence on  $M_0$ ,  $M_1$  and  $\delta$ , and reveals two accuracy regimes. If

the target accuracy  $\varepsilon \geq \varepsilon_{\text{tr}}$ , the method remains in the far regime and converges linearly, requiring  $\mathcal{O}(B + \log \ell_0 + B \log_+ \frac{f(x_0) - f^*}{\varepsilon})$  iterations. If  $\varepsilon < \varepsilon_{\text{tr}}$ , the method first reaches  $\varepsilon_{\text{tr}}$  at a linear rate, after which the complexity is determined by  $M_0$  and  $\delta$ , matching the rate of inexact cubic Newton under (H). The additional term  $\log \ell_0$  is independent of  $\varepsilon$  and arises from bounding the number of overregularized iterations (see the proof sketch). Let us introduce one of the main lemmas.

**Lemma 3.2.** *Suppose that  $f$  is convex, (IH) holds, and either (GH) or (LH) holds. Let  $(y, \tau) := \text{MSBACKTRACK}(x, \eta)$ . If at least one trial is rejected (equivalently  $\tau > \eta$ ), then*

$$\|s\| > \min \left\{ \frac{\rho}{2}, \frac{1}{6\sqrt{M_1}} \right\} \quad \text{or} \quad \tau \leq 8\delta + 24M_0\|s\|$$

This shows that for iterations with rejected trials we have two regimes: either regularization matches the scale of inexact cubic Newton step (set  $M_0 = L$  under (H) and see example after Definition 2.8) or the step is determined by generalized smoothness constants  $M_1, \rho$  and is sufficiently large.

**Proof sketch of Theorem 3.1.** Denote  $\Delta_k = f(x_k) - f(x^*)$ ,  $g_k = \|\nabla f(x_k)\|$ , and  $s_k = x_{k+1} - x_k$ . By MS test we show  $\tau_k\|s_k\|/2 \leq g_{k+1} \leq 3\tau_k\|s_k\|/2$  and using convexity

$$f(x_k) - f(x_{k+1}) \geq 2g_{k+1}^2/(3\tau_k), \quad g_{k+1} \leq 2g_k/\sqrt{3}.$$

Thus, the method is monotone and progress of one step depends on how large the accepted regularization  $\tau_k$  can be relative to  $g_{k+1}$ . We call iteration  $k$  controlled if  $\tau_k \leq \Theta_c(g_{k+1})$ , where  $\Theta_c(g) := \max \{8\delta + \sqrt{48M_0g}, 4g/\rho, 12\sqrt{M_1g}\}$ . The proof is based on separate counting of controlled and uncontrolled iterations.

First, we show, that if at least one trial was rejected the iteration is controlled. By Lemma 3.2 and MS relation we have  $\tau_k \leq \Theta_c(g_{k+1})$ . Let  $U_k := \max \{\tau_{k-1}, \Theta_c(\|\nabla f(x_k)\|)\}$ . Then by the definition of  $U_k$  we get  $U_{k+1} \leq (2/\sqrt{3})U_k$  on every controlled iteration.

An uncontrolled iteration has no rejected trials and therefore accepts  $\tau_k = \eta_k$ . By the update rule for  $\eta_k$ , we obtain  $U_{k+1} = \tau_k \leq \tau_{k-1}/2 \leq U_k/2$ . Therefore  $\Theta_c(g_m) \leq U_0(2/\sqrt{3})^{m_c} 2^{-(m-m_c)}$  after  $m$  iteration of which  $m_c$  are controlled. Taking the logarithm yields

$$m \leq 2m_c + \log_2(U_0/\Theta_c(g_m)).$$

Thus, to show the convergence, it is enough to count the number of controlled steps. For controlled step, using the boundness of  $\tau_k$ , convexity and monotonicity we show recurrence

$$\Delta_k - \Delta_{k+1} \gtrsim \frac{\Delta_{k+1}}{B + \sqrt{M_0 D^3 / \Delta_{k+1} + \delta D^2 / \Delta_{k+1}}}.$$

When  $B$  dominates in denominator ( $\Delta_{k+1} \geq \varepsilon_{\text{tr}}$ ) we get linear convergence, otherwise the rate is determined by  $M_0$  and  $\delta$  terms. To accumulate this progress, we introduce a scalar potential that increases by at least one on every controlled iteration and does not decrease on the others. This gives us a bound

$$m_c \lesssim B \log_+ \Delta_0 / \max\{\varepsilon, \varepsilon_{\text{tr}}\} + \sqrt{M_0 D^3 / \varepsilon} + \delta D^2 / \varepsilon$$

as long as  $\Delta_m > \varepsilon$ . Bounding the logarithmic term in upper bound on  $m$  gives us the rate (4). The complete proof is given in Appendix C.

### 3.2 Strongly convex analysis

In this section, we denote  $\delta_k := \|H(x_k) - \nabla^2 f(x_k)\| \leq \delta$  and  $r_0 := \min \{\rho/2, 1/(6\sqrt{M_1})\}$ .

**Theorem 3.3.** *Suppose that  $f$  is  $\mu$ -strongly convex, (IH) holds, and either (GH) or (LH) holds. Let*

$$\bar{\tau} := \max \left\{ \eta_0, 8\delta + 48M_0\sqrt{2(f(x_0) - f^*)/\mu}, 2(f(x_0) - f^*)/r_0^2 \right\}.$$

*Every completed iteration  $k$  of Algorithm 2 satisfies  $\tau_k \leq \bar{\tau}$  and*

$$\tau_k \leq 8 \max_{0 \leq j \leq k} 2^{j-k} \delta_j + \bar{\tau} \left(1 - \frac{2\mu}{3\bar{\tau} + 2\mu}\right)^k, \quad f(x_{k+1}) - f^* \leq \left(1 - \frac{2\mu}{3\bar{\tau} + 2\mu}\right)^2 (f(x_k) - f^*).$$

Theorem 3.3 guarantees global linear convergence for any sequence  $\delta_k \leq \delta$  and shows that Hessian errors do not accumulate. If  $\delta_k \rightarrow 0$ , then  $\tau_k \rightarrow 0$ , so the contraction factor in convergence rate goes to zero and the method converges superlinearly. In particular, for exact Hessians, we obtain global superlinear convergence as established for cubic Newton in Kamzolov et al. (2025a). This theorem also covers lazy Hessian updates (Doikov et al., 2023), where an exact Hessian is reused for a fixed number of iterations. Indeed, for any  $m \geq 1$ , the choice  $H(x_k) = \nabla^2 f(x_{m\lfloor k/m \rfloor})$  satisfies  $\delta_k \rightarrow 0$ .

The rate in Theorem 3.3 bounds the number of outer iterations and, equivalently, Hessian computations. To bound the first-order oracle calls, let  $N$  be the first index satisfying  $f(x_N) - f^* \leq \varepsilon$ . By Corollary C.12 it can be bounded as  $\mathcal{O}(N + \log_{2,+} \left( \frac{\|\nabla f(x_0)\|}{\eta_0} \max \left\{ \frac{M_0}{\mu} + \frac{\delta}{\sqrt{\mu\varepsilon}}, \frac{1}{r_0} \right\} \right))$ . For inexact Hessians, this bound guarantees only linear convergence in gradient evaluations, even when  $\delta_k \rightarrow 0$  gives superlinear convergence in Hessian evaluations. For  $\delta = 0$ , the additional logarithmic term is independent of  $\varepsilon$ , and the gradient count is  $\mathcal{O}(N + 1)$ . Thus, for exact Hessians, the method converges superlinearly in terms of both gradient and Hessian evaluations.

Our next goal is to recover the local quadratic convergence typical for second-order methods with exact Hessians. We show that Algorithm 2 preserves local quadratic and global superlinear convergence even with inexact Hessians. Both rates hold in terms of gradient and Hessian evaluations under a stronger accuracy assumption than (IH) (Bellavia et al., 2020; 2022; Chen et al., 2022; Semenov et al., 2026, Examples 7, 8):

$$\delta_k \leq \nu \|\nabla f(x_k)\|, \quad k \geq 0. \quad (5)$$

**Theorem 3.4.** *Suppose that  $f$  is  $\mu$ -strongly convex, either (GH) or (LH) holds, and (5) holds for some  $\nu > 0$ . Algorithm 2 converges globally superlinearly. Every completed iteration  $k$  satisfies*

$$\tau_k \leq \max \left\{ \frac{2^{-k}\eta_0}{\|\nabla f(x_0)\|}, \frac{24M_0}{\mu} + 8\nu, \frac{2}{r_0} \right\} \|\nabla f(x_k)\|, \quad f(x_{k+1}) - f^* \leq \left(1 - \frac{2\mu}{3\tau_k + 2\mu}\right)^2 (f(x_k) - f^*).$$

*Let  $C := \frac{9(\|\nabla^2 f(x^*)\| + \mu)}{2\mu^2} \left(8\nu + \frac{24M_0}{\mu}\right)^2$ . Let for some  $K_0 \geq 1$ ,  $f(x_{K_0}) - f^* \leq \min \left\{ \frac{\mu r_0^2}{8}, \frac{1}{2C} \right\}$ . Then, after finitely many additional iterations independent of  $\varepsilon$ , the method converges quadratically:*

$$f(x_{k+1}) - f^* \leq C (f(x_k) - f^*)^2, \quad k \geq K \geq K_0.$$

By Corollary C.14,  $N \geq 1$  completed iterations require at most  $N$  Hessian evaluations and  $\mathcal{O}(N + \log_{2,+} \left( \frac{\|\nabla f(x_0)\|}{\eta_0} \max \left\{ \frac{M_0}{\mu} + \nu, \frac{1}{r_0} \right\} \right))$  gradient evaluations. The logarithmic term is independent of  $\varepsilon$ , so the global superlinear iteration complexity also holds for gradient evaluations. Since our method converges globally, the index  $K_0$  is reachable. After iteration  $K$ , which is defined in Theorem C.15 and independent of  $\varepsilon$ , the method requires  $\mathcal{O}(\log \log(1/\varepsilon))$  additional Hessian and gradient evaluations suffice to reach  $f(x) - f^* \leq \varepsilon$ .

---

**Algorithm 3** Adaptive damped A-NPE with an inexact Hessian

---

```

1: Input:  $x_0 \in \mathbb{R}^d$  and  $\eta_0 > 0$ .
2:  $(w_1, \eta_1) := \text{MSBACKTRACK}(x_0, \eta_0)$ .
3: Set  $A_1 := 1/\eta_1$ ,  $z_1 := x_0 - A_1 \nabla f(w_1)$ .
4: for  $k \geq 1$  do
5:   Set  $a'_k := \frac{1+\sqrt{1+4\eta_k A_k}}{2\eta_k}$ ,  $A'_k := A_k + a'_k$ ,  $v_k := \frac{A_k w_k + a'_k z_k}{A'_k}$ .
6:    $(y_k, \tau_k) := \text{MSBACKTRACK}(v_k, \eta_k)$ .
7:   Set  $\gamma_k := \frac{\eta_k}{\tau_k}$ ,  $a_{k+1} := \gamma_k a'_k$ ,  $A_{k+1} := A_k + a_{k+1}$ .
8:   Set  $w_{k+1} := \frac{(1-\gamma_k)A_k w_k + \gamma_k A'_k y_k}{A_{k+1}}$ ,  $z_{k+1} := z_k - a_{k+1} \nabla f(y_k)$ .
9:   if  $\tau_k = \eta_k$  then
10:     Set  $\eta_{k+1} := \eta_k/2$ .
11:   else
12:     Set  $\eta_{k+1} := 2\eta_k$ .

```

---

## 4 Acceleration under Generalized Smoothness

### 4.1 Suboptimal Acceleration

Before moving to the optimal acceleration scheme, we briefly discuss generalization of accelerated methods with the suboptimal rate  $\mathcal{O}(1/k^3)$  under (H). Inspired by Tyurin (2026), in Appendix D.2 we propose an adaptive Nesterov-type accelerated method with exact Hessians. The main challenge is that under (LH), the regularization parameter  $\mathcal{M}_k$  must dominate  $L(w_{k+1})$  at the output  $w_{k+1}$  of the cubic step (1), even though this point is unknown when  $\mathcal{M}_k$  is chosen. We refer to the proof sketch in Appendix D for the key idea behind resolving this difficulty. Furthermore, this acceleration scheme uses the cubic model, which has two parameters:  $\mathcal{M}$  and  $\hat{\delta}$ . To avoid the difficulty of adapting two parameters simultaneously, in Appendix E we also propose an adaptive A-NPE method with inexact Hessians under (LH), which attains the rate  $\mathcal{O}(1/k^3 + \delta/k^2)$  under (H).

### 4.2 Adaptive Damped A-NPE Method

In this section, we present the adaptive damped A-NPE method, listed as Algorithm 3. The method is parameter-free and attains the optimal convergence rate for convex optimization with inexact Hessians in the classical Lipschitz-Hessian setting. It combines backtracking (Algorithm 1) with acceleration scheme of Carmon et al. (2022). The parameter  $\eta_k$  is the guess regularization parameter for MSBACKTRACK procedure, which returns  $\tau_k \geq \eta_k$ . If  $\tau_k = \eta_k$  the method performs full step with  $A_{k+1} = A_k + a'_k$ , otherwise the update of  $A_{k+1}$  is damped  $A_{k+1} = A_k + (\eta_k/\tau_k)a'_k$ . The next guess is set to  $\eta_k/2$  when the first trial is accepted and to  $2\eta_k$  otherwise.

As in Algorithm 2, each outer iteration uses at most one Hessian, which is reused throughout backtracking. We again maintain one regularization parameter, which adapts to both Hessian inexactness and local smoothness, without requiring them as an input. The intuition behind this acceleration scheme can be found in the proof sketch below. While the outer scheme of Algorithm 3 matches the method of Carmon et al. (2022), our convergence analysis is different. We discuss the key differences below.

For the convergence bounds of Algorithm 3, define

$$B := \max \left\{ 1, (6\sqrt{M_1}R)^{2/3}, (2R/\rho)^{2/3} \right\}, \quad \varepsilon_{\text{tr}} := \max \left\{ 48M_0R^3/B^{7/2}, 16\delta R^2/B^2 \right\},$$

where  $\rho = +\infty$  under (GH). Let  $\bar{\tau}_0 := \max \left\{ 4\delta, 2\sqrt{L(x_0)\|\nabla f(x_0)\|}, 2\|\nabla f(x_0)\|/\rho \right\}$ .

**Theorem 4.1.** *Suppose that  $f$  is convex, (IH) holds, and either (GH) or (LH) holds. For any  $\varepsilon > 0$ , the method generates an iterate  $w_N$  satisfying  $f(w_N) - f^* \leq \varepsilon$  within*

$$\mathcal{O} \left( B \left( 1 + \log_+ \frac{\max\{\eta_0, \bar{\tau}_0\} R^2}{\max\{\varepsilon, \varepsilon_{\text{tr}}\}} \right) + \left( \frac{M_0 R^3}{\varepsilon} \right)^{2/7} + \sqrt{\frac{\delta R^2}{\varepsilon}} \right) \quad (6)$$

iterations. Moreover, there exists iterate  $K \geq 1$  with  $K = \mathcal{O} \left( B \max \left\{ 1, \log \frac{\max\{\eta_0, \bar{\tau}_0\} R^2}{\varepsilon_{\text{tr}}} \right\} \right)$  such that

$$f(w_k) - f^* = \mathcal{O} \left( \max\{\eta_0, \bar{\tau}_0\} R^2 \exp \left( -\frac{c(k-1)}{B} \right) \right), \quad 1 \leq k < K, \quad (7)$$

$$f(w_{K+n}) - f^* = \mathcal{O} \left( \frac{M_0 R^3}{(B+n)^{7/2}} + \frac{\delta R^2}{(B+n)^2} \right), \quad n = 0, 1, \dots \quad (8)$$

Theorem 4.1 bounds outer iterations, which equals the number of inexact Hessian evaluations. In Appendix F.4, we show that the number of gradient evaluations is of the same order as (6) up to two additive terms  $\mathcal{O}(\log_+ \frac{\bar{\tau}_0}{\eta_0})$  and  $\tilde{\mathcal{O}}(B(1 + \log_+ \frac{\max\{\eta_0, \bar{\tau}_0\} R^2}{\varepsilon})^{2/3})$ , where  $\tilde{\mathcal{O}}$  hides logarithmic factor independent of  $\varepsilon$ . By (7) and (8), the method first converges linearly, and then converges in Lipschitz Hessian regime.

Under (H) the complexity is  $\mathcal{O}(1 + \log_+ \frac{\max\{\eta_0, \bar{\tau}_0\}}{LR+\delta} + (\frac{LR^3}{\varepsilon})^{2/7} + (\frac{\delta R^2}{\varepsilon})^{1/2})$ , matching the optimal method of Chen et al. (2026) and the corresponding lower bound (Agafonov et al., 2024b) up to an additive term independent of  $\varepsilon$ . Gradient computations have additional term  $\mathcal{O}(\log_+ \frac{\bar{\tau}_0}{\eta_0})$ . To the best of our knowledge, this is the first second-order method to simultaneously achieve full adaptivity, optimal complexity, and robustness to inexact Hessians. The complete proof is given in Appendix F.

**Proof sketch.** Let  $s_i := y_i - v_i$  and  $\Lambda(A) := \max \left\{ 32\delta, (96M_0R)^{4/7} A^{-3/7}, B^2/A \right\}$ . By the algorithm construction, the regularization guess  $\eta_i$  is driven toward  $\Lambda(A_i)$ .  $\Lambda(A)$  serves as an ideal regularization scale, which will allow us to recover the optimal Lipschitz-Hessian rate near the solution and a geometric rate far from it.

We next use the standard Monteiro-Svaiter potential argument. Let  $\mathcal{E}_k = A_k(f(w_k) - f^*) + 1/2\|z_k - x^*\|^2$ . Then,  $3/8 \sum_{i=1}^{k-1} A'_i \eta_i \|s_i\|^2 \leq \mathcal{E}_1 \leq R^2/2$ . We first count two types of iterations:

$$\begin{aligned} \mathcal{S}_{k-1}^+ &:= \{i \in \{1, \dots, k-1\} : \tau_i > \eta_i, \eta_i \geq \Lambda(A_i)/2\}, \\ \mathcal{S}_{k-1}^- &:= \{i \in \{1, \dots, k-1\} : \tau_i = \eta_i, \eta_i \leq 2\Lambda(A_i)\}. \end{aligned}$$

On  $\mathcal{S}_{k-1}^+$ , at least one trial is rejected and  $\eta_i$  is sufficiently large. Lemma 3.2, together with  $A'_i \geq A_i$  lower-bounds the corresponding terms in the potential estimate:

$$\frac{3R^2}{16\sqrt{A_k\Lambda(A_k)}} |\mathcal{S}_{k-1}^+| \leq \frac{3}{8} \sum_{i \in \mathcal{S}_{k-1}^+} A'_i \eta_i \|s_i\|^2 \leq \frac{R^2}{2} \Rightarrow |\mathcal{S}_{k-1}^+| \leq \frac{8}{3} \sqrt{A_k\Lambda(A_k)}.$$

Here, the condition on  $\eta_i$  yields sufficient potential decrease, similarly to how controlled iterations in the basic method yield sufficient function decrease. Note, that substituting  $A_k \asymp R^2/\varepsilon$  and definition of  $\Lambda(A_k)$  into  $\sqrt{A_k\Lambda(A_k)}$  recovers the desired rate (6).

On  $\mathcal{S}_{k-1}^-$ , no trial is rejected, so  $\gamma_i = 1$  and  $A_{i+1} = A_i + a'_i$ . The upper bound on  $\eta_i = \tau_i$  prevents overregularization. Using the update in line 7, we have

$$\frac{a'_i}{A_i} \geq \frac{1}{\sqrt{2A_i\Lambda(A_i)}} \Rightarrow \frac{1}{3} \leq \sqrt{A_i\Lambda(A_i)} \log \frac{A_{i+1}}{A_i} \leq \int_{A_i}^{A_{i+1}} \sqrt{\frac{\Lambda(t)}{t}} dt. \quad (9)$$

Thus, on these iterations the upper bound on  $\eta_i$  guarantees fast growth of  $A_i$ . Summing the last inequality over  $\mathcal{S}_{k-1}^-$  and calculating integral gives the upper bound (6). Note that  $\Lambda$  is chosen to match the progress on  $\mathcal{S}_{k-1}^>$  and  $\mathcal{S}_{k-1}^-$ .

Iterations that are not in  $\mathcal{S}_{k-1}^> \cup \mathcal{S}_{k-1}^-$  improve regularization guess. If  $\tau_i = \eta_i$  and  $\eta_i > 2\Lambda(A_i)$ , the algorithm halves  $\eta_i$ . Otherwise,  $\tau_i > \eta_i$  and  $\eta_i < \Lambda(A_i)/2$ , and it doubles  $\eta_i$ . The last challenge is to bound the number of such iterations. Similarly to uncontrolled iterations in the basic method, we show

$$k - 1 \leq 2(|\mathcal{S}_{k-1}^>| + |\mathcal{S}_{k-1}^-|) + \log(A_k \Lambda(A_1)).$$

Combining iteration counts yields

$$k \lesssim B \log_+ \min\{A_k, A_{\text{tr}}\}/A_1 + \sqrt{A_k \Lambda(A_k)},$$

where  $A_{\text{tr}} := R^2/(2\varepsilon_{\text{tr}})$ . Since  $f(w_k) - f^* \leq R^2/(2A_k)$ , it suffices to reach  $A_\varepsilon := R^2/(2\varepsilon)$ . For  $\varepsilon \geq \varepsilon_{\text{tr}}$ , we have  $A_\varepsilon \leq A_{\text{tr}}$  and  $\sqrt{A_\varepsilon \Lambda(A_\varepsilon)} = B$ , so the complexity depends logarithmically on the target accuracy. For  $\varepsilon < \varepsilon_{\text{tr}}$ , the logarithmic term is constant, while

$$\sqrt{A_\varepsilon \Lambda(A_\varepsilon)} \simeq (M_0 R^3/\varepsilon)^{2/7} + \sqrt{\delta R^2/\varepsilon}.$$

### 4.3 Related Proof Techniques

Algorithmically, our method builds upon the optimal MS acceleration of [Carmon et al. \(2022\)](#). Recently, [Chen et al. \(2026\)](#) established the optimal rate for inexact Hessians using the schedule of [Adil et al. \(2026\)](#), but their method is non-adaptive. Below, we discuss the key challenges of extending their analyses to an adaptive method under (LH) and (IH).

**Comparison with [Carmon et al. \(2022\)](#).** They split iterations into two sets:  $\mathcal{S}^\uparrow = \{i : \tau_i > \eta_i\}$ , on which the potential bound  $\sum_{i \in \mathcal{S}^\uparrow} A'_i \eta_i \|s_i\|^2 \lesssim R^2$  is used, and  $\mathcal{S}^\downarrow = \{i : \tau_i = \eta_i\}$ , on which the growth of  $A_i$  is given by the weight identity  $\eta_i (a'_i)^2 = A'_i$ . In the latter case, they use the bound of type  $\sqrt{A_{i+1}} - \sqrt{A_i} \geq 1/(2\sqrt{\eta_i})$ , which is enough for polynomial rate. The first difference is that we use log form in (9) to get geometric far rate.

From this point the proofs follow different ideas. [Carmon et al. \(2022\)](#) transfer the growth of  $A_i$  from  $\mathcal{S}^\downarrow$  to neighboring iterations of  $\mathcal{S}^\uparrow$ . The reason is that on  $\mathcal{S}^\uparrow$  they have  $\|s_i\| \gtrsim \tau_i/L$  because of rejection, which turns the potential bound into  $\sum_{i \in \mathcal{S}^\uparrow} A'_i \eta_i^3 \lesssim L^2 R^2$ . Next, since both bounds are on  $\mathcal{S}^\uparrow$ , they use reverse Hölder inequality to remove  $\eta_i$  and get the rate  $A_k \gtrsim k^{7/2}/(LR)$ .

Because of assumptions (LH) and (IH) we do not have such lower bound on  $\|s_i\|$ . Instead, Lemma 3.2 gives three regimes for rejected steps: it is proportional to  $\tau_i$  for  $\delta \lesssim \tau_i \lesssim M_0 r_0$ , it is bounded below only by  $r_0$  for larger  $\tau_i$ , and it may be arbitrary small for  $\tau_i \lesssim \delta$ . Thus, we introduce  $\Lambda(A)$ , whose three terms correspond to these three regimes, and follow the proof of Theorem 3.1 instead.  $\Lambda(A_i)$  plays the role of  $\Theta_c(g)$ , iterations of  $\mathcal{S}_{k-1}^>$  and  $\mathcal{S}_{k-1}^-$  (because of additional bound on guess  $\eta_i$ ) make progress in the potential or show the growth of  $A_i$ . The remaining iterations move  $\eta_i$  toward  $\Lambda(A_i)$ , so their number is bounded by the multiplier on number of steps with progress and additional logarithm (similarly to uncontrolled iterations in the basic method).

**Comparison with [Chen et al. \(2026\)](#).** The cubic step is used with explicit regularization ( $\tau_i = 2\delta + M_0 \|s_i\|$ ), and the guess is  $\eta_i \simeq \delta + (M_0 R)^{4/7} A_i^{-3/7}$  ([Chen et al., 2026](#), Lemma 3.1 and Theorem 3.1). Their acceleration scheme keeps  $\eta_i$  fixed until  $A_i$  doubles and then resets it. Under (H) this matches our  $\Lambda(A)$  up to constants for  $A \geq A_{\text{tr}}$ . However, the analysis is based on the fact that between two resets the guess is a known constant of order at least  $\delta$ , while an adaptive guess changes at every iteration and can be arbitrarily small. We resolve this challenge by counting only iterations with  $\eta_i \asymp \Lambda(A_i)$  and bounding the remaining as described above.

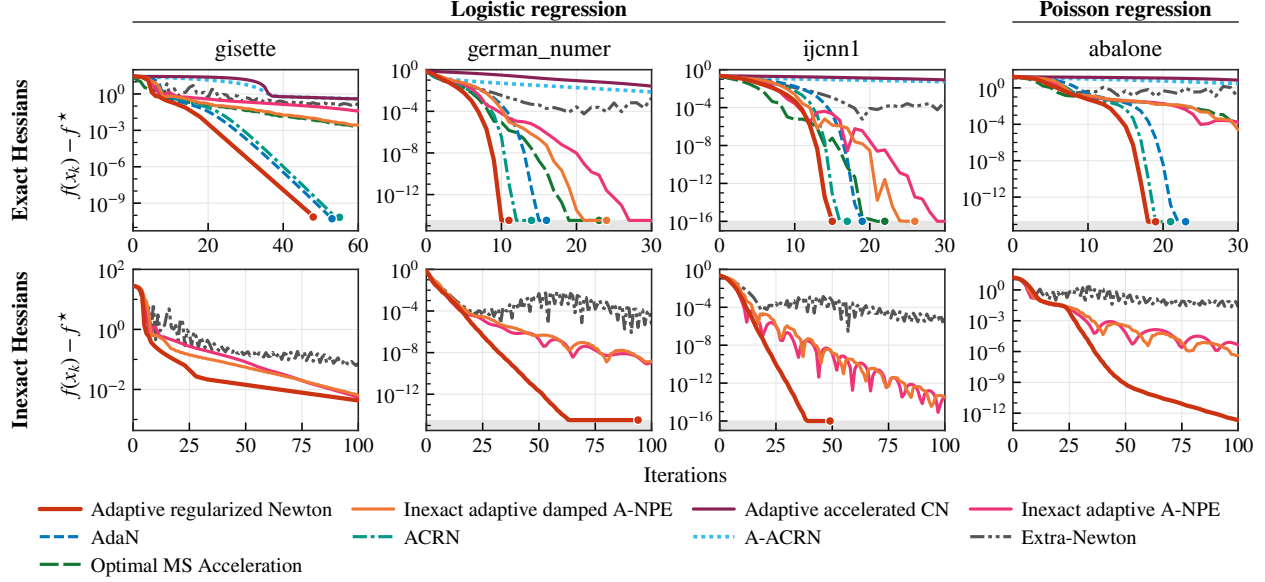


Figure 1: Top row: exact Hessians. Bottom row: Hessians computed on a fixed random 10% of the samples, with exact gradients.

## 5 Experiments

We compare our methods with adaptive second-order baselines on logistic and Poisson regression (Figure 1). We use datasets from LIBSVM (Chang & Lin, 2011) `gisette`, `german_number` and `ijcnn1` for logistic regression, and `abalone` for Poisson regression. Methods start from  $x_0 = -\mathbf{1}$ . We run Algorithms 2, 3, 5 and 6 against AdaN (Mishchenko, 2023), the adaptive cubic regularized Newton (ACRN), its accelerated version (A-ACRN) (Kamzolov et al., 2025b), Extra-Newton (Antonakopoulos et al., 2022), and Optimal MS Acceleration (Carmon et al., 2022)

**Parameters.** All methods are adaptive, so we do not tune them. For a fair comparison, Algorithms 2, 3 and 6 start backtracking with  $\eta_0 = \sqrt{L(x_0)\|\nabla f(x_0)\|}$ , Optimal MS Acceleration takes  $\lambda'_0 = \eta_0$ ,  $\sigma = 1/2$ , and AdaN takes  $H_0 = L(x_0)/2$ , which it doubles before its first trial. For the step-size parameters of Extra-Newton, we take  $\gamma = 2\|x_0\|$  as a proxy for the diameter  $D$ , following Antonakopoulos et al. (2022, Remark 3.1), and  $\beta_0 = L(x_0)\gamma^2\|\nabla f(x_0)\|$ . Thus, these six methods perform the same first trial step. ACRN, A-ACRN and Algorithm 5 start from the standard theoretical choice of the cubic regularization parameter,  $\mathcal{M}_0 = L(x_0)$ .

**Inexact Hessians.** We consider a centralized distributed setup with  $f(x) = \frac{1}{m} \sum_{j=1}^m f_j(x)$ , where  $f_j(x)$  is the local loss stored on machine  $j$ . The server also stores local data and computes the Hessian  $H(x) = \nabla^2 f_1(x)$ . Under the  $\beta$ -similarity this Hessian satisfies (IH) with  $\delta = \beta$  (Zhang & Lin, 2015; Agafonov et al., 2021). To perform a step, the server collects the gradients from the workers and uses the full gradient with its local Hessian. In our experiments,  $H(x)$  is computed from a fixed uniformly sampled subset of  $\lceil n/10 \rceil$  observations, where  $n$  is the number of training samples. We compare only parameter-free methods that allow inexact Hessians.

**Results.** Under exact Hessians, our Adaptive regularized Newton converges faster on all four problems, with ACRN close behind. Among the accelerated methods, our A-NPE methods (Algorithms 3 and 6) perform best, and the optimal Algorithm 3 usually outperforms the suboptimal Algorithm 6. As in the logistic regression experiments of Carmon et al. (2022), the non-accelerated methods outperform the accelerated ones. However, on toy problems satisfying (LH), optimal acceleration

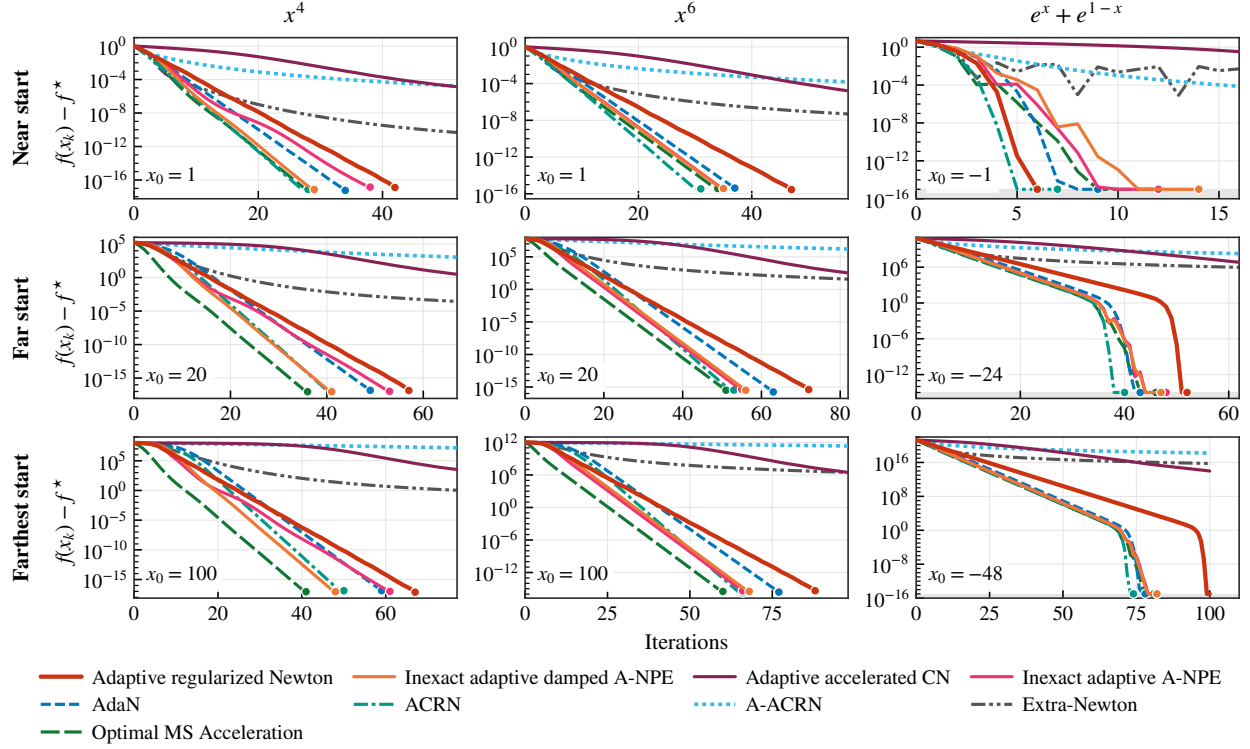


Figure 2: One-dimensional functions with exact Hessians, from the nearest start (top row) to the farthest (bottom row). Each panel ends ten iterations after Algorithm 2 stops.

outperforms non-accelerated methods and suboptimal accelerated ones when starting far from the solution (see below). With inexact Hessians, our three methods outperform Extra-Newton, and Algorithm 2 is again the best. Experimental details and additional results are provided in Appendix G.

**One-dimensional functions satisfying (LH).** We also run all the methods with exact Hessians on three one-dimensional functions from Gorbunov et al. (2025) and Tyurin (2026) (Figure 2):  $x^4$ ,  $x^6$  and  $e^x + e^{1-x}$ . Up to scaling and a shift,  $x^4$  and  $e^x + e^{1-x}$  are Examples 2.4 and 2.2, which satisfy (LH) but not (GH). We start  $x^4$  and  $x^6$  from  $x_0 \in \{1, 20, 100\}$  and  $e^x + e^{1-x}$  from  $x_0 \in \{-1, -24, -48\}$ , compute derivatives in closed form, and set all parameters from  $L(x_0) = M_0 + M_1|f'(x_0)|$  by the rules above, where (TD) holds with  $(M_0, M_1) = (16, 2)$ ,  $(48, 12)$  and  $(0, 1)$ , respectively. Every run has 100 iterations.

Six methods converge from every start: Algorithms 2, 3 and 6, AdaN, ACRN and Optimal MS Acceleration (Carmon et al., 2022). On  $x^4$  and  $x^6$  they converge linearly, as expected at a degenerate minimizer. On  $e^x + e^{1-x}$  they converge in a few iterations once close to  $x^*$ , after a far phase whose length grows linearly with  $|x_0 - x^*|$ . Unlike in Figure 1, acceleration pays off here: from far starts, the accelerated Monteiro–Svaiter-type methods, Algorithms 3 and 6 and Optimal MS Acceleration, converge faster than the non-accelerated Algorithm 2.

## References

Deeksha Adil, Brian Bullins, Arun Jambulapati, and Aaron Sidford. Convex optimization with  $\$p\$-norm oracles. In *37th International Conference on Algorithmic Learning Theory*, 2026. URL$

<https://openreview.net/forum?id=gZax9QXNyR>.

Artem Agafonov, Pavel Dvurechensky, Gesualdo Scutari, Alexander Gasnikov, Dmitry Kamzolov, Aleksandr Lukashevich, and Amir Daneshmand. An accelerated second-order method for distributed stochastic optimization. In *2021 60th IEEE Conference on Decision and Control (CDC)*, pp. 2407–2413, 2021. ISBN 2576-2370. doi: 10.1109/CDC45484.2021.9683400. URL <https://doi.org/10.1109/CDC45484.2021.9683400>.

Artem Agafonov, Dmitry Kamzolov, Pavel Dvurechensky, Alexander Gasnikov, and Martin Takáč. Inexact tensor methods and their application to stochastic convex optimization. *Optimization Methods and Software*, 39(1):42–83, 2024a. doi: 10.1080/10556788.2023.2261604. URL <https://doi.org/10.1080/10556788.2023.2261604>.

Artem Agafonov, Dmitry Kamzolov, Alexander Gasnikov, Ali Kavis, Kimon Antonakopoulos, Volkan Cevher, and Martin Takáč. Advancing the lower bounds: an accelerated, stochastic, second-order method with optimal adaptation to inexactness. In *The Twelfth International Conference on Learning Representations*, 2024b. URL <https://openreview.net/forum?id=otU31x3fus>.

Kimon Antonakopoulos, Ali Kavis, and Volkan Cevher. Extra-Newton: A first approach to noise-adaptive accelerated second-order methods. In S Koyejo, S Mohamed, A Agarwal, D Belgrave, K Cho, and A Oh (eds.), *Advances in Neural Information Processing Systems*, volume 35, pp. 29859–29872. Curran Associates, Inc., 2022. URL [https://proceedings.neurips.cc/paper\\_files/paper/2022/file/c10804702be5a0cca89331315413f1a2-Paper-Conference.pdf](https://proceedings.neurips.cc/paper_files/paper/2022/file/c10804702be5a0cca89331315413f1a2-Paper-Conference.pdf).

Yossi Arjevani, Ohad Shamir, and Ron Shiff. Oracle complexity of second-order methods for smooth convex optimization. *Mathematical Programming*, 178:327–360, 2019. ISSN 1436-4646. doi: 10.1007/s10107-018-1293-1. URL <https://doi.org/10.1007/s10107-018-1293-1>.

S Bellavia, G Gurioli, B Morini, and Ph.L. Toint. Adaptive regularization for nonconvex optimization using inexact function values and randomly perturbed derivatives. *Journal of Complexity*, 68: 101591, 2022. ISSN 0885-064X. doi: <https://doi.org/10.1016/j.jco.2021.101591>. URL <https://www.sciencedirect.com/science/article/pii/S0885064X21000467>.

Stefania Bellavia, Nataša Krejić, and Nataša Krklec Jerinkić. Subsampled inexact Newton methods for minimizing large sums of convex functions. *IMA Journal of Numerical Analysis*, 40(4): 2309–2341, 10 2020. ISSN 0272-4979. doi: 10.1093/imanum/drz027. URL <https://doi.org/10.1093/imanum/drz027>.

Ekaterina Borodich and Dmitry Kovalev. Nesterov finds GRAAL: Optimal and adaptive gradient method for convex optimization. In C. Vondrick, B. Hariharan, C. Raffel, L. Pinto, D. Yang, and A. Faust (eds.), *International Conference on Learning Representations*, volume 2026, pp. 95533–95558, 2026. URL [https://proceedings.iclr.cc/paper\\_files/paper/2026/file/9a419ce12db8db2e35ee68c1f58c1a36-Paper-Conference.pdf](https://proceedings.iclr.cc/paper_files/paper/2026/file/9a419ce12db8db2e35ee68c1f58c1a36-Paper-Conference.pdf).

Yair Carmon, Danielle Hausler, Arun Jambulapati, Yujia Jin, and Aaron Sidford. Optimal and adaptive Monteiro-Svaiter acceleration. In S Koyejo, S Mohamed, A Agarwal, D Belgrave, K Cho, and A Oh (eds.), *Advances in Neural Information Processing Systems*, volume 35, pp. 20338–20350. Curran Associates, Inc., 2022. URL [https://proceedings.neurips.cc/paper\\_files/paper/2022/file/7ff97417474268e6b5a38bcbfae04944-Paper-Conference.pdf](https://proceedings.neurips.cc/paper_files/paper/2022/file/7ff97417474268e6b5a38bcbfae04944-Paper-Conference.pdf).

- Coralia Cartis, Nicholas I M Gould, and Philippe L Toint. Adaptive cubic regularisation methods for unconstrained optimization. Part I: motivation, convergence and numerical results. *Mathematical Programming*, 127:245–295, 2011. ISSN 1436-4646. doi: 10.1007/s10107-009-0286-5. URL <https://doi.org/10.1007/s10107-009-0286-5>.
- Chih-Chung Chang and Chih-Jen Lin. LIBSVM: A library for support vector machines. *ACM Trans. Intell. Syst. Technol.*, 2(3), May 2011. ISSN 2157-6904. doi: 10.1145/1961189.1961199. URL <https://doi.org/10.1145/1961189.1961199>.
- Lesi Chen, Chengchang Liu, Luo Luo, John Lui, and Jingzhao Zhang. Optimal convex optimization with inexact second-order oracles. *arXiv preprint arXiv:2607.24520*, 2026.
- Xi Chen, Bo Jiang, Tianyi Lin, and Shuzhong Zhang. Accelerating adaptive cubic regularization of Newton’s method via random sampling. *Journal of Machine Learning Research*, 23(90):1–38, 2022. URL <http://jmlr.org/papers/v23/20-910.html>.
- Ziyi Chen, Yi Zhou, Yingbin Liang, and Zhaosong Lu. Generalized-smooth nonconvex optimization is as efficient as smooth nonconvex optimization. In Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett (eds.), *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 5396–5427. PMLR, 23–29 Jul 2023. URL <https://proceedings.mlr.press/v202/chen23ar.html>.
- Michael Crawshaw, Mingrui Liu, Francesco Orabona, Wei Zhang, and Zhenxun Zhuang. Robustness to unbounded smoothness of generalized SignSGD. In S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh (eds.), *Advances in Neural Information Processing Systems*, volume 35, pp. 9955–9968. Curran Associates, Inc., 2022. doi: 10.52202/068431-0723. URL [https://proceedings.neurips.cc/paper\\_files/paper/2022/file/40924475a9bf768bdac3725e67745283-Paper-Conference.pdf](https://proceedings.neurips.cc/paper_files/paper/2022/file/40924475a9bf768bdac3725e67745283-Paper-Conference.pdf).
- Nikita Doikov and Yurii Nesterov. Gradient regularization of Newton method with Bregman distances. *Mathematical Programming*, 2023. ISSN 1436-4646. doi: 10.1007/s10107-023-01943-7. URL <https://doi.org/10.1007/s10107-023-01943-7>.
- Nikita Doikov, El Mahdi Chayti, and Martin Jaggi. Second-order optimization with lazy Hessians. In Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett (eds.), *Proceedings of the 40th International Conference on Machine Learning*, volume 202, pp. 8138–8161. PMLR, 9 2023. URL <https://proceedings.mlr.press/v202/doikov23a.html>.
- Nikita Doikov, Konstantin Mishchenko, and Yurii Nesterov. Super-Universal regularized Newton method. *SIAM Journal on Optimization*, 34:27–56, 2024. doi: 10.1137/22M1519444. URL <https://doi.org/10.1137/22M1519444>.
- Shaoying Fang, Naoki Marumo, and Akiko Takeda. Parameter-free cubic-regularized newton method: Sharp complexity and generalized smoothness. *arXiv preprint arXiv:2607.10741*, 2026.
- Saeed Ghadimi, Han Liu, and Tong Zhang. Second-order methods with cubic regularization under inexact information. *arXiv preprint arXiv:1710.05782*, 2017.
- Eduard Gorbunov, Nazarii Tupitsa, Sayantan Choudhury, Alen Aliev, Peter Richtarik, Samuel Horváth, and Martin Takáč. Methods for convex  $(L_0, L_1)$ -smooth optimization:

- Clipping, acceleration, and adaptivity. In Y. Yue, A. Garg, N. Peng, F. Sha, and R. Yu (eds.), *International Conference on Learning Representations*, volume 2025, pp. 97261–97305, 2025. URL [https://proceedings.iclr.cc/paper\\_files/paper/2025/file/f21a5fedfbc55406ddaf51cf90d494a-Paper-Conference.pdf](https://proceedings.iclr.cc/paper_files/paper/2025/file/f21a5fedfbc55406ddaf51cf90d494a-Paper-Conference.pdf).
- G. N. Grapiglia and Yu. Nesterov. Tensor methods for minimizing convex functions with Hölder continuous higher-order derivatives. *SIAM Journal on Optimization*, 30(4):2750–2779, 2020. doi: 10.1137/19M1259432. URL <https://doi.org/10.1137/19M1259432>.
- Sadok Jerad. Fast adaptive tensor methods under local smoothness. *arXiv preprint arXiv:2606.30225*, 2026.
- Dmitry Kamzolov, Artem Agafonov, Dmitry Pasechnyuk, Alexander Gasnikov, and Martin Takáč. OPTAMI: Global superlinear convergence of high-order methods. In *The Thirteenth International Conference on Learning Representations*, 2025a. URL <https://openreview.net/forum?id=Cpr6Wv2tfr>.
- Dmitry Kamzolov, Klea Ziu, Artem Agafonov, and Martin Takáč. Accelerated adaptive cubic regularized quasi-Newton methods. *Journal of Optimization Theory and Applications*, 208:27, 2025b. ISSN 1573-2878. doi: 10.1007/s10957-025-02804-3. URL <https://doi.org/10.1007/s10957-025-02804-3>.
- Jonas Moritz Kohler and Aurelien Lucchi. Sub-sampled cubic regularization for non-convex optimization. In Doina Precup and Yee Whye Teh (eds.), *Proceedings of the 34th International Conference on Machine Learning*, volume 70, pp. 1895–1904. PMLR, 5 2017. URL <https://proceedings.mlr.press/v70/kohler17a.html>.
- Anastasia Koloskova, Hadrien Hendrikx, and Sebastian U Stich. Revisiting gradient clipping: Stochastic bias and tight convergence guarantees. In Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett (eds.), *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 17343–17363. PMLR, 23–29 Jul 2023. URL <https://proceedings.mlr.press/v202/koloskova23a.html>.
- Dmitry Kovalev and Alexander Gasnikov. The first optimal acceleration of high-order methods in smooth convex optimization. In S Koyejo, S Mohamed, A Agarwal, D Belgrave, K Cho, and A Oh (eds.), *Advances in Neural Information Processing Systems*, volume 35, pp. 35339–35351. Curran Associates, Inc., 2022. URL [https://proceedings.neurips.cc/paper\\_files/paper/2022/file/e56f394bbd4f0ec81393d767caa5a31b-Paper-Conference.pdf](https://proceedings.neurips.cc/paper_files/paper/2022/file/e56f394bbd4f0ec81393d767caa5a31b-Paper-Conference.pdf).
- Haochuan Li, Jian Qian, Yi Tian, Alexander Rakhlin, and Ali Jadbabaie. Convex and non-convex optimization under generalized smoothness. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine (eds.), *Advances in Neural Information Processing Systems*, volume 36, pp. 40238–40271. Curran Associates, Inc., 2023. doi: 10.52202/075280-1749. URL [https://proceedings.neurips.cc/paper\\_files/paper/2023/file/7e8bb8d17bb1cb24dfe972a2f8ff2500-Paper-Conference.pdf](https://proceedings.neurips.cc/paper_files/paper/2023/file/7e8bb8d17bb1cb24dfe972a2f8ff2500-Paper-Conference.pdf).
- Konstantin Mishchenko. Regularized Newton method with global  $\mathcal{O}\left(\frac{1}{k^2}\right)$  convergence. *SIAM Journal on Optimization*, 33:1440–1462, 2023. doi: 10.1137/22M1488752. URL <https://doi.org/10.1137/22M1488752>.

- Renato D C Monteiro and B F Svaiter. An accelerated hybrid proximal extragradient method for convex optimization and its implications to second-order methods. *SIAM Journal on Optimization*, 23:1092–1125, 2013. doi: 10.1137/110833786. URL <https://doi.org/10.1137/110833786>.
- Yurii Nesterov. Accelerating the cubic regularization of Newton’s method on convex problems. *Mathematical Programming*, 112:159–181, 2008. ISSN 1436-4646. doi: 10.1007/s10107-006-0089-x. URL <https://doi.org/10.1007/s10107-006-0089-x>.
- Yurii Nesterov. *Lectures on Convex Optimization*. Springer Cham, 2 edition, 2018. ISBN 978-3-319-91577-7. doi: 10.1007/978-3-319-91578-4.
- Yurii Nesterov and Boris T Polyak. Cubic regularization of Newton method and its global performance. *Mathematical Programming*, 108:177–205, 2006. doi: 10.1007/s10107-006-0706-8. URL <https://doi.org/10.1007/s10107-006-0706-8>.
- Andrei Semenov, Martin Jaggi, and Nikita Doikov. Gradient-normalized smoothness for optimization with approximate Hessians. In C. Vondrick, B. Hariharan, C. Raffel, L. Pinto, D. Yang, and A. Faust (eds.), *International Conference on Learning Representations*, volume 2026, pp. 13974–14029, 2026. URL [https://proceedings.iclr.cc/paper\\_files/paper/2026/file/1759a83b007f0685c3fbc460fa1b6395-Paper-Conference.pdf](https://proceedings.iclr.cc/paper_files/paper/2026/file/1759a83b007f0685c3fbc460fa1b6395-Paper-Conference.pdf).
- Alexander Tyurin. Toward a unified theory of gradient descent under generalized smoothness. In Aarti Singh, Maryam Fazel, Daniel Hsu, Simon Lacoste-Julien, Felix Berkenkamp, Tegan Maharaj, Kiri Wagstaff, and Jerry Zhu (eds.), *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 60493–60514. PMLR, 13–19 Jul 2025. URL <https://proceedings.mlr.press/v267/tyurin25a.html>.
- Alexander Tyurin. Near-optimal convergence of accelerated gradient methods under generalized and  $(L_0, L_1)$ -smoothness. In *Forty-third International Conference on Machine Learning*, 2026. URL <https://openreview.net/forum?id=LOT4pmqkMg>.
- Daniil Vankov, Anton Rodomanov, Angelia Nedich, Lalitha Sankar, and Sebastian Stich. Optimizing  $(L_0, L_1)$ -smooth functions by gradient methods. In Y. Yue, A. Garg, N. Peng, F. Sha, and R. Yu (eds.), *International Conference on Learning Representations*, volume 2025, pp. 15953–15979, 2025. URL [https://proceedings.iclr.cc/paper\\_files/paper/2025/file/28a9255e2a0e7062e8e0af715f2e5b86-Paper-Conference.pdf](https://proceedings.iclr.cc/paper_files/paper/2025/file/28a9255e2a0e7062e8e0af715f2e5b86-Paper-Conference.pdf).
- Chenghan Xie, Chenxi Li, Chuwen Zhang, Qi Deng, Dongdong Ge, and Yinyu Ye. Trust region methods for nonconvex stochastic optimization beyond Lipschitz smoothness. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pp. 16049–16057, 2024.
- Peng Xu, Fred Roosta, and Michael W Mahoney. Newton-type methods for non-convex optimization under inexact Hessian information. *Mathematical Programming*, 184:35–70, 2020. ISSN 1436-4646. doi: 10.1007/s10107-019-01405-z. URL <https://doi.org/10.1007/s10107-019-01405-z>.
- Jingzhao Zhang, Tianxing He, Suvrit Sra, and Ali Jadbabaie. Why gradient clipping accelerates training: A theoretical justification for adaptivity. In *International Conference on Learning Representations*, 2020. URL <https://openreview.net/forum?id=BJgnXpVYwS>.
- Yuchen Zhang and Xiao Lin. DiSCO: Distributed optimization for self-concordant empirical loss. In Francis Bach and David Blei (eds.), *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pp. 362–370, Lille, France, 07–09 Jul 2015. PMLR. URL <https://proceedings.mlr.press/v37/zhangb15.html>.

## Contents

|                                                                    |           |
|--------------------------------------------------------------------|-----------|
| <b>A Taylor Remainder Lemmas</b>                                   | <b>19</b> |
| <b>B Proofs for the Relations between Assumptions and Examples</b> | <b>19</b> |
| <b>C Proof of Algorithm 2</b>                                      | <b>23</b> |
| C.1 Step Estimates . . . . .                                       | 23        |
| C.2 Iteration Count . . . . .                                      | 27        |
| C.3 Proof of Theorem 3.1 . . . . .                                 | 29        |
| C.4 Far and Near Convergence Rates . . . . .                       | 31        |
| C.5 Oracle complexity . . . . .                                    | 34        |
| C.6 Strongly Convex Analysis . . . . .                             | 36        |
| <b>D Nesterov-type Acceleration</b>                                | <b>44</b> |
| D.1 Proof of Nesterov-type Acceleration under (GH) . . . . .       | 46        |
| D.2 Adaptive Method under (LH) with Exact Hessian . . . . .        | 54        |
| <b>E Suboptimal A-NPE Method</b>                                   | <b>61</b> |
| E.1 Auxiliary Lemmas . . . . .                                     | 62        |
| E.2 Step Estimates . . . . .                                       | 64        |
| E.3 Explicit Choice of Regularization Parameter . . . . .          | 66        |
| E.4 Main-Loop Backtracking . . . . .                               | 68        |
| E.5 Proof of Theorem E.1 . . . . .                                 | 69        |
| E.6 Oracle Complexity . . . . .                                    | 73        |
| <b>F Proof of Algorithm 3</b>                                      | <b>73</b> |
| F.1 Initialization and Backtracking . . . . .                      | 73        |
| F.2 Step Estimates . . . . .                                       | 74        |
| F.3 Proof of Theorem 4.1 . . . . .                                 | 76        |
| F.4 Oracle Complexity . . . . .                                    | 82        |
| <b>G Extra Experiments</b>                                         | <b>85</b> |
| G.1 Experiment Details . . . . .                                   | 85        |
| G.2 Sensitivity to Initial Parameters . . . . .                    | 85        |

## A Taylor Remainder Lemmas

The following two lemmas record the Taylor remainder bounds used throughout our analysis.

**Lemma A.1.** *Suppose that (GH) holds. Then, for all  $x, s \in \mathbb{R}^d$ ,*

$$\begin{aligned} \|\nabla f(x+s) - \nabla f(x) - \nabla^2 f(x)s\| &\leq \frac{L(x)}{2} \|s\|^2, \\ \left| f(x+s) - f(x) - \langle \nabla f(x), s \rangle - \frac{1}{2} \langle \nabla^2 f(x)s, s \rangle \right| &\leq \frac{L(x)}{6} \|s\|^3. \end{aligned}$$

**Lemma A.2.** *Suppose that (LH) holds. Then, for all  $x, s \in \mathbb{R}^d$  satisfying  $\|s\| \leq \rho$ ,*

$$\begin{aligned} \|\nabla f(x+s) - \nabla f(x) - \nabla^2 f(x)s\| &\leq \frac{L(x)}{2} \|s\|^2, \\ \left| f(x+s) - f(x) - \langle \nabla f(x), s \rangle - \frac{1}{2} \langle \nabla^2 f(x)s, s \rangle \right| &\leq \frac{L(x)}{6} \|s\|^3. \end{aligned}$$

*Proof.* We follow [Nesterov \(2018, Lemma 1.2.4\)](#). For  $t \in [0, 1]$ , (GH), or (LH) with  $\|ts\| \leq \rho$ , gives  $\|\nabla^2 f(x+ts) - \nabla^2 f(x)\| \leq tL(x)\|s\|$ . Hence,

$$\begin{aligned} \|\nabla f(x+s) - \nabla f(x) - \nabla^2 f(x)s\| &= \left\| \int_0^1 \left( \nabla^2 f(x+ts) - \nabla^2 f(x) \right) s dt \right\| \\ &\leq \int_0^1 tL(x)\|s\|^2 dt = \frac{L(x)}{2} \|s\|^2. \end{aligned}$$

Since  $\|ts\| \leq \|s\|$ , the same bound holds for the step  $ts$ :

$$\|\nabla f(x+ts) - \nabla f(x) - t\nabla^2 f(x)s\| \leq \frac{L(x)}{2} t^2 \|s\|^2, \quad t \in [0, 1].$$

Using  $f(x+s) - f(x) = \int_0^1 \langle \nabla f(x+ts), s \rangle dt$  and  $\int_0^1 t dt = \frac{1}{2}$ , we obtain

$$\begin{aligned} &\left| f(x+s) - f(x) - \langle \nabla f(x), s \rangle - \frac{1}{2} \langle \nabla^2 f(x)s, s \rangle \right| \\ &= \left| \int_0^1 \langle \nabla f(x+ts) - \nabla f(x) - t\nabla^2 f(x)s, s \rangle dt \right| \leq \int_0^1 \frac{L(x)}{2} t^2 \|s\|^3 dt = \frac{L(x)}{6} \|s\|^3. \end{aligned}$$

□

## B Proofs for the Relations between Assumptions and Examples

*Proof of Example 2.1.* Direct differentiation gives

$$f'(x) = 2x + \int_0^x \sin(t^2) dt, \quad f''(x) = 2 + \sin(x^2), \quad f^{(3)}(x) = 2x \cos(x^2).$$

Thus,  $f''(x) \geq 1$ , so  $f$  is 1-strongly convex. Moreover,  $f'(0) = 0$  and

$$|f'(x)| = \left| \int_0^x f''(t) dt \right| \geq |x|.$$

If  $|y - x| \leq 1$ , then

$$|f''(y) - f''(x)| = |\sin(y^2) - \sin(x^2)| \leq |y^2 - x^2| \leq (2|x| + 1)|y - x| \leq (2 + 2|f'(x)|)|y - x|.$$

If  $|y - x| > 1$ , then

$$|f''(y) - f''(x)| \leq 2 \leq 2|y - x| \leq (2 + 2|f'(x)|) |y - x|.$$

Hence, (GH) holds with  $M_0 = M_1 = 2$ . On the other hand, for  $x_n = \sqrt{2\pi n}$ ,

$$|f^{(3)}(x_n)| = 2\sqrt{2\pi n} \rightarrow +\infty.$$

Condition (H) would imply  $|f^{(3)}(x)| \leq L$  by taking the directional limit in (H). Therefore, (H) does not hold.  $\square$

*Proof of Example 2.2.* We have

$$f'(x) = \sinh x, \quad f''(x) = \cosh x, \quad f^{(3)}(x) = \sinh x.$$

Since  $f''(x) \geq 1$ , the function is 1-strongly convex. Let  $h = y - x$  and  $|h| \leq \rho$ . For every  $t \in [0, 1]$ ,

$$\begin{aligned} |\sinh(x + th)| &\leq |\sinh x| \cosh(th) + \cosh x |\sinh(th)| \\ &\leq |\sinh x| \cosh \rho + \cosh x \sinh \rho \leq \sinh \rho + e^\rho |\sinh x|, \end{aligned}$$

where we used  $\cosh x \leq 1 + |\sinh x|$ . Therefore,

$$|f''(y) - f''(x)| \leq |h| \int_0^1 |\sinh(x + th)| dt \leq (\sinh \rho + e^\rho |f'(x)|) |y - x|.$$

This proves (LH). If (GH) held, then the choice  $x = 0$  would give

$$\cosh y - 1 \leq M_0 |y|, \quad y \in \mathbb{R},$$

which is impossible as  $y \rightarrow +\infty$ .  $\square$

*Proof of Example 2.3.* We have

$$f'(x) = e^x - 1, \quad f''(x) = f^{(3)}(x) = e^x.$$

Let  $h = y - x$  and  $|h| \leq \rho$ . The mean value theorem gives

$$|f''(y) - f''(x)| = e^x |e^h - 1| \leq e^\rho e^x |h| \leq e^\rho (1 + |f'(x)|) |y - x|.$$

Hence, (LH) holds with  $M_0 = M_1 = e^\rho$ . If (GH) held, then the choice  $x = 0$  would give

$$e^y - 1 \leq M_0 |y|, \quad y \in \mathbb{R},$$

which is impossible as  $y \rightarrow +\infty$ .  $\square$

*Proof of Example 2.4.* We have

$$\nabla f(x) = (\mu + \|x\|^2) x, \quad \nabla^2 f(x) = (\mu + \|x\|^2) I + 2xx^\top.$$

Hence,  $\nabla^2 f(x) \succeq \mu I$ . Let  $s = y - x$  and  $r = \|s\| \leq \rho$ . Using

$$yy^\top - xx^\top = xs^\top + sx^\top + ss^\top,$$

we obtain

$$\begin{aligned}\|\nabla^2 f(y) - \nabla^2 f(x)\| &\leq \left| \|y\|^2 - \|x\|^2 \right| + 2\|yy^\top - xx^\top\| \\ &\leq (2\|x\| + r)r + 2(2\|x\|r + r^2) = (6\|x\| + 3r)r.\end{aligned}$$

For every  $t \geq 0$ ,

$$4 + 2t^3 - 6t = 2(t-1)^2(t+2) \geq 0.$$

Since  $\|\nabla f(x)\| = (\mu + \|x\|^2)\|x\| \geq \|x\|^3$ , we have

$$6\|x\| \leq 4 + 2\|\nabla f(x)\|.$$

Consequently,

$$\|\nabla^2 f(y) - \nabla^2 f(x)\| \leq (3\rho + 4 + 2\|\nabla f(x)\|)\|y - x\|,$$

which proves (LH) with the stated constants. At  $x = 0$ ,

$$\|\nabla^2 f(y) - \nabla^2 f(0)\| = 3\|y\|^2, \quad \nabla f(0) = 0.$$

Thus, (GH) would require  $3\|y\|^2 \leq M_0\|y\|$  for all  $y \in \mathbb{R}^d$ , which is impossible.  $\square$

*Proof of Example 2.5.* Let  $\mu_i(x) := e^{\langle a_i, x \rangle}$ . We have

$$\nabla f(x) = \sum_{i=1}^n (\mu_i(x) - b_i) a_i, \quad \nabla^2 f(x) = \sum_{i=1}^n \mu_i(x) a_i a_i^\top \succeq 0.$$

Since  $[a_i]_1 = 1$  and  $b_i \geq 0$ , the first coordinate of  $\nabla f(x)$  equals  $\sum_{i=1}^n \mu_i(x) - \|b\|_1$ . Hence,

$$\sum_{i=1}^n \mu_i(x) \leq \|b\|_1 + \|\nabla f(x)\|. \quad (10)$$

Let  $s = y - x$  and  $\|s\| \leq \rho$ . Then  $|\langle a_i, s \rangle| \leq c\|s\| \leq c\rho$ . Using  $|e^u - 1| \leq e^{|u|}|u|$  for  $u \in \mathbb{R}$ , we obtain

$$\begin{aligned}\|\nabla^2 f(y) - \nabla^2 f(x)\| &= \left\| \sum_{i=1}^n \mu_i(x) (e^{\langle a_i, s \rangle} - 1) a_i a_i^\top \right\| \leq c^3 e^{c\rho} \|s\| \sum_{i=1}^n \mu_i(x) \\ &\stackrel{(10)}{\leq} \left( \|b\|_1 c^3 e^{c\rho} + c^3 e^{c\rho} \|\nabla f(x)\| \right) \|y - x\|.\end{aligned}$$

This proves (LH) with the stated constants. Let  $e_1$  be the first coordinate vector. Since  $\langle a_i, e_1 \rangle = 1$ , we have  $\nabla^2 f(x + te_1) = e^t \nabla^2 f(x)$  for  $t \in \mathbb{R}$  and

$$\|\nabla^2 f(x)\| \geq \left\langle \nabla^2 f(x) e_1, e_1 \right\rangle = \sum_{i=1}^n \mu_i(x) > 0.$$

If (GH) held, then, for all  $t > 0$ ,

$$(e^t - 1) \|\nabla^2 f(x)\| = \|\nabla^2 f(x + te_1) - \nabla^2 f(x)\| \leq (M_0 + M_1 \|\nabla f(x)\|) t,$$

which is impossible as  $t \rightarrow +\infty$ .  $\square$

*Proof of Example 2.6.* Let  $a_i^\top$  be the rows of  $A$ ,  $r := Ax - b$ , and  $\psi(u) := |u|^{p-2}u$  for  $u \in \mathbb{R}$ . We have

$$\nabla f(x) = \sum_{i=1}^n \psi(r_i) a_i, \quad \nabla^2 f(x) = (p-1) \sum_{i=1}^n |r_i|^{p-2} a_i a_i^\top \succeq 0.$$

First, we bound  $\|r\|_p$  by  $\|\nabla f(x)\|$ . Let  $P$  be the orthogonal projector onto the range of  $A^\top$ . The gradient  $\nabla f(x) = A^\top \psi(r)$  belongs to this range, and  $\|Az\| \geq \sigma_+(A)\|z\|$  for all  $z$  in this range. Therefore, Hölder's inequality and  $\|v\| \leq \sqrt{n}\|v\|_\infty \leq \sqrt{n}\|v\|_p$  for  $v \in \mathbb{R}^n$  give

$$\begin{aligned} \|r\|_p^p - \|r\|_p^{p-1} \|b\|_p &\leq \sum_{i=1}^n \psi(r_i) (r_i + b_i) = \langle \nabla f(x), x \rangle = \langle \nabla f(x), Px \rangle \leq \frac{\|\nabla f(x)\| \|APx\|}{\sigma_+(A)} \\ &= \frac{\|\nabla f(x)\| \|r + b\|}{\sigma_+(A)} \leq \frac{\sqrt{n}}{\sigma_+(A)} \|\nabla f(x)\| (\|r\|_p + \|b\|_p). \end{aligned}$$

If  $\|r\|_p > \max\{1, 2\|b\|_p\}$ , then  $\|r\|_p - \|b\|_p > \|r\|_p/2$  and  $\|r\|_p + \|b\|_p < 3\|r\|_p/2$ . In this case, the previous display gives  $\|r\|_p^{p-1} < \frac{3\sqrt{n}}{\sigma_+(A)} \|\nabla f(x)\| = \frac{M_1}{M_0} \|\nabla f(x)\|$ , and  $\|r\|_p > 1$  implies

$$\begin{aligned} (\|r\|_p + \|A\|\rho)^{p-3} &\leq (1 + \|A\|\rho)^{p-3} \|r\|_p^{p-3} \leq (1 + \|A\|\rho)^{p-3} \|r\|_p^{p-1} \\ &< (1 + 2\|b\|_p + \|A\|\rho)^{p-3} \frac{M_1}{M_0} \|\nabla f(x)\|. \end{aligned}$$

Otherwise,  $\|r\|_p \leq 1 + 2\|b\|_p$  and  $(\|r\|_p + \|A\|\rho)^{p-3} \leq (1 + 2\|b\|_p + \|A\|\rho)^{p-3}$ . In both cases, the definition of  $M_0$  gives

$$(p-1)(p-2)\|A\|^3 (\|r\|_p + \|A\|\rho)^{p-3} \leq M_0 + M_1 \|\nabla f(x)\|. \quad (11)$$

Now let  $s = y - x$  and  $\|s\| \leq \rho$ . Then  $Ay - b = r + As$  and  $|\langle a_i, s \rangle| \leq \|A\| \|s\| \leq \|A\| \rho$ . By the mean value theorem and  $|r_i| \leq \|r\|_p$ ,

$$\begin{aligned} \left| |r_i + \langle a_i, s \rangle|^{p-2} - |r_i|^{p-2} \right| &\leq (p-2) (|r_i| + |\langle a_i, s \rangle|)^{p-3} |\langle a_i, s \rangle| \\ &\leq (p-2) (\|r\|_p + \|A\|\rho)^{p-3} \|A\| \|s\|. \end{aligned}$$

Since  $\|\sum_{i=1}^n w_i a_i a_i^\top\| \leq \max_i |w_i| \|A^\top A\|$  for  $w \in \mathbb{R}^n$ , we obtain

$$\begin{aligned} \|\nabla^2 f(y) - \nabla^2 f(x)\| &= (p-1) \left\| \sum_{i=1}^n \left( |r_i + \langle a_i, s \rangle|^{p-2} - |r_i|^{p-2} \right) a_i a_i^\top \right\| \\ &\leq (p-1)(p-2)\|A\|^3 (\|r\|_p + \|A\|\rho)^{p-3} \|s\| \stackrel{(11)}{\leq} (M_0 + M_1 \|\nabla f(x)\|) \|y - x\|. \end{aligned}$$

This proves (LH). Since  $A \neq 0$ , there is  $j$  with  $a_j \neq 0$ . For  $t \geq |b_j|/\|a_j\|^2$ ,

$$\|\nabla^2 f(ta_j)\| \|a_j\|^2 \geq \left\langle \nabla^2 f(ta_j) a_j, a_j \right\rangle \geq (p-1) (t\|a_j\|^2 - |b_j|)^{p-2} \|a_j\|^4.$$

If (GH) held, then the choice  $x = 0$  and  $y = ta_j$  would give

$$\|\nabla^2 f(ta_j)\| \leq \|\nabla^2 f(0)\| + (M_0 + M_1 \|\nabla f(0)\|) t \|a_j\|, \quad t > 0.$$

Since  $p-2 > 1$ , this contradicts the previous display as  $t \rightarrow +\infty$ .  $\square$

*Pointwise consequence of the pairwise conditions.* For any  $x \in \mathbb{R}^d$ ,  $\|u\| = 1$ , and all sufficiently small  $|t|$ , either pairwise condition gives

$$\frac{\|\nabla^2 f(x + tu) - \nabla^2 f(x)\|}{|t|} \leq M_0 + M_1 \|\nabla f(x)\|.$$

Taking  $t \rightarrow 0$  and then the supremum over  $\|u\| = 1$  proves (TD).  $\square$

The pointwise first-order condition used in Proposition 2.7 is denoted by

$$\|\nabla^2 f(x)\| \leq L_0 + L_1 \|\nabla f(x)\|, \quad x \in \mathbb{R}^d. \quad (\text{FO})$$

*Proof of Proposition 2.7.* Let  $s = y - x$ ,  $r = \|s\| \leq \rho$ , and  $g(t) = \|\nabla f(x + ts)\|$ . The function  $g$  is absolutely continuous. For almost every  $t \in [0, 1]$ , condition (FO) gives

$$g'(t) \leq r \|\nabla^2 f(x + ts)\| \leq r(L_0 + L_1 g(t)).$$

Grönwall's inequality yields

$$g(t) \leq e^{L_1 r t} (g(0) + L_0 r t) \leq e^{L_1 \rho} (\|\nabla f(x)\| + L_0 \rho).$$

Using (TD) along the segment from  $x$  to  $y$ , we obtain

$$\|\nabla^2 f(y) - \nabla^2 f(x)\| \leq r \int_0^1 (\widehat{M}_0 + \widehat{M}_1 g(t)) dt \leq (\widehat{M}_0 + \widehat{M}_1 L_0 \rho e^{L_1 \rho} + \widehat{M}_1 e^{L_1 \rho} \|\nabla f(x)\|) r.$$

This is (LH) with the stated constants.  $\square$

## C Proof of Algorithm 2

We consider completed iterations before a zero-gradient return.

For  $g \geq 0$  and  $k \geq 0$ , recall

$$\begin{aligned} \Theta_c(g) &:= \max \left\{ 8\delta + \sqrt{48M_0g}, \frac{4g}{\rho}, 12\sqrt{M_1g} \right\}, \\ U_k &:= \max \{ \tau_{k-1}, \Theta_c(\|\nabla f(x_k)\|) \}, \quad \tau_{-1} := 2\eta_0. \end{aligned} \quad (12)$$

### C.1 Step Estimates

**Lemma C.1.** *Suppose that  $f$  is convex. For every accepted step of Algorithm 2, set  $s_k := x_{k+1} - x_k$ . Then*

$$\frac{\tau_k}{2} \|s_k\| \leq \|\nabla f(x_{k+1})\| \leq \frac{3\tau_k}{2} \|s_k\|, \quad (13)$$

$$f(x_k) - f(x_{k+1}) \geq \max \left\{ \frac{\tau_k}{2} \|s_k\|^2, \frac{2\|\nabla f(x_{k+1})\|^2}{3\tau_k} \right\}, \quad (14)$$

$$\|\nabla f(x_{k+1})\| \leq \frac{2}{\sqrt{3}} \|\nabla f(x_k)\|. \quad (15)$$

*Proof.* By the MS condition in (2)

$$\begin{aligned}\|\nabla f(x_{k+1})\| &\geq \tau_k \|s_k\| - \|\nabla f(x_{k+1}) + \tau_k s_k\| \geq \frac{\tau_k}{2} \|s_k\|, \\ \|\nabla f(x_{k+1})\| &\leq \|\nabla f(x_{k+1}) + \tau_k s_k\| + \tau_k \|s_k\| \leq \frac{3\tau_k}{2} \|s_k\|.\end{aligned}$$

we prove (13). Squaring the MS condition and expanding the left-hand side gives

$$\|\nabla f(x_{k+1}) + \tau_k s_k\|^2 = \|\nabla f(x_{k+1})\|^2 + 2\tau_k \langle \nabla f(x_{k+1}), s_k \rangle + \tau_k^2 \|s_k\|^2 \leq \frac{\tau_k^2}{4} \|s_k\|^2. \quad (16)$$

Next, by convexity, (16), and (13),

$$\begin{aligned}f(x_k) - f(x_{k+1}) &\geq -\langle \nabla f(x_{k+1}), s_k \rangle \geq \frac{\|\nabla f(x_{k+1})\|^2}{2\tau_k} + \frac{3\tau_k}{8} \|s_k\|^2 \\ &\geq \frac{1}{2\tau_k} \left( \frac{\tau_k}{2} \|s_k\| \right)^2 + \frac{3\tau_k}{8} \|s_k\|^2 = \frac{\tau_k}{2} \|s_k\|^2.\end{aligned}$$

Using instead  $\|s_k\| \geq 2\|\nabla f(x_{k+1})\|/(3\tau_k)$  from (13), we obtain

$$f(x_k) - f(x_{k+1}) \geq \frac{\|\nabla f(x_{k+1})\|^2}{2\tau_k} + \frac{3\tau_k}{8} \left( \frac{2\|\nabla f(x_{k+1})\|}{3\tau_k} \right)^2 = \frac{2\|\nabla f(x_{k+1})\|^2}{3\tau_k}.$$

This proves (14). For (15), convexity at both endpoints gives

$$\langle \nabla f(x_k), s_k \rangle \leq f(x_{k+1}) - f(x_k) \leq \langle \nabla f(x_{k+1}), s_k \rangle.$$

Next, (16) gives

$$\|\nabla f(x_{k+1})\|^2 + \frac{3}{4}\tau_k^2 \|s_k\|^2 \leq -2\tau_k \langle \nabla f(x_{k+1}), s_k \rangle \leq -2\tau_k \langle \nabla f(x_k), s_k \rangle \leq 2\tau_k \|\nabla f(x_k)\| \|s_k\|.$$

Rearranging and completing the square gives

$$\begin{aligned}\|\nabla f(x_{k+1})\|^2 &\leq 2\tau_k \|\nabla f(x_k)\| \|s_k\| - \frac{3}{4}\tau_k^2 \|s_k\|^2 \\ &= \frac{4}{3} \|\nabla f(x_k)\|^2 - \frac{3}{4} \left( \tau_k \|s_k\| - \frac{4}{3} \|\nabla f(x_k)\| \right)^2 \leq \frac{4}{3} \|\nabla f(x_k)\|^2.\end{aligned}$$

□

**Lemma C.2.** Suppose that  $f$  is convex, (IH) holds, and either (GH) or (LH) holds. Set

$$\bar{\tau}(x) := \begin{cases} \max \left\{ 4\delta, 2\sqrt{L(x)\|\nabla f(x)\|} \right\}, & \text{under (GH),} \\ \max \left\{ 4\delta, 2\sqrt{L(x)\|\nabla f(x)\|}, \frac{2\|\nabla f(x)\|}{\rho} \right\}, & \text{under (LH).} \end{cases} \quad (17)$$

Every trial of MSBACKTRACK with  $\tau \geq \bar{\tau}(x)$  satisfies both tests in (2). Consequently, initialized with  $\tau = \eta > 0$ , the algorithm terminates after at most

$$1 + \left\lceil \log_2 \max \left\{ 1, \frac{\bar{\tau}(x)}{\eta} \right\} \right\rceil$$

trials, and its accepted parameter satisfies

$$\eta \leq \tau \leq \max \{ \eta, 2\bar{\tau}(x) \}.$$

*Proof.* Fix a trial parameter  $\tau \geq \bar{\tau}(x)$ . By convexity and (IH),  $H(x) \succeq -\delta I$ . Since  $\tau \geq 4\delta$ ,

$$H(x) + \tau I \succeq (\tau - \delta)I \succeq \frac{3\tau}{4}I \succ 0.$$

Thus PD( $x, \tau$ ) holds, and the linear system  $(H(x) + \tau I)s = -\nabla f(x)$  has a unique solution satisfying

$$\|s\| = \|(H(x) + \tau I)^{-1} \nabla f(x)\| \leq \frac{\|\nabla f(x)\|}{\tau - \delta} \leq \frac{2\|\nabla f(x)\|}{\tau}.$$

Under (LH), the definition of  $\bar{\tau}(x)$  also gives  $\|s\| \leq \rho$ . The gradient remainder bound in Lemma A.1 or Lemma A.2, together with (IH) and the linear system, yields

$$\begin{aligned} \|\nabla f(x + s) + \tau s\| &= \|\nabla f(x + s) - \nabla f(x) - H(x)s\| \\ &\leq \frac{L(x)}{2} \|s\|^2 + \delta \|s\| \leq \left( \frac{L(x)\|\nabla f(x)\|}{\tau} + \delta \right) \|s\| \\ &\leq \left( \frac{\tau}{4} + \frac{\tau}{4} \right) \|s\| = \frac{\tau}{2} \|s\|. \end{aligned}$$

Hence MS( $x, \tau, s$ ) also holds. The trial parameters are  $2^j \eta$ ,  $j = 0, 1, \dots$ , so the first index with  $2^j \eta \geq \bar{\tau}(x)$  gives the stated trial bound. If no trial is rejected, the accepted parameter is  $\tau = \eta$ . Otherwise,  $\tau/2$  was rejected, so  $\tau/2 < \bar{\tau}(x)$  and  $\tau < 2\bar{\tau}(x)$ .  $\square$

**Lemma C.3.** Suppose that  $f$  is convex, (IH) holds, and either (GH) or (LH) holds. Set

$$r_0 := \begin{cases} \frac{1}{6\sqrt{M_1}}, & \text{under (GH),} \\ \min \left\{ \frac{\rho}{2}, \frac{1}{6\sqrt{M_1}} \right\}, & \text{under (LH),} \end{cases} \quad (18)$$

where  $1/\sqrt{M_1} = +\infty$  when  $M_1 = 0$ . Suppose that MSBACKTRACK, initialized at  $x$  with  $\tau = \eta > 0$ , returns  $(y, \tau)$ . Set  $s := y - x$ . If at least one trial is rejected (equivalently  $\tau > \eta$ ), then

$$\|s\| > r_0 \quad \text{or} \quad \tau \leq 8\delta + 24M_0\|s\|. \quad (19)$$

If, in addition,  $\tau \geq 16\delta$ , then

$$\|s\| \geq \min \left\{ r_0, \frac{\tau}{48M_0} \right\}. \quad (20)$$

*Proof.* We prove (19) by considering three cases.

First, if  $\tau < 4\delta$ , then  $\tau \leq 8\delta + 24M_0\|s\|$ , so the second inequality holds.

Second, if  $\tau \geq 4\delta$  and  $\|s\| > r_0$ , the first inequality holds.

It remains to consider the case  $\tau \geq 4\delta$  and  $\|s\| \leq r_0$ . Since  $\tau > \eta$ , the trial with parameter  $\tau/2$  was rejected. Convexity and (IH) give

$$H(x) + \frac{\tau}{2}I \succeq \left( \frac{\tau}{2} - \delta \right) I \succeq \frac{\tau}{4}I \succ 0, \quad (21)$$

so this trial failed the MS test. Its step  $s_{1/2}$  and the accepted step  $s$  satisfy

$$\begin{aligned} \left( H(x) + \frac{\tau}{2}I \right) s_{1/2} &= -\nabla f(x), \\ (H(x) + \tau I) s &= -\nabla f(x). \end{aligned}$$

Subtracting these equations gives

$$s_{1/2} - s = \frac{\tau}{2} \left( H(x) + \frac{\tau}{2} I \right)^{-1} s, \quad \|s_{1/2} - s\| \leq 2\|s\|, \quad (22)$$

where the last inequality follows from (21). For every  $t \in [0, 1]$ , (22) gives

$$\begin{aligned} \|x + ts_{1/2} - y\| &= \|t(s_{1/2} - s) - (1 - t)s\| \\ &\leq t\|s_{1/2} - s\| + (1 - t)\|s\| \leq (1 + t)\|s\| \leq 2\|s\|. \end{aligned}$$

Under (LH), (18) gives

$$\|x - y\| = \|s\| \leq r_0 \leq \frac{\rho}{2}, \quad \|x + ts_{1/2} - y\| \leq 2\|s\| \leq 2r_0 \leq \rho.$$

Thus, applying (GH) or (LH) at  $y$  gives

$$\begin{aligned} \|\nabla^2 f(x + ts_{1/2}) - \nabla^2 f(x)\| &\leq \|\nabla^2 f(x + ts_{1/2}) - \nabla^2 f(y)\| + \|\nabla^2 f(x) - \nabla^2 f(y)\| \\ &\leq L(y) (\|x + ts_{1/2} - y\| + \|x - y\|) \leq 3L(y)\|s\|. \end{aligned} \quad (23)$$

Since the trial with parameter  $\tau/2$  fails the MS test, we have

$$\frac{\tau}{4} \|s_{1/2}\| < \|\nabla f(x + s_{1/2}) + \frac{\tau}{2} s_{1/2}\|. \quad (24)$$

The linear system for  $s_{1/2}$  gives

$$\frac{\tau}{2} s_{1/2} = -\nabla f(x) - H(x)s_{1/2}.$$

Therefore,

$$\begin{aligned} \nabla f(x + s_{1/2}) + \frac{\tau}{2} s_{1/2} &= \nabla f(x + s_{1/2}) - \nabla f(x) - H(x)s_{1/2} \\ &= \int_0^1 \left( \nabla^2 f(x + ts_{1/2}) - H(x) \right) s_{1/2} dt. \end{aligned} \quad (25)$$

Therefore, by (24), (25), (23), and (IH),

$$\frac{\tau}{4} \|s_{1/2}\| < \left\| \int_0^1 \left( \nabla^2 f(x + ts_{1/2}) - H(x) \right) s_{1/2} dt \right\| \leq (\delta + 3L(y)\|s\|) \|s_{1/2}\|.$$

Since this trial fails MS, its step  $s_{1/2}$  is nonzero. The accepted MS test in (2) gives

$$\|\nabla f(y)\| \leq \|\nabla f(y) + \tau s\| + \tau\|s\| \leq \frac{3\tau}{2} \|s\|.$$

Dividing by  $\|s_{1/2}\|$  and using this bound, we obtain

$$\begin{aligned} \frac{\tau}{4} &< \delta + 3L(y)\|s\| = \delta + 3M_0\|s\| + 3M_1\|\nabla f(y)\|\|s\| \\ &\leq \delta + 3M_0\|s\| + \frac{9}{2}M_1\tau\|s\|^2 \leq \delta + 3M_0\|s\| + \frac{\tau}{8}. \end{aligned}$$

The last inequality follows from  $\|s\| \leq r_0$  and (18). Indeed, if  $M_1 > 0$ , then

$$M_1\|s\|^2 \leq M_1 r_0^2 \leq M_1 \left( \frac{1}{6\sqrt{M_1}} \right)^2 = \frac{1}{36}.$$

If  $M_1 = 0$ , then  $M_1\|s\|^2 = 0$ . Thus, in both cases,

$$\frac{9}{2}M_1\tau\|s\|^2 \leq \frac{\tau}{8}.$$

Rearranging proves (19). Finally, if  $\tau \geq 16\delta$ , (19) gives

$$\|s\| \geq \min \left\{ r_0, \frac{\tau - 8\delta}{24M_0} \right\} \geq \min \left\{ r_0, \frac{\tau}{48M_0} \right\},$$

which proves (20).  $\square$

By (12) and (18),

$$\Theta_c(g) = \max \left\{ 8\delta + \sqrt{48M_0g}, \frac{2g}{r_0} \right\}, \quad g \geq 0,$$

where  $1/(+\infty) = 0$ .

**Lemma C.4.** *Suppose that  $f$  is convex, (IH) holds, and either (GH) or (LH) holds. If iteration  $k$  of Algorithm 2 rejects at least one trial, then*

$$\tau_k \leq \Theta_c(\|\nabla f(x_{k+1})\|). \quad (26)$$

*Proof.* Set  $s_k := x_{k+1} - x_k$  and apply Lemma C.3 with  $x = x_k$ ,  $y = x_{k+1}$ ,  $\eta = \eta_k$ , and  $\tau = \tau_k$ . If  $\|s_k\| > r_0$ , then (13) and (12) give

$$\tau_k < \frac{2\|\nabla f(x_{k+1})\|}{r_0} \leq \Theta_c(\|\nabla f(x_{k+1})\|).$$

If  $\|s_k\| \leq r_0$ , (19) and (13) imply

$$\begin{aligned} \tau_k &\leq 8\delta + 24M_0\|s_k\| \leq 8\delta + \frac{48M_0\|\nabla f(x_{k+1})\|}{\tau_k}, \\ \tau_k &\leq 4\delta + \sqrt{16\delta^2 + 48M_0\|\nabla f(x_{k+1})\|} \leq 8\delta + \sqrt{48M_0\|\nabla f(x_{k+1})\|}, \end{aligned}$$

which proves (26).  $\square$

## C.2 Iteration Count

We call a completed iteration  $k$  of Algorithm 2 controlled if

$$\tau_k \leq \Theta_c(\|\nabla f(x_{k+1})\|). \quad (27)$$

By Lemma C.4, every iteration with a rejection is controlled. An iteration accepted at its first trial can belong to either type.

**Lemma C.5.** *Under the assumptions of Lemma C.4, suppose that  $m$  iterations of Algorithm 2 are completed and  $f(x_m) > f^*$ . If  $m_c$  of these iterations are controlled, then*

$$m \leq \frac{5}{4}m_c + \log_2 \frac{U_0}{\Theta_c(\|\nabla f(x_m)\|)}. \quad (28)$$

*Proof.* By (12),  $\Theta_c(\|\nabla f(x_k)\|) \leq U_k$  for every  $k \geq 0$ . The function  $\Theta_c$  is nondecreasing, and (12) gives  $\Theta_c(ag) \leq a\Theta_c(g)$  for  $g \geq 0$  and  $a \geq 1$ .

If iteration  $k$  is controlled, then  $\tau_k \leq \Theta_c(\|\nabla f(x_{k+1})\|)$  by (27). The definition of  $U_{k+1}$  and (15) give

$$U_{k+1} = \Theta_c(\|\nabla f(x_{k+1})\|) \leq \frac{2}{\sqrt{3}}\Theta_c(\|\nabla f(x_k)\|) \leq \frac{2}{\sqrt{3}}U_k.$$

If iteration  $k$  is not controlled, then  $\tau_k > \Theta_c(\|\nabla f(x_{k+1})\|)$ , so  $U_{k+1} = \tau_k$  by definition. By Lemma C.4, this iteration has no rejected trials, so  $\tau_k = \eta_k$ . For  $k \geq 1$ , line 4 of Algorithm 2, applied at iteration  $k-1$ , gives  $\eta_k \leq \tau_{k-1}/2$ . Hence,

$$U_{k+1} = \tau_k = \eta_k \leq \frac{\tau_{k-1}}{2} \leq \frac{U_k}{2}.$$

For  $k=0$ , the same bound holds because  $U_1 = \tau_0 = \eta_0 \leq U_0/2$ . Applying these bounds over the  $m_c$  controlled iterations and  $m - m_c$  remaining iterations gives

$$\Theta_c(\|\nabla f(x_m)\|) \leq U_m \leq U_0 \left(\frac{2}{\sqrt{3}}\right)^{m_c} 2^{-(m-m_c)}.$$

Taking logarithms and rearranging, we obtain

$$m \leq \left(1 + \log_2 \frac{2}{\sqrt{3}}\right) m_c + \log_2 \frac{U_0}{\Theta_c(\|\nabla f(x_m)\|)}.$$

Since  $(2/\sqrt{3})^4 = 16/9 < 2$ , this proves (28).  $\square$

To count the controlled iterations, assume  $D < +\infty$ . By (14), every accepted point remains in  $\mathcal{L}(x_0)$ , which also contains  $x^*$ . Hence, by convexity,

$$\begin{aligned} \|x_k - x^*\| &\leq \|x_k - x_0\| + \|x_0 - x^*\| \leq 2D, \\ f(x_k) - f^* &\leq \langle \nabla f(x_k), x_k - x^* \rangle \leq 2D \|\nabla f(x_k)\|. \end{aligned} \tag{29}$$

By (3) and (18),  $B = \max\{1, D/r_0\}$ , where  $D/(+\infty) = 0$ . For  $t > 0$ , define

$$\Phi(t) := \max \left\{ \frac{6D}{r_0}, 12\sqrt{6}\sqrt{\frac{M_0 D^3}{t}} + \frac{48\delta D^2}{t} \right\}. \tag{30}$$

The function  $\Phi$  is positive and nonincreasing, and  $t(1 + \Phi(t))$  is nondecreasing.

**Lemma C.6.** *For the prefix and count of Lemma C.5, assume in addition that  $0 < D < +\infty$ . Then*

$$m_c \leq \int_{f(x_m)-f^*}^{f(x_0)-f^*} \frac{1 + \Phi(t)}{t} dt. \tag{31}$$

*Proof.* For a controlled iteration, we have

$$\begin{aligned} f(x_k) - f(x_{k+1}) &\stackrel{(14)}{\geq} \frac{2\|\nabla f(x_{k+1})\|^2}{3\tau_k} \stackrel{(27)}{\geq} \frac{2\|\nabla f(x_{k+1})\|^2}{3\Theta_c(\|\nabla f(x_{k+1})\|)} \\ &\stackrel{(12)}{=} \min \left\{ \frac{2\|\nabla f(x_{k+1})\|^2}{3(8\delta + \sqrt{48M_0}\|\nabla f(x_{k+1})\|)}, \frac{r_0\|\nabla f(x_{k+1})\|}{3} \right\}. \end{aligned}$$

By convexity and Cauchy-Schwarz,

$$f(x_{k+1}) - f^* \leq \langle \nabla f(x_{k+1}), x_{k+1} - x^* \rangle \leq \|\nabla f(x_{k+1})\| \|x_{k+1} - x^*\| \stackrel{(29)}{\leq} 2D \|\nabla f(x_{k+1})\|.$$

Substituting  $\|\nabla f(x_{k+1})\| \geq \frac{f(x_{k+1}) - f^*}{2D}$  gives

$$\begin{aligned} f(x_k) - f(x_{k+1}) &\geq \min \left\{ \frac{(f(x_{k+1}) - f^*)^2}{48\delta D^2 + 12\sqrt{6}\sqrt{M_0}D^3(f(x_{k+1}) - f^*)}, \frac{r_0(f(x_{k+1}) - f^*)}{6D} \right\} \\ &= \frac{f(x_{k+1}) - f^*}{\Phi(f(x_{k+1}) - f^*)}. \end{aligned}$$

By (14),  $f(x_{k+1}) - f^* \leq f(x_k) - f^*$ . For every  $t \in [f(x_{k+1}) - f^*, f(x_k) - f^*]$ , the monotonicity of  $t(1 + \Phi(t))$  gives

$$\frac{1 + \Phi(t)}{t} \geq \frac{(f(x_{k+1}) - f^*)(1 + \Phi(f(x_{k+1}) - f^*))}{t^2}.$$

Thus, we obtain

$$\begin{aligned} \int_{f(x_{k+1}) - f^*}^{f(x_k) - f^*} \frac{1 + \Phi(t)}{t} dt &\geq (f(x_{k+1}) - f^*)(1 + \Phi(f(x_{k+1}) - f^*)) \int_{f(x_{k+1}) - f^*}^{f(x_k) - f^*} \frac{dt}{t^2} \\ &= (f(x_{k+1}) - f^*)(1 + \Phi(f(x_{k+1}) - f^*)) \left( \frac{1}{f(x_{k+1}) - f^*} - \frac{1}{f(x_k) - f^*} \right) \\ &= (1 + \Phi(f(x_{k+1}) - f^*)) \frac{f(x_k) - f(x_{k+1})}{f(x_k) - f^*} \geq \frac{1 + \Phi(f(x_{k+1}) - f^*)}{\Phi(f(x_{k+1}) - f^*) + 1} = 1. \end{aligned}$$

Summing over the controlled iterations proves (31).  $\square$

### C.3 Proof of Theorem 3.1

**Theorem C.7.** Suppose that  $f$  is convex, (IH) holds, and either (GH) or (LH) holds. Define

$$B := \max \left\{ 1, \frac{2D}{\rho}, 6\sqrt{M_1}D \right\}, \quad \varepsilon_{\text{tr}} := \max \left\{ \frac{M_0 D^3}{B^2}, \frac{\delta D^2}{B} \right\}, \quad (32)$$

where  $\rho = +\infty$  under (GH). For every  $\varepsilon > 0$ , Algorithm 2 generates an iterate  $x_N$  satisfying  $f(x_N) - f^* \leq \varepsilon$  within

$$N \leq 2 + \log_2 \max \left\{ 1, \frac{2D \|\nabla f(x_0)\|}{\varepsilon_{\text{tr}}}, \frac{\eta_0 D^2}{\sqrt{6}B\varepsilon_{\text{tr}}} \right\} + 100B \log_+ \frac{f(x_0) - f^*}{\max\{\varepsilon, \varepsilon_{\text{tr}}\}} + 80\sqrt{\frac{M_0 D^3}{\varepsilon}} + 64\frac{\delta D^2}{\varepsilon}. \quad (33)$$

iterations.

*Proof.* Each search terminates by Lemma C.2. Consider a prefix  $x_0, \dots, x_m$  with  $f(x_m) - f^* > \varepsilon$ . By (29),  $\|\nabla f(x_m)\| > \varepsilon/(2D)$ . Combining (28) and (31) gives

$$\begin{aligned} m &\leq \frac{5}{4} \int_{f(x_m) - f^*}^{f(x_0) - f^*} \frac{1 + \Phi(t)}{t} dt + \log_2 \frac{U_0}{\Theta_c(\|\nabla f(x_m)\|)} \\ &\leq \frac{5}{4} \int_{\varepsilon}^{f(x_0) - f^*} \frac{1 + \Phi(t)}{t} dt + \log_{2,+} \frac{U_0}{\Theta_c(\varepsilon/(2D))}. \end{aligned} \quad (34)$$

Let  $N$  be the first index with  $f(x_N) - f^* \leq \varepsilon$ . If  $N \geq 1$ , then  $f(x_{N-1}) - f^* > \varepsilon$ . Therefore,

$$N \leq 1 + \frac{5}{4} \int_{\varepsilon}^{f(x_0) - f^*} \frac{1 + \Phi(t)}{t} dt + \log_{2,+} \frac{U_0}{\Theta_c(\varepsilon/(2D))}. \quad (35)$$

We begin with bounding integral term in (35).

By (32) and (30),  $1 + \Phi(t) \leq 79B$  for  $t \geq \varepsilon_{\text{tr}}$ . For  $0 < t \leq \varepsilon_{\text{tr}}$ , we have

$$\max \left\{ \sqrt{\frac{M_0 D^3}{t}}, \frac{\delta D^2}{t} \right\} \geq B \geq \max \left\{ 1, \frac{D}{r_0} \right\}, 1 + \Phi(t) \leq (12\sqrt{6} + 1) \sqrt{\frac{M_0 D^3}{t}} + 49 \frac{\delta D^2}{t}.$$

If  $\varepsilon_{\text{tr}} \leq \varepsilon < f(x_0) - f^*$ , the whole integration interval lies above  $\varepsilon_{\text{tr}}$ , so

$$\int_{\varepsilon}^{f(x_0) - f^*} \frac{1 + \Phi(t)}{t} dt \leq 79B \int_{\varepsilon}^{f(x_0) - f^*} \frac{dt}{t} = 79B \log \frac{f(x_0) - f^*}{\varepsilon}.$$

If  $0 < \varepsilon < \varepsilon_{\text{tr}} \leq f(x_0) - f^*$ , split the integral as

$$\begin{aligned} \int_{\varepsilon}^{f(x_0) - f^*} \frac{1 + \Phi(t)}{t} dt &= \int_{\varepsilon}^{\varepsilon_{\text{tr}}} \frac{1 + \Phi(t)}{t} dt + \int_{\varepsilon_{\text{tr}}}^{f(x_0) - f^*} \frac{1 + \Phi(t)}{t} dt \\ &\leq (12\sqrt{6} + 1) \sqrt{M_0 D^3} \int_{\varepsilon}^{\varepsilon_{\text{tr}}} \frac{dt}{t^{3/2}} + 49\delta D^2 \int_{\varepsilon}^{\varepsilon_{\text{tr}}} \frac{dt}{t^2} + 79B \int_{\varepsilon_{\text{tr}}}^{f(x_0) - f^*} \frac{dt}{t} \\ &= (24\sqrt{6} + 2) \sqrt{M_0 D^3} \left( \frac{1}{\sqrt{\varepsilon}} - \frac{1}{\sqrt{\varepsilon_{\text{tr}}}} \right) + 49\delta D^2 \left( \frac{1}{\varepsilon} - \frac{1}{\varepsilon_{\text{tr}}} \right) + 79B \log \frac{f(x_0) - f^*}{\varepsilon_{\text{tr}}} \\ &\leq (24\sqrt{6} + 2) \sqrt{\frac{M_0 D^3}{\varepsilon}} + 49 \frac{\delta D^2}{\varepsilon} + 79B \log \frac{f(x_0) - f^*}{\varepsilon_{\text{tr}}}. \end{aligned}$$

If  $0 < \varepsilon < f(x_0) - f^* < \varepsilon_{\text{tr}}$ , the whole interval lies below  $\varepsilon_{\text{tr}}$ , so

$$\begin{aligned} \int_{\varepsilon}^{f(x_0) - f^*} \frac{1 + \Phi(t)}{t} dt &\leq (12\sqrt{6} + 1) \sqrt{M_0 D^3} \int_{\varepsilon}^{f(x_0) - f^*} \frac{dt}{t^{3/2}} + 49\delta D^2 \int_{\varepsilon}^{f(x_0) - f^*} \frac{dt}{t^2} \\ &= (24\sqrt{6} + 2) \sqrt{M_0 D^3} \left( \frac{1}{\sqrt{\varepsilon}} - \frac{1}{\sqrt{f(x_0) - f^*}} \right) + 49\delta D^2 \left( \frac{1}{\varepsilon} - \frac{1}{f(x_0) - f^*} \right) \\ &\leq (24\sqrt{6} + 2) \sqrt{\frac{M_0 D^3}{\varepsilon}} + 49 \frac{\delta D^2}{\varepsilon}. \end{aligned}$$

Here  $\log_+ \frac{f(x_0) - f^*}{\varepsilon_{\text{tr}}} = 0$  by definition.

Combining the three cases yields

$$\int_{\varepsilon}^{f(x_0) - f^*} \frac{1 + \Phi(t)}{t} dt \leq 79B \log_+ \frac{f(x_0) - f^*}{\max\{\varepsilon, \varepsilon_{\text{tr}}\}} + (24\sqrt{6} + 2) \sqrt{\frac{M_0 D^3}{\varepsilon}} + 49 \frac{\delta D^2}{\varepsilon}. \quad (36)$$

Now, we need to bound the last term in (35).

For  $g_2 \geq g_1 > 0$ , (12) gives  $\Theta_c(g_2) \leq (g_2/g_1)\Theta_c(g_1)$ . Together with monotonicity of  $\Theta_c$ , this yields

$$\log_{2,+} \frac{U_0}{\Theta_c(\varepsilon/(2D))} \leq \log_{2,+} \frac{U_0}{\Theta_c(\varepsilon_{\text{tr}}/(2D))} + \log_{2,+} \frac{\varepsilon_{\text{tr}}}{\varepsilon}.$$

If  $\varepsilon \geq \varepsilon_{\text{tr}}$ , then  $\log_{2,+}(\varepsilon_{\text{tr}}/\varepsilon) = 0$ . For  $0 < \varepsilon < \varepsilon_{\text{tr}}$ , we have

$$\begin{aligned}\log_2 \frac{\varepsilon_{\text{tr}}}{\varepsilon} &\leq 3\sqrt{\frac{\varepsilon_{\text{tr}}}{\varepsilon}} = \frac{3}{B}\sqrt{\frac{M_0 D^3}{\varepsilon}}, & \text{if } \varepsilon_{\text{tr}} = \frac{M_0 D^3}{B^2}, \\ \log_2 \frac{\varepsilon_{\text{tr}}}{\varepsilon} &\leq 2\frac{\varepsilon_{\text{tr}}}{\varepsilon} = \frac{2}{B}\frac{\delta D^2}{\varepsilon}, & \text{if } \varepsilon_{\text{tr}} = \frac{\delta D^2}{B}.\end{aligned}$$

Since  $B \geq 1$ ,

$$\log_{2,+} \frac{U_0}{\Theta_c(\varepsilon/(2D))} \leq \log_{2,+} \frac{U_0}{\Theta_c(\varepsilon_{\text{tr}}/(2D))} + 3\sqrt{\frac{M_0 D^3}{\varepsilon}} + 2\frac{\delta D^2}{\varepsilon}. \quad (37)$$

Substituting (36) and (37) into (35), and using

$$\frac{5}{4} \cdot 79 \leq 100, \quad \frac{5}{4}(24\sqrt{6} + 2) + 3 \leq 80, \quad \frac{5}{4} \cdot 49 + 2 \leq 64,$$

we obtain

$$N \leq 2 + \log_{2,+} \frac{U_0}{\Theta_c(\varepsilon_{\text{tr}}/(2D))} + 100B \log_+ \frac{f(x_0) - f^*}{\max\{\varepsilon, \varepsilon_{\text{tr}}\}} + 80\sqrt{\frac{M_0 D^3}{\varepsilon}} + 64\frac{\delta D^2}{\varepsilon}.$$

Recall that  $\Theta_c$  is nondecreasing and  $\Theta_c(ag) \leq a\Theta_c(g)$  for  $a \geq 1$ . Thus,

$$\Theta_c(\|\nabla f(x_0)\|) \leq \Theta_c\left(\max\left\{1, \frac{2D\|\nabla f(x_0)\|}{\varepsilon_{\text{tr}}}\right\} \frac{\varepsilon_{\text{tr}}}{2D}\right) \leq \max\left\{1, \frac{2D\|\nabla f(x_0)\|}{\varepsilon_{\text{tr}}}\right\} \Theta_c\left(\frac{\varepsilon_{\text{tr}}}{2D}\right).$$

If  $\varepsilon_{\text{tr}} = M_0 D^3/B^2$

$$\Theta_c\left(\frac{\varepsilon_{\text{tr}}}{2D}\right) \geq \sqrt{\frac{24M_0\varepsilon_{\text{tr}}}{D}} = \frac{2\sqrt{6}B\varepsilon_{\text{tr}}}{D^2}.$$

Otherwise  $\varepsilon_{\text{tr}} = \delta D^2/B$

$$\Theta_c\left(\frac{\varepsilon_{\text{tr}}}{2D}\right) \geq 8\delta = \frac{8B\varepsilon_{\text{tr}}}{D^2} \geq \frac{2\sqrt{6}B\varepsilon_{\text{tr}}}{D^2}.$$

Since  $U_0 = \max\{2\eta_0, \Theta_c(\|\nabla f(x_0)\|)\}$ , these estimates give

$$\begin{aligned}\log_{2,+} \frac{U_0}{\Theta_c(\varepsilon_{\text{tr}}/(2D))} &= \log_2 \max\left\{1, \frac{\Theta_c(\|\nabla f(x_0)\|)}{\Theta_c(\varepsilon_{\text{tr}}/(2D))}, \frac{2\eta_0}{\Theta_c(\varepsilon_{\text{tr}}/(2D))}\right\} \\ &\leq \log_2 \max\left\{1, \frac{2D\|\nabla f(x_0)\|}{\varepsilon_{\text{tr}}}, \frac{\eta_0 D^2}{\sqrt{6}B\varepsilon_{\text{tr}}}\right\}.\end{aligned} \quad (38)$$

Substituting (38) into the bound for  $N$  proves (33). □

#### C.4 Far and Near Convergence Rates

**Corollary C.8.** *Under the assumptions of Theorem C.7, suppose that  $0 < D < +\infty$ . Define*

$$\ell_0 := \max\left\{1, \frac{2D\|\nabla f(x_0)\|}{\varepsilon_{\text{tr}}}, \frac{\eta_0 D^2}{\sqrt{6}B\varepsilon_{\text{tr}}}\right\}.$$

For every  $k \geq 0$  with  $f(x_k) - f^* > \varepsilon_{\text{tr}}$ ,

$$f(x_k) - f^* \leq (f(x_0) - f^*) \exp\left(-\frac{(k - \log_2 \ell_0)_+}{100B}\right), \quad (39)$$

where we write  $(t)_+ := \max\{0, t\}$ . If  $f(x_k) - f^* \leq \varepsilon_{\text{tr}}$ , the subsequent iterates satisfy

$$f(x_{k+n}) - f^* \leq \frac{7396M_0D^3}{(B + (n - \log_2 \ell_0)_+)^2} + \frac{140\delta D^2}{B + (n - \log_2 \ell_0)_+}, \quad n = 0, 1, \dots, \quad (40)$$

where  $t_+ := \max\{t, 0\}$ .

*Proof.* If  $f(x_k) - f^* > \varepsilon_{\text{tr}}$ , (34) gives

$$k \leq \frac{5}{4} \int_{f(x_k)-f^*}^{f(x_0)-f^*} \frac{1 + \Phi(t)}{t} dt + \log_2 \frac{U_0}{\Theta_c(\|\nabla f(x_k)\|)}.$$

For  $t \geq \varepsilon_{\text{tr}}$ , definitions (32) and (30) give

$$\Phi(t) \leq \max \left\{ 6B, 12\sqrt{6} \sqrt{\frac{M_0D^3}{\varepsilon_{\text{tr}}}} + 48 \frac{\delta D^2}{\varepsilon_{\text{tr}}} \right\} \leq \max \left\{ 6B, (12\sqrt{6} + 48)B \right\} = (12\sqrt{6} + 48)B.$$

Therefore,  $1 + \Phi(t) \leq 79B$ . By (29), monotonicity of  $\Theta_c$ , and (38),

$$\log_2 \frac{U_0}{\Theta_c(\|\nabla f(x_k)\|)} \leq \log_{2,+} \frac{U_0}{\Theta_c(\varepsilon_{\text{tr}}/(2D))} \leq \log_2 \ell_0.$$

Substituting these bounds yields

$$\begin{aligned} k &\leq \frac{5}{4} \cdot 79B \int_{f(x_k)-f^*}^{f(x_0)-f^*} \frac{dt}{t} + \log_2 \ell_0 \\ &= \frac{395}{4} B \log \frac{f(x_0) - f^*}{f(x_k) - f^*} + \log_2 \ell_0 \leq 100B \log \frac{f(x_0) - f^*}{f(x_k) - f^*} + \log_2 \ell_0. \end{aligned}$$

Rearranging and using  $f(x_k) \leq f(x_0)$  when  $k \leq \log_2 \ell_0$  proves (39).

Now suppose that  $f(x_k) - f^* \leq \varepsilon_{\text{tr}}$ . Let  $n_{\text{tr}} \leq k$  be the first index with  $f(x_{n_{\text{tr}}}) - f^* \leq \varepsilon_{\text{tr}}$ . For  $n \geq 0$  with  $f(x_{n_{\text{tr}}+n}) > f^*$ , the first inequality in (34), applied from iteration  $n_{\text{tr}}$ , gives

$$n \leq \frac{5}{4} \int_{f(x_{n_{\text{tr}}+n})-f^*}^{f(x_{n_{\text{tr}}})-f^*} \frac{1 + \Phi(t)}{t} dt + \log_2 \frac{U_{n_{\text{tr}}}}{\Theta_c(\|\nabla f(x_{n_{\text{tr}}+n})\|)}. \quad (41)$$

For  $0 < t \leq \varepsilon_{\text{tr}}$ , the definitions of  $\varepsilon_{\text{tr}}$  and  $B$  give

$$\max \left\{ \sqrt{\frac{M_0D^3}{t}}, \frac{\delta D^2}{t} \right\} \geq B = \max \left\{ 1, \frac{D}{r_0} \right\}.$$

Hence, by the definition of  $\Phi$  in (30),

$$1 + \Phi(t) \leq 1 + 12\sqrt{6} \sqrt{\frac{M_0D^3}{t}} + 48 \frac{\delta D^2}{t} \leq (12\sqrt{6} + 1) \sqrt{\frac{M_0D^3}{t}} + 49 \frac{\delta D^2}{t}.$$

Since  $0 < f(x_{n_{\text{tr}}+n}) - f^* \leq f(x_{n_{\text{tr}}}) - f^* \leq \varepsilon_{\text{tr}}$ ,

$$\begin{aligned} \int_{f(x_{n_{\text{tr}}+n})-f^*}^{f(x_{n_{\text{tr}}})-f^*} \frac{1+\Phi(t)}{t} dt &\leq (12\sqrt{6}+1)\sqrt{M_0 D^3} \int_{f(x_{n_{\text{tr}}+n})-f^*}^{f(x_{n_{\text{tr}}})-f^*} \frac{dt}{t^{3/2}} + 49\delta D^2 \int_{f(x_{n_{\text{tr}}+n})-f^*}^{f(x_{n_{\text{tr}}})-f^*} \frac{dt}{t^2} \\ &= (24\sqrt{6}+2)\sqrt{M_0 D^3} \left( \frac{1}{\sqrt{f(x_{n_{\text{tr}}+n})-f^*}} - \frac{1}{\sqrt{f(x_{n_{\text{tr}}})-f^*}} \right) \\ &\quad + 49\delta D^2 \left( \frac{1}{f(x_{n_{\text{tr}}+n})-f^*} - \frac{1}{f(x_{n_{\text{tr}}})-f^*} \right) \\ &\leq (24\sqrt{6}+2)\sqrt{\frac{M_0 D^3}{f(x_{n_{\text{tr}}+n})-f^*}} + 49\frac{\delta D^2}{f(x_{n_{\text{tr}}+n})-f^*}. \end{aligned}$$

To bound the logarithm in (41), first express  $U_{n_{\text{tr}}}$  through  $\Theta_c$ . If there is no controlled iteration before  $n_{\text{tr}}$ , the proof of Lemma C.5 gives  $U_{n_{\text{tr}}} \leq U_0$ . Otherwise, let  $i < n_{\text{tr}}$  be the last controlled iteration. Then

$$U_{n_{\text{tr}}} \leq U_{i+1} = \Theta_c(\|\nabla f(x_{i+1})\|).$$

Next, by  $f(x_0) - f^* > \varepsilon_{\text{tr}} = \max\{M_0 D^3/B^2, \delta D^2/B\}$

$$\delta \leq \frac{B(f(x_0) - f^*)}{D^2}, \quad M_0 \leq \frac{B^2(f(x_0) - f^*)}{D^3}.$$

By the definition of  $B$ ,

$$\max\left\{\frac{4}{\rho}, 12\sqrt{M_1}\right\} = \frac{2}{D} \max\left\{\frac{2D}{\rho}, 6\sqrt{M_1}D\right\} \leq \frac{2B}{D}.$$

If  $\|\nabla f(x_{i+1})\| < (f(x_0) - f^*)/D$ , then  $B \geq 1$  gives  $\|\nabla f(x_{i+1})\| < B \frac{f(x_0) - f^*}{D}$ .

Otherwise, since iteration  $i$  is controlled, by (14), (27), and the definition of  $\Theta_c$

$$\begin{aligned} \|\nabla f(x_{i+1})\| &\leq \frac{3}{2}(f(x_0) - f^*) \max\left\{\frac{8\delta}{\|\nabla f(x_{i+1})\|} + \sqrt{\frac{48M_0}{\|\nabla f(x_{i+1})\|}}, \frac{4}{\rho}, 12\sqrt{M_1}\right\} \\ &\leq \frac{3(8 + \sqrt{48})B(f(x_0) - f^*)}{2D} \leq \frac{23B(f(x_0) - f^*)}{D}. \end{aligned}$$

Thus,

$$\|\nabla f(x_{i+1})\| \leq 23B \frac{f(x_0) - f^*}{D} \leq 23B \|\nabla f(x_0)\|.$$

Substituting into  $\Theta_c$  and using its monotonicity and  $\Theta_c(ag) \leq a\Theta_c(g)$  for  $a \geq 1$ , we obtain

$$U_{n_{\text{tr}}} \leq \Theta_c(\|\nabla f(x_{i+1})\|) \leq \Theta_c(23B \|\nabla f(x_0)\|) \leq 23B \Theta_c(\|\nabla f(x_0)\|) \leq 23BU_0.$$

Using this bound,  $\|\nabla f(x_{n_{\text{tr}}+n})\| \geq (f(x_{n_{\text{tr}}+n}) - f^*)/(2D)$ , and (37) with (38), we obtain

$$\begin{aligned} \log_2 \frac{U_{n_{\text{tr}}}}{\Theta_c(\|\nabla f(x_{n_{\text{tr}}+n})\|)} &\leq \log_2(23B) + \log_{2,+} \frac{U_0}{\Theta_c((f(x_{n_{\text{tr}}+n}) - f^*)/(2D))} \\ &\leq \log_2 \ell_0 + \log_2(23B) + 3\sqrt{\frac{M_0 D^3}{f(x_{n_{\text{tr}}+n}) - f^*}} + 2\frac{\delta D^2}{f(x_{n_{\text{tr}}+n}) - f^*}. \end{aligned}$$

Substitution into (41) yields

$$n \leq \log_2 \ell_0 + \log_2(23B) + 80\sqrt{\frac{M_0 D^3}{f(x_{n_{\text{tr}}+n}) - f^*}} + 64\frac{\delta D^2}{f(x_{n_{\text{tr}}+n}) - f^*}. \quad (42)$$

The definition of  $\varepsilon_{\text{tr}}$  and  $f(x_{n_{\text{tr}}+n}) - f^* \leq \varepsilon_{\text{tr}}$  imply

$$B \leq \max \left\{ \sqrt{\frac{M_0 D^3}{f(x_{n_{\text{tr}}+n}) - f^*}}, \frac{\delta D^2}{f(x_{n_{\text{tr}}+n}) - f^*} \right\}, \quad \log_2(23B) \leq 5B.$$

Inequality (42) therefore gives

$$\begin{aligned} B + (n - \log_2 \ell_0)_+ &\leq 6B + 80\sqrt{\frac{M_0 D^3}{f(x_{n_{\text{tr}}+n}) - f^*}} + 64\frac{\delta D^2}{f(x_{n_{\text{tr}}+n}) - f^*} \\ &\leq 86\sqrt{\frac{M_0 D^3}{f(x_{n_{\text{tr}}+n}) - f^*}} + 70\frac{\delta D^2}{f(x_{n_{\text{tr}}+n}) - f^*}. \end{aligned}$$

Multiplying by  $f(x_{n_{\text{tr}}+n}) - f^*$  and using  $2ab \leq a^2 + b^2$ , we obtain

$$\begin{aligned} (B + (n - \log_2 \ell_0)_+) (f(x_{n_{\text{tr}}+n}) - f^*) &\leq 86\sqrt{M_0 D^3 (f(x_{n_{\text{tr}}+n}) - f^*)} + 70\delta D^2 \\ &\leq \frac{(B + (n - \log_2 \ell_0)_+) (f(x_{n_{\text{tr}}+n}) - f^*)}{2} + \frac{3698M_0 D^3}{B + (n - \log_2 \ell_0)_+} + 70\delta D^2, \\ f(x_{n_{\text{tr}}+n}) - f^* &\leq \frac{7396M_0 D^3}{(B + (n - \log_2 \ell_0)_+)^2} + \frac{140\delta D^2}{B + (n - \log_2 \ell_0)_+}. \end{aligned}$$

Since  $f(x_{k+n}) \leq f(x_{n_{\text{tr}}+n})$ , this proves (40).  $\square$

### C.5 Oracle complexity

**Corollary C.9.** *Let  $N := \min \{k \geq 0 : f(x_k) - f^* \leq \varepsilon\}$  and  $g_0 = \|\nabla f(x_0)\|$ . The total number of rejected trials of Algorithm 2 before obtaining  $x_N$ , including failed PD tests, is at most*

$$\frac{5}{4}(N - 1) + 4 + \log_{2,+} \frac{Bg_0}{D\eta_0} + \log_{2,+} \frac{\varepsilon_{\text{tr}}}{\varepsilon}. \quad (43)$$

Thus, the total number of gradient evaluations and rejected trials is bounded by

$$\mathcal{O}\left(1 + \log \ell_0 + \log_+ \frac{Bg_0}{D\eta_0} + B \log_+ \frac{f(x_0) - f^*}{\max\{\varepsilon, \varepsilon_{\text{tr}}\}} + \sqrt{\frac{M_0 D^3}{\varepsilon}} + \frac{\delta D^2}{\varepsilon}\right). \quad (44)$$

*Proof.* Let  $b_j$  be the number of rejected trials on iteration  $j$ . Then,  $b_0 = \log_2 \frac{\tau_0}{\eta_0}$  and  $\tau_j = 2^{b_j} \eta_j$ . Let  $g_j = \|\nabla f(x_j)\|$ . Therefore, for  $j \geq 1$  by line 4 of Algorithm 2

$$b_j = \log_2 \frac{\tau_j}{\eta_j} = \log_2 \frac{2\tau_j}{\tau_{j-1} \min\{1, g_j/g_{j-1}\}} = 1 + \log_2 \frac{\tau_j}{\tau_{j-1}} + \log_2 \max\left\{1, \frac{g_{j-1}}{g_j}\right\}.$$

Next, let us sum the number of rejected trials to obtain  $x_N$

$$\sum_{j=0}^{N-1} b_j = \log_2 \frac{\tau_0}{\eta_0} + (N - 1) + \log_2 \frac{\tau_{N-1}}{\tau_0} + \sum_{j=1}^{N-1} \log_{2,+} \frac{g_{j-1}}{g_j}.$$

Let us bound the last sum

$$\sum_{j=1}^{N-1} \log_{2,+} \frac{g_{j-1}}{g_j} = \sum_{j=1}^{N-1} \left( \log_{2,+} \frac{g_j}{g_{j-1}} + \log_2 \frac{g_{j-1}}{g_j} \right) = \sum_{j=1}^{N-1} \log_{2,+} \frac{g_j}{g_{j-1}} + \log_2 \frac{g_0}{g_{N-1}}$$

Therefore,

$$\sum_{j=0}^{N-1} b_j = N - 1 + \log_2 \frac{g_0 \tau_{N-1}}{\eta_0 g_{N-1}} + \sum_{j=1}^{N-1} \log_{2,+} \frac{g_j}{g_{j-1}}. \quad (45)$$

By (15), we have  $g_j \leq (2/\sqrt{3})g_{j-1}$ . Therefore,

$$\sum_{j=1}^{N-1} \log_{2,+} \frac{g_j}{g_{j-1}} \leq (N-1) \log_{2,+} \frac{2}{\sqrt{3}} \leq \frac{N-1}{4}. \quad (46)$$

Now, it remains to bound  $\log_{2,+}(\tau_{N-1}/g_{N-1})$ . By Lemma C.2, every accepted  $\tau$  satisfies

$$\tau_j \leq \max \left\{ \eta_j, 8\delta, 4\sqrt{L(x_j)g_j}, \frac{4g_j}{\rho} \right\}.$$

Therefore,

$$\frac{\tau_j}{g_j} \leq \max \left\{ \frac{\eta_j}{g_j}, \frac{8\delta}{g_j}, 4\sqrt{\frac{M_0}{g_j} + M_1}, \frac{4}{\rho} \right\}. \quad (47)$$

By the definitions (32) of  $B$  and  $\varepsilon_{\text{tr}}$

$$\delta \leq \frac{B\varepsilon_{\text{tr}}}{D^2}, \quad M_0 \leq \frac{B^2\varepsilon_{\text{tr}}}{D^3}, \quad M_1 \leq \frac{B^2}{36D^2}, \quad \frac{1}{\rho} \leq \frac{B}{2D}.$$

Next, by definition,  $N$  is the first iteration, such that  $f(x_N) - f^* \leq \varepsilon$ . Then for  $j < N$  we have  $g_j \geq \varepsilon/(2D)$ . Therefore,

$$\begin{aligned} \frac{8\delta}{g_j} &\leq \frac{16\delta D}{\varepsilon} \leq \frac{16B}{D} \frac{\varepsilon_{\text{tr}}}{\varepsilon}, \quad \frac{4}{\rho} \leq \frac{2B}{D} \leq \frac{16B}{D} \\ 4\sqrt{\frac{M_0}{g_j} + M_1} &\leq 4\sqrt{\frac{2M_0 D}{\varepsilon} + M_1} \leq \frac{4B}{D} \sqrt{2\frac{\varepsilon_{\text{tr}}}{\varepsilon} + \frac{1}{36}} \leq \frac{16B}{D} \max \left\{ 1, \frac{\varepsilon_{\text{tr}}}{\varepsilon} \right\}, \end{aligned}$$

Moreover, for  $j \geq 1$ , the update for  $\eta_j$  gives

$$\frac{\eta_j}{g_j} = \frac{\tau_{j-1}}{2g_{j-1}} \min \left\{ 1, \frac{g_{j-1}}{g_j} \right\} \leq \frac{\tau_{j-1}}{2g_{j-1}}.$$

Substituting these estimates into (47) gives for  $1 \leq j < N$

$$\frac{\tau_0}{g_0} \leq \max \left\{ \frac{\eta_0}{g_0}, \frac{16B}{D} \max \left\{ 1, \frac{\varepsilon_{\text{tr}}}{\varepsilon} \right\} \right\}, \quad \frac{\tau_j}{g_j} \leq \max \left\{ \frac{\tau_{j-1}}{2g_{j-1}}, \frac{16B}{D} \max \left\{ 1, \frac{\varepsilon_{\text{tr}}}{\varepsilon} \right\} \right\}.$$

Induction therefore gives

$$\frac{\tau_j}{g_j} \leq \max \left\{ \frac{\eta_0}{g_0}, \frac{16B}{D} \max \left\{ 1, \frac{\varepsilon_{\text{tr}}}{\varepsilon} \right\} \right\}, \quad 0 \leq j < N. \quad (48)$$

Finally, by combining (45), (46), and (48), we obtain

$$\begin{aligned} \sum_{j=0}^{N-1} b_j &\leq \frac{5}{4}(N-1) + \log_2 \max \left\{ 1, \frac{16Bg_0}{D\eta_0} \max \left\{ 1, \frac{\varepsilon_{\text{tr}}}{\varepsilon} \right\} \right\} \\ &\leq \frac{5}{4}(N-1) + 4 + \log_{2,+} \frac{Bg_0}{D\eta_0} + \log_{2,+} \frac{\varepsilon_{\text{tr}}}{\varepsilon}. \end{aligned}$$

This proves (43).

Each trial requires at most one new gradient evaluation. Hence, the total number of gradient evaluations is at most  $1 + N + \sum_{j=0}^{N-1} b_j$ . Next, by the definition of  $\varepsilon_{\text{tr}}$ , we have  $\max \left\{ \sqrt{\frac{M_0 D^3}{\varepsilon_{\text{tr}}}}, \frac{\delta D^2}{\varepsilon_{\text{tr}}} \right\} = B \geq 1$ . Using  $\log_2 t \leq \min \{2\sqrt{t}, t\}$  for  $t \geq 1$ , we obtain  $\log_{2,+} \frac{\varepsilon_{\text{tr}}}{\varepsilon} \leq 2\sqrt{\frac{M_0 D^3}{\varepsilon}} + \frac{\delta D^2}{\varepsilon}$ . Combining these bounds with (4) proves (44).  $\square$

## C.6 Strongly Convex Analysis

Let  $f$  be strongly convex  $\mu$ -strongly convex for  $\mu > 0$ :

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\mu}{2} \|y - x\|^2, \quad x, y \in \mathbb{R}^d. \quad (49)$$

This implies

$$\frac{\mu}{2} \|x - x^*\|^2 \leq f(x) - f^* \leq \frac{\|\nabla f(x)\|^2}{2\mu}. \quad (50)$$

In particular, strong convexity with monotonicity (Lemma C.1) imply

$$\|x_k - x^*\| \leq \sqrt{\frac{2(f(x_k) - f^*)}{\mu}} \leq \sqrt{\frac{2(f(x_0) - f^*)}{\mu}}. \quad (51)$$

We also denote  $\delta_k := \|\nabla^2 f(x_k) - H(x_k)\| \leq \delta$  for all  $h \geq 0$ .

### C.6.1 One Step Estimates

**Lemma C.10.** *Under (49), every completed iteration of Algorithm 2 satisfies*

$$f(x_{k+1}) - f^* \leq \left( \frac{3\tau_k}{3\tau_k + 2\mu} \right)^2 (f(x_k) - f^*), \quad (52)$$

$$\|s_k\| \leq \frac{\|\nabla f(x_k)\|}{\mu + \tau_k/2} \leq \frac{\|\nabla f(x_k)\|}{\mu}. \quad (53)$$

*Proof.* By (16),

$$-\langle \nabla f(x_{k+1}), s_k \rangle \geq \frac{\|\nabla f(x_{k+1})\|^2}{2\tau_k} + \frac{3\tau_k}{8} \|s_k\|^2.$$

Apply (49) at  $x_{k+1}$ , then use (13) and (50):

$$\begin{aligned}
f(x_k) - f(x_{k+1}) &\geq -\langle \nabla f(x_{k+1}), s_k \rangle + \frac{\mu}{2} \|s_k\|^2 \geq \frac{\|\nabla f(x_{k+1})\|^2}{2\tau_k} + \left( \frac{3\tau_k}{8} + \frac{\mu}{2} \right) \|s_k\|^2 \\
&\geq \left( \frac{1}{2\tau_k} + \frac{4}{9\tau_k^2} \left( \frac{3\tau_k}{8} + \frac{\mu}{2} \right) \right) \|\nabla f(x_{k+1})\|^2 = \left( \frac{2}{3\tau_k} + \frac{2\mu}{9\tau_k^2} \right) \|\nabla f(x_{k+1})\|^2 \\
&\geq \left( \frac{4\mu}{3\tau_k} + \frac{4\mu^2}{9\tau_k^2} \right) (f(x_{k+1}) - f^*).
\end{aligned}$$

The last inequality uses (50) at  $x_{k+1}$ :

$$\|\nabla f(x_{k+1})\|^2 \geq 2\mu(f(x_{k+1}) - f^*).$$

Adding  $f(x_{k+1}) - f^*$  to the bound for  $f(x_k) - f(x_{k+1})$  and completing the square gives

$$\begin{aligned}
f(x_k) - f^* &\geq \left( 1 + \frac{4\mu}{3\tau_k} + \frac{4\mu^2}{9\tau_k^2} \right) (f(x_{k+1}) - f^*) \\
&= \left( 1 + \frac{2\mu}{3\tau_k} \right)^2 (f(x_{k+1}) - f^*),
\end{aligned}$$

which is (52). For (53), recall that  $s_k = x_{k+1} - x_k$ . Strong convexity gives

$$\mu \|s_k\|^2 \leq \langle \nabla f(x_{k+1}) - \nabla f(x_k), s_k \rangle.$$

The accepted MS test (2) gives

$$\|\nabla f(x_{k+1}) + \tau_k s_k\| \leq \frac{\tau_k}{2} \|s_k\|.$$

Hence, by Cauchy-Schwarz,

$$\begin{aligned}
\langle \nabla f(x_{k+1}), s_k \rangle &= \langle \nabla f(x_{k+1}) + \tau_k s_k, s_k \rangle - \tau_k \|s_k\|^2 \\
&\leq \|\nabla f(x_{k+1}) + \tau_k s_k\| \|s_k\| - \tau_k \|s_k\|^2 \leq \frac{\tau_k}{2} \|s_k\|^2 - \tau_k \|s_k\|^2 = -\frac{\tau_k}{2} \|s_k\|^2.
\end{aligned}$$

Combining the two bounds and applying Cauchy-Schwarz once more gives

$$\begin{aligned}
\mu \|s_k\|^2 &\leq -\frac{\tau_k}{2} \|s_k\|^2 - \langle \nabla f(x_k), s_k \rangle, \\
\left( \mu + \frac{\tau_k}{2} \right) \|s_k\|^2 &\leq -\langle \nabla f(x_k), s_k \rangle \leq \|\nabla f(x_k)\| \|s_k\|.
\end{aligned}$$

Dividing by  $\|s_k\| > 0$  and using  $\tau_k > 0$  gives

$$\|s_k\| \leq \frac{\|\nabla f(x_k)\|}{\mu + \tau_k/2} \leq \frac{\|\nabla f(x_k)\|}{\mu}.$$

□

### C.6.2 Global Linear Convergence with Inexact Hessians

We recall definition of  $r_0$  from (18):

$$r_0 = \begin{cases} \frac{1}{6\sqrt{M_1}}, & \text{under (GH),} \\ \min \left\{ \frac{\rho}{2}, \frac{1}{6\sqrt{M_1}} \right\}, & \text{under (LH).} \end{cases}$$

**Theorem C.11.** *Suppose that (49) and (IH) hold, and either (GH) or (LH) holds. Define*

$$\bar{\tau} := \max \left\{ \eta_0, 8\delta + 48M_0 \sqrt{\frac{2(f(x_0) - f^*)}{\mu}}, \frac{2(f(x_0) - f^*)}{r_0^2} \right\}. \quad (54)$$

*Every completed iteration satisfies  $\tau_k \leq \bar{\tau}$ , and*

$$f(x_k) - f^* \leq \left( \frac{3\bar{\tau}}{3\bar{\tau} + 2\mu} \right)^{2k} (f(x_0) - f^*), \quad k \geq 0. \quad (55)$$

*Proof.* All searches terminate by Lemma C.2. We prove  $\tau_k \leq \bar{\tau}$  by induction.

Suppose that on iteration  $k$  at least one trial is rejected, so  $\tau_k > \eta_k$ . If  $\|s_k\| \leq r_0$ , then (19) and (51) give

$$\begin{aligned} \tau_k &\leq 8\delta_k + 24M_0\|s_k\| \leq 8\delta + 24M_0(\|x_k - x^*\| + \|x_{k+1} - x^*\|) \\ &\leq 8\delta + 48M_0 \sqrt{\frac{2(f(x_0) - f^*)}{\mu}} \leq \bar{\tau}. \end{aligned}$$

If  $\|s_k\| > r_0$ , then (14) gives

$$\frac{\tau_k}{2} r_0^2 < \frac{\tau_k}{2} \|s_k\|^2 \leq f(x_k) - f(x_{k+1}) \leq f(x_0) - f^*,$$

so again  $\tau_k \leq \bar{\tau}$ .

Now, let's assume that in iteration  $k$  there is no rejection. For  $k = 0$ , the accepted value is  $\tau_0 = \eta_0 \leq \bar{\tau}$ . For  $k \geq 1$ , line 4 gives

$$\tau_k = \eta_k \leq \frac{\tau_{k-1}}{2} \leq \bar{\tau}.$$

Thus the induction closes. Since  $0 < \tau_k \leq \bar{\tau}$ ,

$$\left( \frac{3\tau_k}{3\tau_k + 2\mu} \right)^2 = \left( 1 - \frac{2\mu}{3\tau_k + 2\mu} \right)^2 \leq \left( 1 - \frac{2\mu}{3\bar{\tau} + 2\mu} \right)^2 = \left( \frac{3\bar{\tau}}{3\bar{\tau} + 2\mu} \right)^2.$$

Substitution into (52) and iteration prove (55).  $\square$

**Corollary C.12.** *Under the assumptions of Theorem C.11, let  $0 < \varepsilon < f(x_0) - f^*$  and  $N := \min \{k \geq 0 : f(x_k) - f^* \leq \varepsilon\}$ . The total number of gradient evaluations before obtaining  $x_N$  is at most*

$$1 + \frac{9}{4}N + \log_{2,+} \left( \frac{\|\nabla f(x_0)\|}{\eta_0} \max \left\{ \frac{24M_0}{\mu} + \frac{8\delta}{\sqrt{2\mu\varepsilon}}, \frac{2}{r_0} \right\} \right). \quad (56)$$

*Consequently, it is bounded by*

$$1 + \frac{9}{4} \left\lceil \frac{\log((f(x_0) - f^*)/\varepsilon)}{2 \log \left( 1 + \frac{2\mu}{3\bar{\tau}} \right)} \right\rceil + \log_{2,+} \left( \frac{\|\nabla f(x_0)\|}{\eta_0} \max \left\{ \frac{24M_0}{\mu} + \frac{8\delta}{\sqrt{2\mu\varepsilon}}, \frac{2}{r_0} \right\} \right).$$

*For  $\delta = 0$ , the logarithmic term in (56) is independent of  $\varepsilon$ , and the bound holds for any  $N \geq 1$  completed iterations.*

*Proof.* Let  $g_k := \|\nabla f(x_k)\|$  and let  $b_k$  count the rejected trials on iteration  $k$ . By (45) and (46),

$$\sum_{k=0}^{N-1} b_k \leq \frac{5}{4}(N-1) + \log_2 \frac{g_0 \tau_{N-1}}{\eta_0 g_{N-1}}.$$

We prove by induction that

$$\frac{\tau_k}{g_k} \leq \max \left\{ \frac{2^{-k} \eta_0}{g_0}, \frac{24M_0}{\mu} + \frac{8\delta}{g_k}, \frac{2}{r_0} \right\}. \quad (57)$$

Suppose first that at least one trial is rejected. If  $\|s_k\| \leq r_0$ , then (19) and (53) give

$$\tau_k \leq 8\delta + 24M_0 \|s_k\| \leq 8\delta + \frac{24M_0}{\mu} g_k.$$

If  $\|s_k\| > r_0$ , then (53) gives

$$\tau_k \leq \frac{2g_k}{\|s_k\|} < \frac{2g_k}{r_0}.$$

Both estimates imply (57) for every  $k \geq 0$  with a rejection.

For  $k = 0$ , the estimates above apply if a trial is rejected. Otherwise,  $\tau_0 = \eta_0$ , and

$$\frac{\tau_0}{g_0} = \frac{\eta_0}{g_0} \leq \max \left\{ \frac{\eta_0}{g_0}, \frac{24M_0}{\mu} + \frac{8\delta}{g_0}, \frac{2}{r_0} \right\}.$$

Thus (57) holds for  $k = 0$ . Now let  $k \geq 1$  and assume it holds for  $k - 1$ . If a trial is rejected, the estimates above apply again. If there is no rejection, line 4 gives

$$\begin{aligned} \tau_k &= \frac{\tau_{k-1}}{2} \min \left\{ 1, \frac{g_k}{g_{k-1}} \right\} \leq \max \left\{ 2^{-k} \eta_0 \frac{g_k}{g_0}, 4\delta + \frac{12M_0}{\mu} g_k, \frac{g_k}{r_0} \right\} \\ &\leq \max \left\{ 2^{-k} \eta_0 \frac{g_k}{g_0}, 8\delta + \frac{24M_0}{\mu} g_k, \frac{2g_k}{r_0} \right\}. \end{aligned}$$

This proves (57) by induction. By the definition of  $N$ , we have  $f(x_{N-1}) - f^* > \varepsilon$ . By (50),  $g_{N-1} > \sqrt{2\mu\varepsilon}$ . Substituting this into (57), we obtain

$$\begin{aligned} \frac{g_0 \tau_{N-1}}{\eta_0 g_{N-1}} &\leq \max \left\{ 2^{-(N-1)}, \frac{g_0}{\eta_0} \left( \frac{24M_0}{\mu} + \frac{8\delta}{g_{N-1}} \right), \frac{2g_0}{\eta_0 r_0} \right\} \\ &\leq \max \left\{ 1, \frac{g_0}{\eta_0} \left( \frac{24M_0}{\mu} + \frac{8\delta}{\sqrt{2\mu\varepsilon}} \right), \frac{2g_0}{\eta_0 r_0} \right\}. \end{aligned}$$

Each trial passing PD requires at most one new gradient evaluation. The accepted gradient is reused at the next iteration. Hence the total number of gradient evaluations is at most

$$1 + N + \sum_{k=0}^{N-1} b_k,$$

which, together with the preceding bounds, proves (56). □

### C.6.3 Proof of Theorem 3.3

We refine the global bound from Theorem C.11 to show superlinear convergence by retaining the individual Hessian errors  $\delta_k$  instead of replacing them by  $\delta$ .

**Theorem C.13.** *Under the assumptions of Theorem C.11, let  $\bar{\tau}$  be defined by (54) and set*

$$\bar{\tau}_k := \min \left\{ \bar{\tau}, 8 \max_{0 \leq j \leq k} 2^{j-k} \delta_j + \bar{\tau} \left( \frac{3\bar{\tau}}{3\bar{\tau} + 2\mu} \right)^k \right\}, \quad \zeta_k := \left( \frac{3\bar{\tau}_k}{3\bar{\tau}_k + 2\mu} \right)^2.$$

Every completed iteration satisfies

$$\begin{aligned} \tau_k &\leq \bar{\tau}_k, \\ f(x_{k+1}) - f^* &\leq \zeta_k (f(x_k) - f^*), \quad 0 < \zeta_k < 1. \end{aligned} \quad (58)$$

*Proof.* For every  $i \geq 1$ , by (13) and (53),

$$\|\nabla f(x_i)\| \leq \frac{3\tau_{i-1}}{2} \|s_{i-1}\| \leq \frac{3\tau_{i-1}}{\tau_{i-1} + 2\mu} \|\nabla f(x_{i-1})\|.$$

Dividing by  $\|\nabla f(x_{i-1})\| > 0$  and substituting into the update in line 4 gives

$$\frac{\eta_i}{\tau_{i-1}} = \frac{1}{2} \min \left\{ 1, \frac{\|\nabla f(x_i)\|}{\|\nabla f(x_{i-1})\|} \right\} \leq \min \left\{ \frac{1}{2}, \frac{3\tau_{i-1}}{2\tau_{i-1} + 4\mu} \right\} \leq \min \left\{ \frac{1}{2}, \frac{3\bar{\tau}}{2\bar{\tau} + 4\mu} \right\} \leq \frac{3\bar{\tau}}{3\bar{\tau} + 2\mu}, \quad (59)$$

where we used  $\tau_{i-1} \leq \bar{\tau}$  from Theorem C.11. Fix a completed iteration  $k$ .

If no trial was rejected during iterations  $0, \dots, k$ , then  $\tau_i = \eta_i$  for  $i = 0, \dots, k$ . By (59),

$$\tau_k = \eta_0 \prod_{i=1}^k \frac{\eta_i}{\tau_{i-1}} \leq \eta_0 \left( \frac{3\bar{\tau}}{3\bar{\tau} + 2\mu} \right)^k \leq \bar{\tau} \left( \frac{3\bar{\tau}}{3\bar{\tau} + 2\mu} \right)^k.$$

Otherwise, let  $j$  be the last iteration among  $0, \dots, k$  on which a trial is rejected. By (50) and (55),

$$\begin{aligned} \|s_j\| &\leq \|x_j - x^*\| + \|x_{j+1} - x^*\| \\ &\leq \sqrt{\frac{2(f(x_j) - f^*)}{\mu}} + \sqrt{\frac{2(f(x_{j+1}) - f^*)}{\mu}} \leq 2\sqrt{\frac{2(f(x_0) - f^*)}{\mu}} \left( \frac{3\bar{\tau}}{3\bar{\tau} + 2\mu} \right)^j. \end{aligned}$$

Since a trial is rejected on iteration  $j$ , the two cases in the proof of Theorem C.11 give the first bound below. Substituting the step estimate above and (55), we obtain

$$\begin{aligned} \tau_j &\leq \max \left\{ 8\delta_j + 24M_0\|s_j\|, \frac{2(f(x_j) - f^*)}{r_0^2} \right\} \\ &\leq \max \left\{ 8\delta_j + 48M_0\sqrt{\frac{2(f(x_0) - f^*)}{\mu}} \left( \frac{3\bar{\tau}}{3\bar{\tau} + 2\mu} \right)^j, \frac{2(f(x_0) - f^*)}{r_0^2} \left( \frac{3\bar{\tau}}{3\bar{\tau} + 2\mu} \right)^{2j} \right\}. \end{aligned}$$

By (54) we obtain

$$\tau_j \leq 8\delta_j + \bar{\tau} \left( \frac{3\bar{\tau}}{3\bar{\tau} + 2\mu} \right)^j. \quad (60)$$

By the choice of  $j$ , no trial is rejected on iterations  $j + 1, \dots, k$ , so  $\tau_i = \eta_i$  for  $i = j + 1, \dots, k$ . Applying (59), we obtain

$$\prod_{i=j+1}^k \frac{\eta_i}{\tau_{i-1}} \leq \prod_{i=j+1}^k \frac{1}{2} = 2^{j-k}, \quad \prod_{i=j+1}^k \frac{\eta_i}{\tau_{i-1}} \leq \prod_{i=j+1}^k \frac{3\bar{\tau}}{3\bar{\tau} + 2\mu} = \left( \frac{3\bar{\tau}}{3\bar{\tau} + 2\mu} \right)^{k-j}.$$

Substituting (60), we obtain

$$\begin{aligned} \tau_k &= \tau_j \prod_{i=j+1}^k \frac{\eta_i}{\tau_{i-1}} \leq \left( 8\delta_j + \bar{\tau} \left( \frac{3\bar{\tau}}{3\bar{\tau} + 2\mu} \right)^j \right) \prod_{i=j+1}^k \frac{\eta_i}{\tau_{i-1}} \leq 8\delta_j 2^{j-k} + \bar{\tau} \left( \frac{3\bar{\tau}}{3\bar{\tau} + 2\mu} \right)^j \left( \frac{3\bar{\tau}}{3\bar{\tau} + 2\mu} \right)^{k-j} \\ &= 8\delta_j 2^{j-k} + \bar{\tau} \left( \frac{3\bar{\tau}}{3\bar{\tau} + 2\mu} \right)^k \leq 8 \max_{0 \leq i \leq k} 2^{i-k} \delta_i + \bar{\tau} \left( \frac{3\bar{\tau}}{3\bar{\tau} + 2\mu} \right)^k. \end{aligned}$$

Thus, in both cases,

$$\tau_k \leq 8 \max_{0 \leq j \leq k} 2^{j-k} \delta_j + \bar{\tau} \left( \frac{3\bar{\tau}}{3\bar{\tau} + 2\mu} \right)^k. \quad (61)$$

Combining (61) with  $\tau_k \leq \bar{\tau}$  gives  $\tau_k \leq \bar{\tau}_k$ .

Since  $0 < \tau_k \leq \bar{\tau}_k$ ,

$$\left( \frac{3\tau_k}{3\tau_k + 2\mu} \right)^2 = \left( 1 - \frac{2\mu}{3\tau_k + 2\mu} \right)^2 \leq \left( 1 - \frac{2\mu}{3\bar{\tau}_k + 2\mu} \right)^2 = \zeta_k.$$

Substitution into (52) proves (58).  $\square$

#### C.6.4 Global Superlinear Convergence under Relative Hessian Accuracy

We consider Hessian approximations satisfying

$$\delta_k = \|H(x_k) - \nabla^2 f(x_k)\| \leq \nu \|\nabla f(x_k)\|, \quad k \geq 0, \quad (62)$$

for some  $\nu > 0$ .

**Corollary C.14.** *Under the assumptions of Theorem C.11, suppose additionally that (62) holds. The method either returns a minimizer or converges superlinearly in function value:*

$$\frac{f(x_{k+1}) - f^*}{f(x_k) - f^*} \leq \left( \frac{3\tau_k}{3\tau_k + 2\mu} \right)^2 \rightarrow 0.$$

The first  $N \geq 1$  completed iterations require at most  $N$  Hessian evaluations and

$$1 + \frac{9}{4}N + \log_{2,+} \left( \frac{\|\nabla f(x_0)\|}{\eta_0} \max \left\{ \frac{24M_0}{\mu} + 8\nu, \frac{2}{r_0} \right\} \right) \quad (63)$$

gradient evaluations.

*Proof.* Let  $g_k := \|\nabla f(x_k)\|$ . In the proof of (57), retain the current error  $\delta_k$  on iterations with a rejection and use  $\delta_k/g_k \leq \nu$  from (62). The same bounds for the two cases give

$$\frac{\tau_0}{g_0} \leq \max \left\{ \frac{\eta_0}{g_0}, \frac{24M_0}{\mu} + 8\nu, \frac{2}{r_0} \right\}, \quad \frac{\tau_k}{g_k} \leq \max \left\{ \frac{\tau_{k-1}}{2g_{k-1}}, \frac{24M_0}{\mu} + 8\nu, \frac{2}{r_0} \right\}, \quad k \geq 1.$$

Iterating this inequality gives

$$\frac{\tau_k}{g_k} \leq \max \left\{ \frac{2^{-k}\eta_0}{g_0}, \frac{24M_0}{\mu} + 8\nu, \frac{2}{r_0} \right\}. \quad (64)$$

By (14) and (55), for  $k \geq 1$ ,

$$g_k^2 \leq \frac{3\tau_{k-1}}{2} (f(x_{k-1}) - f(x_k)) \leq \frac{3\bar{\tau}}{2} (f(x_{k-1}) - f^*) \leq \frac{3\bar{\tau}(f(x_0) - f^*)}{2} \left( \frac{3\bar{\tau}}{3\bar{\tau} + 2\mu} \right)^{2(k-1)}.$$

Substituting into (64) gives

$$\tau_k \leq \sqrt{\frac{3\bar{\tau}(f(x_0) - f^*)}{2}} \left( \frac{3\bar{\tau}}{3\bar{\tau} + 2\mu} \right)^{k-1} \max \left\{ \frac{2^{-k}\eta_0}{g_0}, \frac{24M_0}{\mu} + 8\nu, \frac{2}{r_0} \right\} \rightarrow 0.$$

Thus, (52) proves superlinear convergence.

Let  $b_k$  count the rejected trials on iteration  $k$ . By (45) and (46),

$$\sum_{k=0}^{N-1} b_k \leq \frac{5}{4}(N-1) + \log_2 \frac{g_0\tau_{N-1}}{\eta_0 g_{N-1}}.$$

By (64),

$$\frac{g_0\tau_{N-1}}{\eta_0 g_{N-1}} \leq \max \left\{ 2^{-(N-1)}, \frac{g_0}{\eta_0} \left( \frac{24M_0}{\mu} + 8\nu \right), \frac{2g_0}{\eta_0 r_0} \right\} \leq \max \left\{ 1, \frac{g_0}{\eta_0} \left( \frac{24M_0}{\mu} + 8\nu \right), \frac{2g_0}{\eta_0 r_0} \right\}.$$

Hence

$$\sum_{k=0}^{N-1} b_k \leq \frac{5}{4}(N-1) + \log_{2,+} \left( \frac{g_0}{\eta_0} \max \left\{ \frac{24M_0}{\mu} + 8\nu, \frac{2}{r_0} \right\} \right).$$

The gradient count is at most  $1 + N + \sum_{k=0}^{N-1} b_k$ , which proves (63).  $\square$

### C.6.5 Local Quadratic Convergence

**Theorem C.15.** *Under the assumptions of Corollary C.14, set*

$$C := \frac{9(\|\nabla^2 f(x^*)\| + \mu)}{2\mu^2} \left( 8\nu + \frac{24M_0}{\mu} \right)^2.$$

*Unless a minimizer is returned earlier, let  $K_0 \geq 1$  satisfy*

$$f(x_{K_0}) - f^* \leq \min \left\{ \frac{\mu r_0^2}{8}, \frac{1}{2C} \right\} \quad (65)$$

*and define*

$$K := K_0 + \left\lceil \log_{2,+} \frac{\tau_{K_0-1}}{2 \left( 8\nu + \frac{24M_0}{\mu} \right) \sqrt{2\mu(f(x_{K_0-1}) - f^*)}} \right\rceil. \quad (66)$$

*Then, for every  $k \geq K$ ,*

$$f(x_{k+1}) - f^* \leq C(f(x_k) - f^*)^2. \quad (67)$$

*The gradient norms and the distances to  $x^*$  also converge quadratically. After a finite initial cost independent of  $\varepsilon$ , attaining  $f(x) - f^* \leq \varepsilon$  requires  $\mathcal{O}(\log \log(1/\varepsilon))$  additional iterations, Hessian evaluations, and gradient evaluations.*

*Proof.* By Theorem C.11, an index  $K_0$  satisfying (65) exists unless a minimizer is returned earlier. Since  $\|\nabla^2 f(x^*)\| \geq \mu$ , the definition of  $C$  gives

$$C \geq \frac{9}{\mu} \left( \frac{24M_0}{\mu} \right)^2 \geq \frac{4M_0^2}{\mu^3}, \quad \frac{1}{2C} \leq \frac{\mu^3}{8M_0^2}.$$

Together with monotonicity and (50), condition (65) gives, for  $k \geq K_0$ ,

$$\begin{aligned} \|x_k - x^*\| &\leq \sqrt{\frac{2(f(x_k) - f^*)}{\mu}} \leq \frac{1}{2} \min \left\{ r_0, \frac{\mu}{M_0} \right\}, \\ \|s_k\| &\leq \|x_k - x^*\| + \|x_{k+1} - x^*\| \leq 2\sqrt{\frac{2(f(x_{K_0}) - f^*)}{\mu}} \leq r_0. \end{aligned}$$

Let  $g_k := \|\nabla f(x_k)\|$ . Since  $\|s_k\| \leq r_0$ , the proof of (64) applies without the term  $2/r_0$ . Starting at  $K_0$ , it gives, for  $j \geq 0$ ,

$$\frac{\tau_{K_0+j}}{g_{K_0+j}} \leq \max \left\{ \frac{2^{-j-1}\tau_{K_0-1}}{g_{K_0-1}}, 8\nu + \frac{24M_0}{\mu} \right\} \leq \max \left\{ \frac{2^{-j-1}\tau_{K_0-1}}{\sqrt{2\mu(f(x_{K_0-1}) - f^*)}}, 8\nu + \frac{24M_0}{\mu} \right\}.$$

By (66), for every  $k \geq K$ ,

$$\tau_k \leq \left( 8\nu + \frac{24M_0}{\mu} \right) g_k. \quad (68)$$

Combining (13) and (53), we obtain

$$g_{k+1} \leq \frac{3\tau_k}{2} \|s_k\| \leq \frac{3\tau_k}{2\mu} g_k \leq \frac{3}{2\mu} \left( 8\nu + \frac{24M_0}{\mu} \right) g_k^2. \quad (69)$$

Thus, the gradient norms converge quadratically. Next, we relate the gradient norm to the function gap. For  $\|z - x^*\| \leq \min \{r_0, \mu/M_0\}$ , either (GH) or (LH) gives

$$\|\nabla^2 f(z)\| \leq \|\nabla^2 f(x^*)\| + M_0\|z - x^*\| \leq \|\nabla^2 f(x^*)\| + \mu,$$

since  $\nabla f(x^*) = 0$  and  $r_0 \leq \rho/2$  under (LH). Thus,  $f$  has  $(\|\nabla^2 f(x^*)\| + \mu)$ -Lipschitz gradient on this ball. Hence, for  $k \geq K_0$ ,

$$\begin{aligned} g_k &\leq (\|\nabla^2 f(x^*)\| + \mu) \|x_k - x^*\|, \\ \left\| x_k - \frac{\nabla f(x_k)}{\|\nabla^2 f(x^*)\| + \mu} - x^* \right\| &\leq \|x_k - x^*\| + \frac{g_k}{\|\nabla^2 f(x^*)\| + \mu} \leq 2\|x_k - x^*\| \leq \min \left\{ r_0, \frac{\mu}{M_0} \right\}. \end{aligned}$$

By Lipschitzness of the gradient, for any  $y$  in the ball,

$$f(y) \leq f(x_k) + \langle \nabla f(x_k), y - x_k \rangle + \frac{\|\nabla^2 f(x^*)\| + \mu}{2} \|y - x_k\|^2.$$

Substituting the point above, we obtain

$$\begin{aligned} f^* &\leq f \left( x_k - \frac{\nabla f(x_k)}{\|\nabla^2 f(x^*)\| + \mu} \right) \leq f(x_k) - \frac{g_k^2}{\|\nabla^2 f(x^*)\| + \mu} + \frac{(\|\nabla^2 f(x^*)\| + \mu) g_k^2}{2(\|\nabla^2 f(x^*)\| + \mu)^2} \\ &= f(x_k) - \frac{g_k^2}{2(\|\nabla^2 f(x^*)\| + \mu)}. \end{aligned}$$

Rearranging gives

$$g_k^2 \leq 2 \left( \|\nabla^2 f(x^*)\| + \mu \right) (f(x_k) - f^*).$$

Substituting this and (68) into (52), for  $k \geq K$  we get

$$\begin{aligned} f(x_{k+1}) - f^* &\leq \left( \frac{3\tau_k}{3\tau_k + 2\mu} \right)^2 (f(x_k) - f^*) \\ &\leq \frac{9}{4\mu^2} \left( 8\nu + \frac{24M_0}{\mu} \right)^2 g_k^2 (f(x_k) - f^*) \leq C(f(x_k) - f^*)^2. \end{aligned}$$

This proves (67). For the distances, strong convexity and (69) give

$$\|x_{k+1} - x^*\| \leq \frac{g_{k+1}}{\mu} \leq \frac{3 \left( \|\nabla^2 f(x^*)\| + \mu \right)^2}{2\mu^2} \left( 8\nu + \frac{24M_0}{\mu} \right) \|x_k - x^*\|^2.$$

By (65),  $C(f(x_K) - f^*) \leq 1/2$ . Iterating (67) gives

$$f(x_{K+n}) - f^* \leq \frac{1}{C} (C(f(x_K) - f^*))^{2^n} \leq \frac{2^{-2^n}}{C}, \quad n \geq 0.$$

Thus,  $f(x_{K+n}) - f^* \leq \varepsilon$  after  $n = \mathcal{O}(\log \log(1/\varepsilon))$  additional iterations. By Corollary C.14, this requires  $\mathcal{O}(K + \log \log(1/\varepsilon))$  Hessian and gradient evaluations in total.  $\square$

## D Nesterov-type Acceleration

We present an accelerated cubic method for the global condition (GH). Fix a positive radius estimate  $\bar{R} \geq R$  and an initial scale  $\Gamma_0 > 0$  satisfying

$$\Gamma_0 \bar{R}^3 \geq 3(f(x_0) - f^*). \quad (70)$$

We discuss how to choose parameters satisfying (70) in Section D.2.

We present the resulting method as Algorithm 4. The acceleration mechanism aggregates the linear models  $\ell_w(x) := f(w) + \langle \nabla f(w), x - w \rangle$ . At each iteration,  $\alpha_k$  is defined by the scalar equation in line 5 of Algorithm 4. This definition is well posed because the mapping  $\alpha \mapsto \alpha^3/(1 - \alpha)$  is continuous and strictly increasing from 0 to  $+\infty$  on  $(0, 1)$ . Hence, for  $\Gamma_k > 0$  and  $\mathcal{M}_k > 0$ , the equation has a unique solution, which can be found by bisection.

**Theorem D.1.** *Assume that  $f$  is convex, (GH) and (70) hold. For every  $\varepsilon > 0$ , Algorithm 4 returns an iterate  $w_N$  satisfying  $f(w_N) - f^* \leq \varepsilon$  for*

$$N = \mathcal{O} \left( \left( \frac{M_0 \bar{R}^3}{\varepsilon} \right)^{1/3} + \left( 1 + \sqrt{M_1 \bar{R}} \right) \log \max \left\{ 1, \min \left\{ \frac{M_1^{3/2} \bar{R}^3 \Gamma_0}{M_0}, \frac{\Gamma_0 \bar{R}^3}{\varepsilon} \right\} \right\} \right). \quad (74)$$

**Proof sketch.** We use the standard accelerated-cubic estimating sequence technique Nesterov (2008). The main obstruction is that under (GH) the regularization coefficient  $\mathcal{M}_k$  must dominate the generalized smoothness parameter  $L(w_{k+1})$  at the output  $w_{k+1}$  of the cubic step (73), although this output is not known when the coefficient is selected. Inspired by Tyurin (2026), we provide an explicit way of choosing regularization parameter  $\mathcal{M}_k$  (71), such that this condition is satisfied. The complete proof is given in Appendix D.1.

---

**Algorithm 4** Accelerated cubic regularization under global generalized Hessian smoothness

---

- 1: **Input:**  $x_0 \in \mathbb{R}^d$ ,  $M_0 > 0$ ,  $M_1 \geq 0$ ,  $\bar{R} \geq R$  with  $\bar{R} > 0$ , and  $\Gamma_0 > 0$  satisfying  $\Gamma_0 \bar{R}^3 \geq 3(f(x_0) - f^*)$
- 2: Set  $A_0 := 1$ ,  $w_0 := x_0$ , and  $z_0 := x_0$ ,  $\psi_0(x) := f(x_0) + \frac{\Gamma_0}{3} \|x - x_0\|^3$ .
- 3: **for**  $k \geq 0$  **do**
- 4:   Set

$$\mathcal{M}_k := \max \left\{ 2M_0, 12\sqrt{2}M_1^{3/2}\bar{R}^3\Gamma_k \right\}. \quad (71)$$

- 5:   Find the unique  $\alpha_k \in (0, 1)$  satisfying  $\frac{\alpha_k^3}{1-\alpha_k} = \frac{\Gamma_k}{72\mathcal{M}_k}$ .
- 6:   Set

$$A_{k+1} := \frac{A_k}{1-\alpha_k}, \quad a_{k+1} := \alpha_k A_{k+1}, \quad \Gamma_{k+1} := (1 - \alpha_k)\Gamma_k. \quad (72)$$

- 7:   Set  $v_k := (1 - \alpha_k)w_k + \alpha_k z_k$ .
- 8:   Compute

$$s_k := \operatorname{argmin}_{s \in \mathbb{R}^d} \left\{ \langle \nabla f(v_k), s \rangle + \frac{1}{2} \langle \nabla^2 f(v_k) s, s \rangle + \frac{15\mathcal{M}_k}{6} \|s\|^3 \right\}, \quad w_{k+1} := v_k + s_k. \quad (73)$$

- 9:   Set  $\psi_{k+1}(x) := \psi_k(x) + a_{k+1}\ell_{w_{k+1}}(x)$  and compute  $z_{k+1} := \operatorname{argmin}_{x \in \mathbb{R}^d} \psi_{k+1}(x)$ .
- 

Let us consider one step of Algorithm 4. For some  $\mathcal{M} > 0$ , let  $A_+(\mathcal{M})$ ,  $w(\mathcal{M})$ , and  $\psi_+(\cdot; \mathcal{M})$  denote the quantities produced by one iteration, and set  $\psi_+^*(\mathcal{M}) = \min_{x \in \mathbb{R}^d} \psi_+(x; \mathcal{M})$ . Assume first that  $\mathcal{M} \geq L(w(\mathcal{M}))$ . Then the standard accelerated-cubic estimating-sequence argument gives

$$A_+(\mathcal{M})f(w(\mathcal{M})) + \frac{A_+(\mathcal{M})}{9} \frac{\|\nabla f(w(\mathcal{M}))\|^{3/2}}{\sqrt{\mathcal{M}}} \leq \psi_+^*(\mathcal{M}) \leq A_+(\mathcal{M})f^* + \frac{2}{3}\Gamma_0\bar{R}^3. \quad (75)$$

Thus, conditioned on proper choice of  $\mathcal{M} \geq L(w(\mathcal{M}))$ , the usual estimating-sequence invariant closes. It remains to choose  $\mathcal{M}$  before  $w(\mathcal{M})$  is known. Consider the virtual family of steps and set  $\mathcal{U}_k = \{\mathcal{M} > 0 : \mathcal{M} < L(w(\mathcal{M}))\}$ . If this set is nonempty, let  $\mathcal{M}_k^\dagger = \sup \mathcal{U}_k$ . Since all sufficiently large  $\mathcal{M}$  are admissible,  $\mathcal{M}_k^\dagger < +\infty$ , and continuity gives  $\mathcal{M}_k^\dagger = L(w(\mathcal{M}_k^\dagger))$ . Let  $g^\dagger = \nabla f(w(\mathcal{M}_k^\dagger))$ . Combining the two sides of (75) at this boundary yields

$$\frac{\|g^\dagger\|^{3/2}}{\sqrt{\mathcal{M}_k^\dagger}} \leq 6\Gamma_k\bar{R}^3, \quad \mathcal{M}_k^\dagger \leq \max \left\{ 2M_0, 12\sqrt{2}M_1^{3/2}\bar{R}^3\Gamma_k \right\} = \mathcal{M}_k.$$

Hence the explicit  $\mathcal{M}_k$  used by the algorithm is admissible. This is why we retain the positive remainder in (75). Although it can be omitted in the standard convergence argument, here it controls the gradient at the unknown output and closes the explicit choice of  $\mathcal{M}$ . For the actual step, admissibility gives

$$A_{k+1}(f(w_{k+1}) - f^*) \leq \psi_{k+1}^* - A_{k+1}f^* \leq \frac{2}{3}\Gamma_0\bar{R}^3.$$

Then, recurrence for  $\Gamma_k$  then gives the rate (74).

## D.1 Proof of Nesterov-type Acceleration under (GH)

### D.1.1 Auxiliary Lemmas

**Lemma D.2** (Nesterov (2008, Lemma 2)). *For any  $a \geq 0$ ,  $g \in \mathbb{R}^d$ , and  $\beta > 0$ ,*

$$\min_{h \in \mathbb{R}^d} \left\{ \frac{\beta}{3!} \|h\|^3 + a \langle g, h \rangle \right\} = -\frac{2\sqrt{2}}{3} \frac{a^{3/2} \|g\|^{3/2}}{\sqrt{\beta}}. \quad (76)$$

**Lemma D.3** (Agafonov et al. (2024a, Lemma 7)). *Let  $h : \mathbb{R}^d \rightarrow \mathbb{R}$  be convex,  $x_0 \in \mathbb{R}^d$ , and  $\theta > 0$ . Define*

$$\varphi(x) := h(x) + \frac{\theta}{3} \|x - x_0\|^3, \quad \bar{x} := \operatorname{argmin}_{x \in \mathbb{R}^d} \varphi(x).$$

*Then, for every  $u \in \mathbb{R}^d$ ,*

$$\varphi(\bar{x} + u) \geq \varphi(\bar{x}) + \frac{\theta}{6} \|u\|^3.$$

**Lemma D.4.** *The weight updates in Algorithm 4 admit the following equivalent representation:*

$$a_{k+1} := A_{k+1} - A_k, \quad \Gamma_k = \frac{\Gamma_0}{A_k}, \quad v_k = \frac{A_k w_k + a_{k+1} z_k}{A_{k+1}}, \quad a_{k+1}^3 = \frac{\Gamma_0 A_{k+1}^2}{72 \mathcal{M}_k}. \quad (77)$$

*Proof.* Since  $A_0 = 1$ , the updates of  $A_k$  and  $\Gamma_k$  (72) give

$$A_{k+1} \Gamma_{k+1} = \frac{A_k}{1 - \alpha_k} (1 - \alpha_k) \Gamma_k = A_k \Gamma_k = \Gamma_0,$$

and hence  $\Gamma_k = \Gamma_0 / A_k$ . Moreover,

$$\begin{aligned} A_{k+1} - A_k &= A_{k+1} - (1 - \alpha_k) A_{k+1} = \alpha_k A_{k+1} = a_{k+1}, \\ A_{k+1} v_k &= (1 - \alpha_k) A_{k+1} w_k + \alpha_k A_{k+1} z_k = A_k w_k + a_{k+1} z_k. \end{aligned}$$

Finally,

$$\frac{\alpha_k^3}{1 - \alpha_k} = \frac{\Gamma_k}{72 \mathcal{M}_k} \iff \alpha_k^3 A_{k+1}^3 = \frac{\Gamma_k A_k A_{k+1}^2}{72 \mathcal{M}_k} \iff a_{k+1}^3 = \frac{\Gamma_0 A_{k+1}^2}{72 \mathcal{M}_k}.$$

□

### D.1.2 Cubic Step Progress

**Lemma D.5.** *Fix  $v \in \mathbb{R}^d$  and  $\mathcal{M} > 0$ , and define*

$$s := \operatorname{argmin}_{h \in \mathbb{R}^d} \left\{ \langle \nabla f(v), h \rangle + \frac{1}{2} \langle \nabla^2 f(v) h, h \rangle + \frac{15\mathcal{M}}{6} \|h\|^3 \right\}. \quad (78)$$

*Set  $w := v + s$ . Suppose that either (GH) holds, or (LH) holds and  $\|s\| \leq \rho$ . If*

$$\mathcal{M} \geq L(w) = M_0 + M_1 \|\nabla f(w)\|,$$

*then*

$$\|\nabla f(w)\| \leq 9\mathcal{M} \|s\|^2, \quad (79)$$

$$\langle \nabla f(w), v - w \rangle \geq 6\mathcal{M} \|s\|^3, \quad (80)$$

$$\langle \nabla f(w), v - w \rangle \geq \frac{2}{9} \frac{\|\nabla f(w)\|^{3/2}}{\sqrt{\mathcal{M}}}. \quad (81)$$

*Proof.* By optimality condition for the cubic subproblem (78)

$$\nabla f(v) + \nabla^2 f(v)s + \frac{15\mathcal{M}}{2}\|s\|s = 0. \quad (82)$$

Under (GH), Lemma A.1 applies at  $w$  with  $v - w = -s$ . Under (LH), Lemma A.2 applies in the same way, since  $\|s\| \leq \rho$ . Together with the corresponding smoothness condition, this gives

$$\begin{aligned} \|\nabla f(v) - \nabla f(w) + \nabla^2 f(w)s\| &\leq \frac{L(w)}{2}\|s\|^2 \leq \frac{\mathcal{M}}{2}\|s\|^2, \\ \|\nabla^2 f(v) - \nabla^2 f(w)\| &\leq L(w)\|s\| \leq \mathcal{M}\|s\|. \end{aligned}$$

By the triangle inequality

$$\begin{aligned} &\|\nabla f(v) - \nabla f(w) + \nabla^2 f(v)s\| \\ &\leq \|\nabla f(v) - \nabla f(w) + \nabla^2 f(w)s\| + \|\nabla^2 f(v) - \nabla^2 f(w)\|\|s\| \leq \frac{3\mathcal{M}}{2}\|s\|^2. \end{aligned}$$

Combining this estimate with (82)

$$\begin{aligned} \|\nabla f(w)\| &= \|\nabla f(w) - (\nabla f(v) + \nabla^2 f(v)s + \frac{15\mathcal{M}}{2}\|s\|s)\| \\ &\leq \|\nabla f(v) - \nabla f(w) + \nabla^2 f(v)s\| + \frac{15\mathcal{M}}{2}\|s\|^2 \leq 9\mathcal{M}\|s\|^2, \end{aligned}$$

which proves (79).

Next,

$$\begin{aligned} \langle \nabla f(w), -s \rangle &\stackrel{(82)}{=} \left\langle \nabla f(v) - \nabla f(w) + \nabla^2 f(v)s, s \right\rangle + \frac{15\mathcal{M}}{2}\|s\|^3, \\ \langle \nabla f(w), -s \rangle &\geq -\frac{3\mathcal{M}}{2}\|s\|^3 + \frac{15\mathcal{M}}{2}\|s\|^3 = 6\mathcal{M}\|s\|^3, \end{aligned}$$

which proves (80). Finally, (79) implies

$$\|s\| \geq \sqrt{\frac{\|\nabla f(w)\|}{9\mathcal{M}}}.$$

Substituting this bound into (80) proves (81).  $\square$

### D.1.3 Estimating Sequence One Step Analysis

**Lemma D.6.** *Suppose that, at the beginning of iteration  $k$ ,*

$$\psi_k^* := \min_{x \in \mathbb{R}^d} \psi_k(x) \geq A_k f(w_k). \quad (83)$$

*Then*

$$\psi_{k+1}^* \geq A_{k+1}f(w_{k+1}) + A_{k+1} \langle \nabla f(w_{k+1}), v_k - w_{k+1} \rangle - \frac{A_{k+1}}{9} \frac{\|\nabla f(w_{k+1})\|^{3/2}}{\sqrt{\mathcal{M}_k}}. \quad (84)$$

*If, in addition, the hypotheses of Lemma D.5 hold for  $v = v_k$  and  $\mathcal{M} = \mathcal{M}_k$ , then*

$$\psi_{k+1}^* \geq A_{k+1}f(w_{k+1}) + \frac{A_{k+1}}{9} \frac{\|\nabla f(w_{k+1})\|^{3/2}}{\sqrt{\mathcal{M}_k}}. \quad (85)$$

*Consequently,*

$$\psi_{k+1}^* \geq A_{k+1}f(w_{k+1}). \quad (86)$$

*Proof.* By the definition of  $\psi_k$ ,

$$\psi_k(x) = f(x_0) + \sum_{i=0}^{k-1} a_{i+1} \ell_{w_{i+1}}(x) + \frac{\Gamma_0}{3} \|x - x_0\|^3.$$

The first two terms are affine and hence convex. Therefore, Lemma D.3 with  $\theta = \Gamma_0$  gives for any  $h \in \mathbb{R}^d$

$$\psi_k(z_k + h) \geq \psi_k^* + \frac{\Gamma_0}{6} \|h\|^3. \quad (87)$$

Next, using assumption of this Lemma (83) and Lemma D.2,

$$\begin{aligned} \psi_{k+1}^* &= \min_{x \in \mathbb{R}^d} \psi_{k+1}(x) = \min_{h \in \mathbb{R}^d} \{\psi_k(z_k + h) + a_{k+1} \ell_{w_{k+1}}(z_k + h)\} \\ &= a_{k+1} f(w_{k+1}) + \min_{h \in \mathbb{R}^d} \{\psi_k(z_k + h) + a_{k+1} \langle \nabla f(w_{k+1}), z_k + h - w_{k+1} \rangle\} \\ &\stackrel{(87)}{\geq} \psi_k^* + a_{k+1} f(w_{k+1}) + a_{k+1} \langle \nabla f(w_{k+1}), z_k - w_{k+1} \rangle \\ &\quad + \min_{h \in \mathbb{R}^d} \left\{ \frac{\Gamma_0}{6} \|h\|^3 + a_{k+1} \langle \nabla f(w_{k+1}), h \rangle \right\} \\ &\stackrel{(83)}{\geq} A_k f(w_k) + a_{k+1} f(w_{k+1}) + a_{k+1} \langle \nabla f(w_{k+1}), z_k - w_{k+1} \rangle \\ &\quad + \min_{h \in \mathbb{R}^d} \left\{ \frac{\Gamma_0}{6} \|h\|^3 + a_{k+1} \langle \nabla f(w_{k+1}), h \rangle \right\} \\ &\stackrel{(76)}{=} A_k f(w_k) + a_{k+1} f(w_{k+1}) + a_{k+1} \langle \nabla f(w_{k+1}), z_k - w_{k+1} \rangle - \frac{2\sqrt{2}}{3} \frac{a_{k+1}^{3/2} \|\nabla f(w_{k+1})\|^{3/2}}{\sqrt{\Gamma_0}}. \end{aligned} \quad (88)$$

Relations (77) imply

$$\begin{aligned} A_{k+1} v_k &= A_k w_k + a_{k+1} z_k, & A_{k+1} w_{k+1} &= A_k w_{k+1} + a_{k+1} w_{k+1}, \\ a_{k+1} (z_k - w_{k+1}) &= A_{k+1} (v_k - w_{k+1}) - A_k (w_k - w_{k+1}). \end{aligned} \quad (89)$$

Consequently,

$$\begin{aligned} &A_k f(w_k) + a_{k+1} f(w_{k+1}) + a_{k+1} \langle \nabla f(w_{k+1}), z_k - w_{k+1} \rangle \\ &\stackrel{(89)}{=} A_{k+1} f(w_{k+1}) + A_{k+1} \langle \nabla f(w_{k+1}), v_k - w_{k+1} \rangle \\ &\quad + A_k (f(w_k) - f(w_{k+1}) - \langle \nabla f(w_{k+1}), w_k - w_{k+1} \rangle) \\ &\geq A_{k+1} f(w_{k+1}) + A_{k+1} \langle \nabla f(w_{k+1}), v_k - w_{k+1} \rangle, \end{aligned} \quad (90)$$

where the inequality follows from convexity of  $f$ . By (77),

$$\frac{2\sqrt{2}}{3} \frac{a_{k+1}^{3/2}}{\sqrt{\Gamma_0}} = \frac{A_{k+1}}{9\sqrt{\mathcal{M}_k}}.$$

Substituting this identity and (90) into (88) proves (84). Under the hypotheses of Lemma D.5, substituting (81) into (84) gives

$$\psi_{k+1}^* \geq A_{k+1} f(w_{k+1}) + \left( \frac{2A_{k+1}}{9\sqrt{\mathcal{M}_k}} - \frac{A_{k+1}}{9\sqrt{\mathcal{M}_k}} \right) \|\nabla f(w_{k+1})\|^{3/2},$$

which proves (85). Dropping the nonnegative second term proves (86).  $\square$

**Lemma D.7.** *For every  $k \geq 0$ ,*

$$\psi_k(x^*) \leq A_k f^* + \Delta_0 + \frac{\Gamma_0 R^3}{3}, \quad (91)$$

where  $\Delta_0 := f(x_0) - f^*$ . Moreover, under (70),

$$\Delta_0 + \frac{\Gamma_0 R^3}{3} \leq \frac{2}{3} \Gamma_0 \bar{R}^3. \quad (92)$$

*Proof.* By convexity,  $\ell_w(x^*) \leq f^*$  for every  $w$ . Since  $A_k = 1 + \sum_{i=0}^{k-1} a_{i+1}$ ,

$$\begin{aligned} \psi_k(x^*) &= f(x_0) + \frac{\Gamma_0}{3} R^3 + \sum_{i=0}^{k-1} a_{i+1} \ell_{w_{i+1}}(x^*) \\ &\leq f(x_0) + \frac{\Gamma_0}{3} R^3 + (A_k - 1) f^* = A_k f^* + \Delta_0 + \frac{\Gamma_0 R^3}{3}. \end{aligned}$$

Finally, (70) gives  $\Delta_0 \leq \Gamma_0 \bar{R}^3/3$ , and  $R \leq \bar{R}$  gives  $\Gamma_0 R^3/3 \leq \Gamma_0 \bar{R}^3/3$ , proving the statement.  $\square$

#### D.1.4 Explicit Choice of Regularization Parameter

Fix an iteration  $k$  and suppose that (83) holds. For  $\mathcal{M} > 0$ , let  $\alpha(\mathcal{M}) \in (0, 1)$  solve

$$\frac{\alpha(\mathcal{M})^3}{1 - \alpha(\mathcal{M})} = \frac{\Gamma_k}{72\mathcal{M}}. \quad (93)$$

Define

$$\begin{aligned} A_+(\mathcal{M}) &:= \frac{A_k}{1 - \alpha(\mathcal{M})}, \quad a(\mathcal{M}) := \alpha(\mathcal{M}) A_+(\mathcal{M}), \\ v(\mathcal{M}) &:= (1 - \alpha(\mathcal{M})) w_k + \alpha(\mathcal{M}) z_k, \end{aligned} \quad (94)$$

and let  $w(\mathcal{M})$  be the output of the cubic subproblem (78) with base point  $v(\mathcal{M})$  and regularization parameter  $\mathcal{M}$ . Finally, set

$$\psi_+(x; \mathcal{M}) := \psi_k(x) + a(\mathcal{M}) \ell_{w(\mathcal{M})}(x), \quad \psi_+^*(\mathcal{M}) := \min_{x \in \mathbb{R}^d} \psi_+(x; \mathcal{M}).$$

We call  $\mathcal{M}$  admissible if  $\mathcal{M} \geq L(w(\mathcal{M}))$ . The proof of Lemma D.6 directly applies to these quantities, since the weight identities in Lemma D.4 remain valid. Thus, every admissible  $\mathcal{M}$  satisfies

$$\psi_+^*(\mathcal{M}) \geq A_+(\mathcal{M}) f(w(\mathcal{M})) + \frac{A_+(\mathcal{M}) \|\nabla f(w(\mathcal{M}))\|^{3/2}}{9 \sqrt{\mathcal{M}}}. \quad (95)$$

It remains to verify admissibility for the explicit choice in Algorithm 4.

**Lemma D.8.** *The mappings  $\mathcal{M} \mapsto \|w(\mathcal{M}) - v(\mathcal{M})\|$  and  $\mathcal{M} \mapsto \mathcal{M} - L(w(\mathcal{M}))$  are continuous on  $(0, +\infty)$ , and*

$$\|w(\mathcal{M}) - v(\mathcal{M})\| \rightarrow 0, \quad \mathcal{M} - L(w(\mathcal{M})) \rightarrow +\infty \quad \text{as } \mathcal{M} \rightarrow +\infty. \quad (96)$$

*Proof.* The mapping  $\alpha \mapsto \alpha^3/(1 - \alpha)$  is continuous and strictly increasing on  $(0, 1)$ . Hence  $\alpha(\mathcal{M})$  and  $v(\mathcal{M})$  are continuous.

Let  $s(\mathcal{M}) := w(\mathcal{M}) - v(\mathcal{M})$ . For any sequence  $\mathcal{M}_j \rightarrow \mathcal{M} > 0$ , the optimality condition of the cubic subproblem and  $\nabla^2 f(v(\mathcal{M}_j)) \succeq 0$  give

$$\|s(\mathcal{M}_j)\|^2 \leq \frac{2\|\nabla f(v(\mathcal{M}_j))\|}{15\mathcal{M}_j}. \quad (97)$$

By (94)  $\alpha(\mathcal{M}_j) \rightarrow \alpha(\mathcal{M})$  as  $\mathcal{M}_j \rightarrow \mathcal{M}$ , therefore  $v(\mathcal{M}_j) \rightarrow v(\mathcal{M})$ . Since  $f$  is twice continuously differentiable  $\nabla f(v(\mathcal{M}_j)) \rightarrow \nabla f(v(\mathcal{M}))$ . Thus  $\{\nabla f(v(\mathcal{M}_j))\}$  is bounded and  $\{s(\mathcal{M}_j)\}$  is bounded as well.

Next, we show that  $s(\mathcal{M}_j) \rightarrow s(\mathcal{M})$ . By Bolzano-Weierstrass theorem bounded sequence  $\{s(\mathcal{M}_j)\}$  has a convergent subsequence  $\{s(\mathcal{M}_{j_i})\} \rightarrow \hat{s}$ . Taking the limit in the optimality condition of cubic subproblem shows that  $\hat{s}$  satisfies this optimality condition corresponding to  $\mathcal{M}$ . Since cubic subproblem is strictly convex, it has unique minimizer and  $\hat{s} = s(\mathcal{M})$ . The same argument applies to any convergent subsequence of  $\{s(\mathcal{M}_j)\}$ . Thus,  $s(\mathcal{M}_j) \rightarrow s(\mathcal{M})$  since  $s(\mathcal{M}_j)$  is bounded sequence with only one accumulation point. Therefore  $w(\mathcal{M}_j) \rightarrow w(\mathcal{M})$  and thus  $w(\mathcal{M})$  is continuous. Since  $L(\cdot)$  is continuous, the mappings  $\|s(\mathcal{M})\|$  and  $\mathcal{M} - L(w(\mathcal{M}))$  are continuous as well.

The estimate (97) gives  $s(\mathcal{M}) \rightarrow 0$  as  $\mathcal{M} \rightarrow +\infty$ , while  $\alpha(\mathcal{M}) \rightarrow 0$  by (93) and  $v(\mathcal{M}) \rightarrow w_k$  by (94). Hence  $w(\mathcal{M}) \rightarrow w_k$ , so  $L(w(\mathcal{M}))$  remains bounded. This gives us (96).  $\square$

**Lemma D.9.** *Suppose that (GH) and (83) hold. Then the parameter*

$$\mathcal{M}_k = \max \left\{ 2M_0, 12\sqrt{2}M_1^{3/2}\bar{R}^3\Gamma_k \right\}$$

*used in Algorithm 4 satisfies*

$$\mathcal{M}_k \geq L(w_{k+1}) = M_0 + M_1\|\nabla f(w_{k+1})\|. \quad (98)$$

*Consequently, estimates (85) and (86) hold for the actual iteration.*

*Proof.* Define

$$\mathcal{U}_k := \{\mathcal{M} > 0 : \mathcal{M} < L(w(\mathcal{M}))\}.$$

If  $\mathcal{U}_k = \emptyset$ , the statement is immediate. Otherwise, let  $\mathcal{M}_k^\dagger := \sup \mathcal{U}_k$ . Lemma D.8 shows that  $\mathcal{M}_k^\dagger < +\infty$ . By continuity,

$$\mathcal{M}_k^\dagger = L(w(\mathcal{M}_k^\dagger)), \quad (99)$$

and every  $\mathcal{M} \geq \mathcal{M}_k^\dagger$  is admissible.

Set

$$w^\dagger := w(\mathcal{M}_k^\dagger), \quad g^\dagger := \nabla f(w^\dagger), \quad A_+^\dagger := A_+(\mathcal{M}_k^\dagger).$$

Applying (95) at the admissible boundary and evaluating  $\psi_+(\cdot; \mathcal{M}_k^\dagger)$  at  $x^*$ , we obtain

$$A_+^\dagger f(w^\dagger) + \frac{A_+^\dagger}{9} \frac{\|g^\dagger\|^{3/2}}{\sqrt{\mathcal{M}_k^\dagger}} \leq \psi_+^*(\mathcal{M}_k^\dagger) \leq \psi_+(x^*; \mathcal{M}_k^\dagger) \leq A_+^\dagger f^* + \Delta_0 + \frac{\Gamma_0 R^3}{3}.$$

Since  $f(w^\dagger) \geq f^*$ ,  $A_+^\dagger \geq A_k$ , and  $\Gamma_k = \Gamma_0/A_k$ , equation (92) gives

$$\frac{\|g^\dagger\|^{3/2}}{\sqrt{\mathcal{M}_k^\dagger}} \leq 6\Gamma_k \bar{R}^3. \quad (100)$$

For every  $g \in \mathbb{R}^d$ ,

$$M_0 + M_1\|g\| \leq \max \left\{ 2M_0, 2\sqrt{2}M_1^{3/2} \frac{\|g\|^{3/2}}{\sqrt{M_0 + M_1\|g\|}} \right\}. \quad (101)$$

Indeed, the first term applies when  $M_1\|g\| \leq M_0$ . Otherwise,  $M_0 + M_1\|g\| \leq 2M_1\|g\|$  and

$$\frac{\|g\|^{3/2}}{\sqrt{M_0 + M_1\|g\|}} \geq \frac{\|g\|}{\sqrt{2M_1}}.$$

Applying this bound with  $g = g^\dagger$  and using (99) and (100), we obtain

$$\mathcal{M}_k^\dagger \leq \max \left\{ 2M_0, 2\sqrt{2}M_1^{3/2} \frac{\|g^\dagger\|^{3/2}}{\sqrt{\mathcal{M}_k^\dagger}} \right\} \leq \max \left\{ 2M_0, 12\sqrt{2}M_1^{3/2} \bar{R}^3 \Gamma_k \right\} = \mathcal{M}_k.$$

Therefore  $\mathcal{M}_k$  is admissible and (98) holds. The last statement follows from Lemma D.6.  $\square$

### D.1.5 Estimating Sequence Bounds

**Lemma D.10.** *Suppose that (GH) holds. For every  $k \geq 0$ , the parameter  $\mathcal{M}_k$  in Algorithm 4 satisfies (98), and*

$$\psi_k^* \geq A_k f(w_k). \quad (102)$$

Moreover,

$$\psi_{k+1}^* \geq A_{k+1} f(w_{k+1}) + \frac{A_{k+1}}{9} \frac{\|\nabla f(w_{k+1})\|^{3/2}}{\sqrt{\mathcal{M}_k}}. \quad (103)$$

*Proof.* For  $k = 0$ ,

$$\psi_0^* = \min_{x \in \mathbb{R}^d} \left\{ f(x_0) + \frac{\Gamma_0}{3} \|x - x_0\|^3 \right\} = f(x_0) = A_0 f(w_0).$$

Suppose that (102) holds at iteration  $k$ . Theorem D.9 gives (98). Hence, Lemma D.6 gives (103) and

$$\psi_{k+1}^* \geq A_{k+1} f(w_{k+1}),$$

which closes the induction.  $\square$

**Lemma D.11.** *Suppose that (GH) holds. For every  $k \geq 0$ ,*

$$f(w_k) - f^* \leq \frac{\Delta_0 + \Gamma_0 R^3/3}{A_k} \leq \frac{2}{3} \Gamma_k \bar{R}^3. \quad (104)$$

Moreover, every cubic output satisfies

$$f(w_{k+1}) - f^* + \frac{1}{9} \frac{\|\nabla f(w_{k+1})\|^{3/2}}{\sqrt{\mathcal{M}_k}} \leq \frac{\Delta_0 + \Gamma_0 R^3/3}{A_{k+1}} \leq \frac{2}{3} \Gamma_{k+1} \bar{R}^3. \quad (105)$$

*Proof.* Combining (102) and (91),

$$A_k f(w_k) \stackrel{(102)}{\leq} \psi_k^* \leq \psi_k(x^*) \stackrel{(91)}{\leq} A_k f^* + \Delta_0 + \frac{\Gamma_0 R^3}{3}.$$

Therefore,

$$f(w_k) - f^* \leq \frac{\Delta_0 + \Gamma_0 R^3/3}{A_k} \stackrel{(92)}{\leq} \frac{2\Gamma_0 \bar{R}^3}{3A_k} \stackrel{(77)}{=} \frac{2}{3} \Gamma_k \bar{R}^3.$$

Similarly,

$$A_{k+1} f(w_{k+1}) + \frac{A_{k+1}}{9} \frac{\|\nabla f(w_{k+1})\|^{3/2}}{\sqrt{\mathcal{M}_k}} \stackrel{(103)}{\leq} \psi_{k+1}^* \leq \psi_{k+1}(x^*) \stackrel{(91)}{\leq} A_{k+1} f^* + \Delta_0 + \frac{\Gamma_0 R^3}{3}.$$

Dividing by  $A_{k+1}$  and using (92) and (77) proves (105).  $\square$

### D.1.6 Convergence Rates

Since  $\alpha_k \in (0, 1)$  and  $\Gamma_0 > 0$ ,

$$0 < \Gamma_{k+1} = (1 - \alpha_k) \Gamma_k < \Gamma_k.$$

Thus,  $\{\Gamma_k\}_{k \geq 0}$  is strictly decreasing. Define the transition level

$$\Gamma_{\text{tr}} := \begin{cases} \frac{M_0}{6\sqrt{2}M_1^{3/2}\bar{R}^3}, & M_1 > 0, \\ +\infty, & M_1 = 0. \end{cases} \quad (106)$$

**Lemma D.12** (Far regime). *Suppose that  $M_1 > 0$  and  $\Gamma_k > \Gamma_{\text{tr}}$ . Then*

$$\alpha_k \geq \frac{1}{12\sqrt[6]{2}(1 + \sqrt{M_1}\bar{R})}, \quad (107)$$

and

$$\Gamma_{k+1} \leq \exp\left(-\frac{1}{12\sqrt[6]{2}(1 + \sqrt{M_1}\bar{R})}\right) \Gamma_k. \quad (108)$$

*Proof.* In this regime,

$$\mathcal{M}_k = 12\sqrt{2}M_1^{3/2}\bar{R}^3\Gamma_k,$$

and hence

$$\frac{\alpha_k^3}{1 - \alpha_k} = \frac{1}{864\sqrt{2}M_1^{3/2}\bar{R}^3}.$$

If  $\alpha_k > 1/2$ , then (107) holds immediately. If  $\alpha_k \leq 1/2$ , then

$$\alpha_k^3 = \frac{1 - \alpha_k}{864\sqrt{2}M_1^{3/2}\bar{R}^3} \geq \frac{1}{1728\sqrt{2}M_1^{3/2}\bar{R}^3}, \quad \alpha_k \geq \frac{1}{12\sqrt[6]{2}\sqrt{M_1}\bar{R}}.$$

Combining the two cases proves (107). Finally,  $1 - \alpha_k \leq \exp(-\alpha_k)$  proves (108).  $\square$

**Lemma D.13** (Near regime). *Suppose that (GH) holds and  $\Gamma_K \leq \Gamma_{\text{tr}}$ . Then  $\mathcal{M}_j = 2M_0$  for every  $j \geq K$ , and*

$$A_{K+n}^{1/3} \geq A_K^{1/3} + \frac{n}{3} \left( \frac{\Gamma_0}{144M_0} \right)^{1/3}, \quad n \geq 1. \quad (109)$$

Consequently,

$$\Gamma_{K+n} \leq \frac{3888M_0}{n^3}, \quad n \geq 1, \quad (110)$$

and

$$f(w_{K+n}) - f^* \leq \frac{2592M_0\bar{R}^3}{n^3}, \quad n \geq 1. \quad (111)$$

*Proof.* The monotonicity of  $\Gamma_k$  and (106) give  $\mathcal{M}_j = 2M_0$  for every  $j \geq K$ . Therefore, (77) gives

$$a_{j+1}^3 = \frac{\Gamma_0 A_{j+1}^2}{144M_0}.$$

Using  $A_{j+1} - A_j = a_{j+1}$ ,

$$A_{j+1}^{1/3} - A_j^{1/3} = \frac{a_{j+1}}{A_{j+1}^{2/3} + A_{j+1}^{1/3}A_j^{1/3} + A_j^{2/3}} \geq \frac{a_{j+1}}{3A_{j+1}^{2/3}} = \frac{1}{3} \left( \frac{\Gamma_0}{144M_0} \right)^{1/3}.$$

Summing over  $j = K, \dots, K+n-1$  proves (109). Hence,

$$A_{K+n} \geq \frac{n^3\Gamma_0}{3888M_0}, \quad \Gamma_{K+n} \stackrel{(77)}{=} \frac{\Gamma_0}{A_{K+n}} \leq \frac{3888M_0}{n^3},$$

which proves (110). Combining (104) and (110) proves (111).  $\square$

**Theorem D.14.** *Suppose that  $f$  is convex, (GH) and (70) hold. For every  $\varepsilon > 0$ , Algorithm 4 returns an iterate  $w_N$  satisfying*

$$f(w_N) - f^* \leq \varepsilon$$

after at most

$$N \leq \left\lceil 12\sqrt[6]{2} \left( 1 + \sqrt{M_1\bar{R}} \right) \log \max \left\{ 1, \min \left\{ \frac{6\sqrt{2}M_1^{3/2}\bar{R}^3\Gamma_0}{M_0}, \frac{2\Gamma_0\bar{R}^3}{3\varepsilon} \right\} \right\} \right\rceil + \left\lceil \left( \frac{2592M_0\bar{R}^3}{\varepsilon} \right)^{1/3} \right\rceil \quad (112)$$

iterations.

*Proof.* Define

$$\Gamma_\varepsilon := \frac{3\varepsilon}{2\bar{R}^3}.$$

By (104), if  $\Gamma_k \leq \Gamma_\varepsilon$ , then

$$f(w_k) - f^* \leq \frac{2}{3}\Gamma_k\bar{R}^3 \leq \frac{2}{3}\Gamma_\varepsilon\bar{R}^3 = \varepsilon.$$

Suppose first that  $M_1 > 0$ , and set

$$N_{\text{far}} := \left\lceil 12\sqrt[6]{2} \left( 1 + \sqrt{M_1\bar{R}} \right) \log \max \left\{ 1, \frac{\Gamma_0}{\max \{\Gamma_{\text{tr}}, \Gamma_\varepsilon\}} \right\} \right\rceil.$$

Suppose that  $\Gamma_k > \max \{\Gamma_{\text{tr}}, \Gamma_\varepsilon\}$  for every  $k = 0, \dots, N_{\text{far}} - 1$ . Then (108) applies at each of these iterations, and

$$\Gamma_{N_{\text{far}}} \leq \exp \left( -\frac{N_{\text{far}}}{12\sqrt[6]{2}(1 + \sqrt{M_1 R})} \right) \Gamma_0 \leq \max \{\Gamma_{\text{tr}}, \Gamma_\varepsilon\}.$$

Therefore, after at most  $N_{\text{far}}$  iterations, either  $\Gamma_k \leq \Gamma_\varepsilon$ , or  $\Gamma_k \leq \Gamma_{\text{tr}}$  and the near regime begins. Moreover,

$$\begin{aligned} \frac{\Gamma_0}{\Gamma_{\text{tr}}} &= \frac{6\sqrt{2}M_1^{3/2}\bar{R}^3\Gamma_0}{M_0}, & \frac{\Gamma_0}{\Gamma_\varepsilon} &= \frac{2\Gamma_0\bar{R}^3}{3\varepsilon}, \\ \frac{\Gamma_0}{\max \{\Gamma_{\text{tr}}, \Gamma_\varepsilon\}} &= \min \left\{ \frac{6\sqrt{2}M_1^{3/2}\bar{R}^3\Gamma_0}{M_0}, \frac{2\Gamma_0\bar{R}^3}{3\varepsilon} \right\}. \end{aligned}$$

If the near regime is needed, Lemma D.13 shows that

$$N_{\text{near}} := \left\lceil \left( \frac{2592M_0\bar{R}^3}{\varepsilon} \right)^{1/3} \right\rceil$$

additional iterations suffice. Hence,  $N \leq N_{\text{far}} + N_{\text{near}}$ , which proves (112).

If  $M_1 = 0$ , then  $\Gamma_{\text{tr}} = +\infty$  and the near regime starts at  $K = 0$ . The first term in (112) is zero, while Lemma D.13 gives the second term.  $\square$

## D.2 Adaptive Method under (LH) with Exact Hessian

Algorithm 4 uses the problem parameters in its regularization coefficient and requires an initial scale satisfying (70). We now give one implementation that does not use  $M_0$ ,  $M_1$ ,  $R$ ,  $\rho$ , or  $f^*$ .

**Initialization step.** Our first goal is to construct a computable initial scale  $\Gamma_0$  satisfying

$$\Gamma_0 R^3 \geq 3(f(x_0) - f^*). \quad (113)$$

without knowing  $R$  and  $f^*$ . We construct them by solving the cubic subproblem (115) with backtracking regularization parameter  $\tilde{\mathcal{M}}$  until condition (116) holds. The whole warm-up backtracking procedure is listed in lines 4-7 of Algorithm 5. The following lemma shows that the backtracking procedure terminates after a finite number of iterations and provides an exact way of satisfying initial condition (113).

**Lemma D.15.** *Suppose that either (GH) or (LH) holds. If  $\nabla f(x_0) = 0$ , Algorithm 5 returns the optimal point  $x_0$ . Otherwise, the initialization backtracking (lines 4-7 of Algorithm 5) satisfies (116) after at most  $\left\lceil \log_2 \max \left\{ 1, \frac{T_0}{\mathcal{M}_0} \right\} \right\rceil$  rejected trials, where*

$$T_0 = \begin{cases} M_0 + 2M_1\|\nabla f(x_0)\|, & \text{under (GH),} \\ \max \left\{ M_0 + 2M_1\|\nabla f(x_0)\|, \frac{2\|\nabla f(x_0)\|}{15\rho^2} \right\}, & \text{under (LH).} \end{cases} \quad (114)$$

For the accepted step,  $\Gamma_0 := \frac{3\|\nabla f(x_0)\|^3}{(f(x_0) - f(\tilde{x}))^2}$  is well defined and satisfies initial condition (113).

For the statement below, define

$$B := \begin{cases} 12\sqrt{2}M_1^{3/2}R^3, & \text{under (GH),} \\ \max \left\{ 12\sqrt{2}M_1^{3/2}R^3, \frac{2R^3}{9\rho^3} \right\}, & \text{under (LH).} \end{cases} \quad (118)$$

---

**Algorithm 5** Fully adaptive accelerated cubic regularization

---

- 1: **Input:**  $x_0 \in \mathbb{R}^d$  and  $\tilde{\mathcal{M}}_0 > 0$ .
- 2: If  $\nabla f(x_0) = 0$ , return  $x_0$ .
- 3: Set  $\tilde{\mathcal{M}} := \tilde{\mathcal{M}}_0$ .
- 4: Compute

$$\tilde{s} := \operatorname{argmin}_{s \in \mathbb{R}^d} \left\{ \langle \nabla f(x_0), s \rangle + \frac{1}{2} \langle \nabla^2 f(x_0) s, s \rangle + \frac{15\tilde{\mathcal{M}}}{6} \|s\|^3 \right\}, \quad \tilde{x} := x_0 + \tilde{s}. \quad (115)$$

- 5: **while** condition

$$\langle \nabla f(\tilde{x}), -\tilde{s} \rangle \geq \frac{2}{9\sqrt{\tilde{\mathcal{M}}}} \|\nabla f(\tilde{x})\|^{3/2}. \quad (116)$$

does not hold **do**

- 6:   Set  $\tilde{\mathcal{M}} := 2\tilde{\mathcal{M}}$ .
- 7:   Recompute  $\tilde{s}$  and  $\tilde{x}$  according to (115).
- 8: Set  $\Gamma_0 := \frac{3\|\nabla f(x_0)\|^3}{(f(x_0) - f(\tilde{x}))^2}$ ,  $A_0 := 1$ ,  $w_0 := x_0$ ,  $z_0 := x_0$ , and  $\psi_0(x) := f(x_0) + \frac{\Gamma_0}{3} \|x - x_0\|^3$ .
- 9: Set  $\mathcal{M}_0 := \Gamma_0$ .
- 10: **for**  $k \geq 0$  **do**
- 11:   Find the unique  $\alpha_k \in (0, 1)$  satisfying  $\frac{\alpha_k^3}{1 - \alpha_k} = \frac{\Gamma_k}{72\mathcal{M}_k}$ .
- 12:   Set  $A_{k+1}$ ,  $a_{k+1}$ , and  $\Gamma_{k+1}$  according to (72).
- 13:   Set  $v_k := (1 - \alpha_k)w_k + \alpha_k z_k$ .
- 14:   Compute  $s_k$  and  $w_{k+1}$  according to (73) with regularization coefficient  $\mathcal{M}_k$ .
- 15:   **while** condition

$$\langle \nabla f(w_{k+1}), v_k - w_{k+1} \rangle \geq \frac{2}{9\sqrt{\mathcal{M}_k}} \|\nabla f(w_{k+1})\|^{3/2}. \quad (117)$$

does not hold **do**

- 16:   Set  $\mathcal{M}_k := 2\mathcal{M}_k$ .
  - 17:   Recompute  $\alpha_k$ ,  $A_{k+1}$ ,  $a_{k+1}$ ,  $\Gamma_{k+1}$ , and  $v_k$  as above.
  - 18:   Recompute  $w_{k+1}$  according to (73) with regularization coefficient  $\mathcal{M}_k$ .
  - 19:   Set  $\psi_{k+1}(x) := \psi_k(x) + a_{k+1}\ell_{w_{k+1}}(x)$  and compute  $z_{k+1} := \operatorname{argmin}_{x \in \mathbb{R}^d} \psi_{k+1}(x)$ .
  - 20:   Set  $\mathcal{M}_{k+1} := \frac{\Gamma_{k+1}}{\Gamma_k} \mathcal{M}_k$ .
- 

**Theorem D.16.** Suppose that  $f$  is convex and either (GH) or (LH) holds. All backtracking loops in Algorithm 5 terminate. For every  $\varepsilon > 0$ , under (GH), the method returns an iterate  $w_N$  satisfying  $f(w_N) - f^* \leq \varepsilon$  for

$$N = \mathcal{O} \left( \left( \frac{M_0 R^3}{\varepsilon} \right)^{1/3} \right) + \tilde{\mathcal{O}} \left( 1 + \sqrt{M_1} R \right). \quad (119)$$

Under (LH), the corresponding bound is

$$N = \mathcal{O} \left( \left( \frac{M_0 R^3}{\varepsilon} \right)^{1/3} \right) + \tilde{\mathcal{O}} \left( 1 + \sqrt{M_1} R + \frac{R}{\rho} \right). \quad (120)$$

The number of rejected initialization trials is bounded as in Lemma D.15. Before returning  $w_N$ , the main backtracking loops reject at most  $\left\lceil \log_2 \max \left\{ 1, B, \frac{4M_0 R^3}{3\varepsilon} \right\} \right\rceil$  trials in total, where  $B$  is defined in (118).

### D.2.1 Proof of Lemma D.15

*Proof.* Suppose that  $\nabla f(x_0) \neq 0$  and fix a trial value  $\tilde{\mathcal{M}} > 0$ . The optimality condition for (115) is

$$\nabla f(x_0) + \nabla^2 f(x_0)\tilde{s} + \frac{15\tilde{\mathcal{M}}}{2}\|\tilde{s}\|\tilde{s} = 0. \quad (121)$$

Since  $f$  is convex,  $\nabla^2 f(x_0) \succeq 0$ . Taking the inner product of (121) with  $\tilde{s}$ , we obtain

$$\frac{15\tilde{\mathcal{M}}}{2}\|\tilde{s}\|^3 \leq -\langle \nabla f(x_0), \tilde{s} \rangle \leq \|\nabla f(x_0)\|\|\tilde{s}\|,$$

and hence

$$\|\tilde{s}\|^2 \leq \frac{2\|\nabla f(x_0)\|}{15\tilde{\mathcal{M}}}. \quad (122)$$

Let  $\tilde{\mathcal{M}} \geq T_0$ , defined in (114). Under (LH), equations (114) and (122) give

$$\|\tilde{s}\|^2 \leq \frac{2\|\nabla f(x_0)\|}{15\tilde{\mathcal{M}}} \leq \rho^2.$$

Therefore, the Taylor remainder at  $x_0$  can be applied under both assumptions. Using (121), we get

$$\begin{aligned} \|\nabla f(\tilde{x})\| &\leq \|\nabla f(\tilde{x}) - \nabla f(x_0) - \nabla^2 f(x_0)\tilde{s}\| + \frac{15\tilde{\mathcal{M}}}{2}\|\tilde{s}\|^2 \\ &\leq \frac{M_0 + M_1\|\nabla f(x_0)\| + 15\tilde{\mathcal{M}}}{2}\|\tilde{s}\|^2 \\ &\leq \left(1 + \frac{M_0 + M_1\|\nabla f(x_0)\|}{15\tilde{\mathcal{M}}}\right)\|\nabla f(x_0)\| \leq \frac{16}{15}\|\nabla f(x_0)\|. \end{aligned}$$

Consequently,

$$M_0 + M_1\|\nabla f(\tilde{x})\| \leq M_0 + 2M_1\|\nabla f(x_0)\| \leq \tilde{\mathcal{M}}.$$

Since  $\|\tilde{s}\| \leq \rho$  under (LH), the hypotheses of Lemma D.5 hold under both assumptions. Therefore, (81) gives

$$\langle \nabla f(\tilde{x}), -\tilde{s} \rangle \geq \frac{2}{9\sqrt{\tilde{\mathcal{M}}}}\|\nabla f(\tilde{x})\|^{3/2},$$

which is condition (116). Hence, every trial with  $\tilde{\mathcal{M}} \geq T_0$  is accepted.

Starting from the input value  $\tilde{\mathcal{M}}_0$ , the  $j$ -th trial uses  $2^j\tilde{\mathcal{M}}_0$ . For

$$j = \left\lceil \log_2 \max \left\{ 1, \frac{T_0}{\tilde{\mathcal{M}}_0} \right\} \right\rceil,$$

we have  $2^j\tilde{\mathcal{M}}_0 \geq T_0$ . Therefore, the backtracking rejects at most  $j$  trials.

It remains to verify (113). If  $\nabla f(\tilde{x}) \neq 0$ , condition (116) and convexity give

$$f(x_0) - f(\tilde{x}) \geq \langle \nabla f(\tilde{x}), x_0 - \tilde{x} \rangle > 0.$$

If  $\nabla f(\tilde{x}) = 0$ , then  $\tilde{x}$  is optimal. Since  $\nabla f(x_0) \neq 0$ , in this case we also have  $f(x_0) - f(\tilde{x}) > 0$ . Moreover, by convexity,

$$0 < f(x_0) - f(\tilde{x}) \leq f(x_0) - f^* \leq \langle \nabla f(x_0), x_0 - x^* \rangle \leq \|\nabla f(x_0)\|R.$$

Thus,

$$0 < \frac{f(x_0) - f(\tilde{x})}{\|\nabla f(x_0)\|} \leq R.$$

Using this estimate one more time, we obtain

$$\begin{aligned} f(x_0) - f^* &\leq \|\nabla f(x_0)\| R = (f(x_0) - f(\tilde{x})) \frac{\|\nabla f(x_0)\| R}{f(x_0) - f(\tilde{x})} \\ &\leq (f(x_0) - f(\tilde{x})) \left( \frac{\|\nabla f(x_0)\| R}{f(x_0) - f(\tilde{x})} \right)^3 = \frac{\Gamma_0 R^3}{3}. \end{aligned}$$

This proves (113).  $\square$

### D.2.2 Proof of Theorem D.16

Fix an iteration  $k$ . For a trial coefficient  $\mathcal{M} > 0$ , Algorithm 5 computes  $\alpha_k$ ,  $A_{k+1}$ ,  $a_{k+1}$ ,  $v_k$ , and  $w_{k+1}$  in the same way as Algorithm 4 with  $\mathcal{M}_k = \mathcal{M}$ . Hence, under the induction assumption (83), Lemma D.6 gives (84) for every trial. If the trial passes (117), which is (81) for  $v = v_k$  and  $\mathcal{M} = \mathcal{M}_k$ , then the proof of Lemma D.6 gives (85) and (86).

We now prove that (117) holds for every sufficiently large trial coefficient. Let  $B$  be defined by (118).

**Lemma D.17.** *Suppose that either (GH) or (LH) holds and that (83) is satisfied. Every trial with  $\mathcal{M} \geq \max\{2M_0, B\Gamma_k\}$  satisfies (117).*

*Proof.* By (113), condition (70) holds with  $\bar{R} = R$ . Under (GH), we set  $\rho = +\infty$ . The trial with coefficient  $\mathcal{M}$  coincides with the virtual step  $w(\mathcal{M})$  defined after (94). Set  $r(\mathcal{M}) := \|w(\mathcal{M}) - v(\mathcal{M})\|$  and

$$\mathcal{U}_k := \{\mathcal{M} > 0 : r(\mathcal{M}) > \rho \text{ or } \mathcal{M} < L(w(\mathcal{M}))\}.$$

For  $\mathcal{M} \notin \mathcal{U}_k$ , the hypotheses of Lemma D.5 hold for  $v = v(\mathcal{M})$ , and (81) gives (117). Thus, it suffices to show that every  $\mathcal{M} \geq \max\{2M_0, B\Gamma_k\}$  satisfies  $\mathcal{M} \notin \mathcal{U}_k$ .

If  $\mathcal{U}_k = \emptyset$ , there is nothing to prove. Otherwise, Lemma D.8 shows that  $\mathcal{U}_k$  is open and bounded above. Let  $\mathcal{M}_k^\dagger := \sup \mathcal{U}_k < +\infty$ . Then  $\mathcal{M} \notin \mathcal{U}_k$  for every  $\mathcal{M} \geq \mathcal{M}_k^\dagger$ . By continuity,

$$r^\dagger \leq \rho, \quad \mathcal{M}_k^\dagger \geq L(w^\dagger), \quad r^\dagger = \rho \text{ or } \mathcal{M}_k^\dagger = L(w^\dagger), \quad (123)$$

where a dagger denotes evaluation at  $\mathcal{M}_k^\dagger$  and  $g^\dagger := \nabla f(w^\dagger)$ .

By (123), the hypotheses of Lemma D.5 hold at  $\mathcal{M}_k^\dagger$ . Hence, (79), (80), and (81) give

$$\begin{aligned} \left\langle g^\dagger, v^\dagger - w^\dagger \right\rangle - \frac{\|g^\dagger\|^{3/2}}{9\sqrt{\mathcal{M}_k^\dagger}} &\geq 6\mathcal{M}_k^\dagger (r^\dagger)^3 - \frac{(9\mathcal{M}_k^\dagger (r^\dagger)^2)^{3/2}}{9\sqrt{\mathcal{M}_k^\dagger}} = 3\mathcal{M}_k^\dagger (r^\dagger)^3, \\ \left\langle g^\dagger, v^\dagger - w^\dagger \right\rangle - \frac{\|g^\dagger\|^{3/2}}{9\sqrt{\mathcal{M}_k^\dagger}} &\geq \frac{\|g^\dagger\|^{3/2}}{9\sqrt{\mathcal{M}_k^\dagger}}. \end{aligned}$$

Lemma D.6 applies to the virtual step, as explained before (95). Therefore, (84) and evaluation of  $\psi_+(\cdot; \mathcal{M}_k^\dagger)$  at  $x^*$  give

$$A_+^\dagger f(w^\dagger) + A_+^\dagger \max \left\{ 3\mathcal{M}_k^\dagger (r^\dagger)^3, \frac{\|g^\dagger\|^{3/2}}{9\sqrt{\mathcal{M}_k^\dagger}} \right\} \leq \psi_+^*(\mathcal{M}_k^\dagger) \leq \psi_+(x^*; \mathcal{M}_k^\dagger) \leq A_+^\dagger f^* + \Delta_0 + \frac{\Gamma_0 R^3}{3}.$$

Since  $f(w^\dagger) \geq f^*$ ,  $A_+^\dagger \geq A_k$ , and  $\Gamma_k = \Gamma_0/A_k$ , equation (92) with  $\bar{R} = R$  gives

$$\max \left\{ 3\mathcal{M}_k^\dagger (r^\dagger)^3, \frac{\|g^\dagger\|^{3/2}}{9\sqrt{\mathcal{M}_k^\dagger}} \right\} \leq \frac{2}{3}\Gamma_k R^3. \quad (124)$$

If  $r^\dagger = \rho$ , then (124) gives

$$\mathcal{M}_k^\dagger \leq \frac{2R^3}{9\rho^3}\Gamma_k \leq B\Gamma_k.$$

If  $\mathcal{M}_k^\dagger = L(w^\dagger)$ , then (101) and (124) give

$$\mathcal{M}_k^\dagger \leq \max \left\{ 2M_0, 2\sqrt{2}M_1^{3/2} \frac{\|g^\dagger\|^{3/2}}{\sqrt{\mathcal{M}_k^\dagger}} \right\} \leq \max \left\{ 2M_0, 12\sqrt{2}M_1^{3/2} R^3 \Gamma_k \right\} \leq \max \{2M_0, B\Gamma_k\}.$$

In both cases, every  $\mathcal{M} \geq \max \{2M_0, B\Gamma_k\}$  satisfies  $\mathcal{M} \geq \mathcal{M}_k^\dagger$ , and hence  $\mathcal{M} \notin \mathcal{U}_k$ .  $\square$

**Lemma D.18.** *Suppose that either (GH) or (LH) holds. Every main backtracking loop in Algorithm 5 terminates. For every accepted iteration  $k \geq 0$ ,*

$$\mathcal{M}_k \leq \max \{4M_0, \max \{1, 2B\} \Gamma_k\}. \quad (125)$$

*For every  $K \geq 1$ , the total number of rejected main trials during iterations  $k = 0, \dots, K-1$  is at most*

$$\left\lceil \log_2 \max \left\{ 1, B, \frac{2M_0}{\Gamma_{K-1}} \right\} \right\rceil. \quad (126)$$

*Proof.* At  $k = 0$ , (83) holds with equality. Suppose that it holds at the beginning of iteration  $k$ . Repeated doubling sends  $\mathcal{M}_k$  to  $+\infty$ . Hence, Lemma D.17 shows that condition (117) is satisfied after finitely many trials. For the accepted trial, (86) gives (83) at iteration  $k+1$ . Thus, every backtracking loop terminates by induction.

It remains to prove (125). At  $k = 0$ , if the first trial is accepted, then

$$\mathcal{M}_0 = \Gamma_0 \leq \max \{1, 2B\} \Gamma_0.$$

If at least one trial is rejected, the coefficient immediately preceding the accepted one is  $\mathcal{M}_0/2$ . Lemma D.17 gives  $\frac{\mathcal{M}_0}{2} < \max \{2M_0, B\Gamma_0\}$ . Therefore,  $\mathcal{M}_0 < \max \{4M_0, 2B\Gamma_0\}$ , and (125) follows.

Suppose that (125) holds at iteration  $k-1$ . If the first trial is accepted at iteration  $k$ , then

$$\mathcal{M}_k = \frac{\Gamma_k}{\Gamma_{k-1}} \mathcal{M}_{k-1} \leq \frac{\Gamma_k}{\Gamma_{k-1}} \max \{4M_0, \max \{1, 2B\} \Gamma_{k-1}\} \leq \max \{4M_0, \max \{1, 2B\} \Gamma_k\},$$

where we used  $\Gamma_k \leq \Gamma_{k-1}$ . If at least one trial is rejected, the same argument as for  $k = 0$  proves the bound. This closes the induction.

Finally, the initialization and the update of  $\mathcal{M}_k$  in Algorithm 5 imply that the ratio  $\mathcal{M}_k/\Gamma_k$  is initially equal to one, remains unchanged between accepted iterations, and doubles after every rejected trial. Since  $\Gamma_k$  is decreasing, for every  $k = 0, \dots, K-1$ ,

$$\max \left\{ B, \frac{2M_0}{\Gamma_k} \right\} \leq \max \left\{ B, \frac{2M_0}{\Gamma_{K-1}} \right\}.$$

By Lemma D.17, every trial satisfying

$$\frac{\mathcal{M}_k}{\Gamma_k} \geq \max \left\{ B, \frac{2M_0}{\Gamma_{K-1}} \right\}$$

is accepted. After  $J$  rejected main trials, the current ratio is  $2^J$ . This proves (126).  $\square$

**Lemma D.19.** *Suppose that  $f$  is convex and either (GH) or (LH) holds. For every accepted iterate of Algorithm 5,*

$$f(w_k) - f^* \leq \frac{2}{3} \Gamma_k R^3. \quad (127)$$

*Proof.* The induction in the proof of Lemma D.18 shows that (83) holds for every accepted iteration. The argument in the proof of Lemma D.11, applied with  $\bar{R} = R$ , proves (127).  $\square$

For the scalar analysis, set

$$\hat{B} := \max \{1, 2B\}, \quad \Gamma_{\text{tr}}^{\text{ad}} := \frac{4M_0}{\hat{B}}. \quad (128)$$

The following two lemmas separate the convergence of the fully adaptive method into the far and near regimes. This separation is used only in the analysis and does not require the algorithm to know the transition level.

**Lemma D.20** (Far regime). *Suppose that  $f$  is convex, either (GH) or (LH) holds, and  $\Gamma_k > \Gamma_{\text{tr}}^{\text{ad}}$ . Then*

$$\alpha_k \geq \frac{1}{2 + (144\hat{B})^{1/3}}, \quad (129)$$

and

$$\Gamma_{k+1} \leq \exp \left( -\frac{1}{2 + (144\hat{B})^{1/3}} \right) \Gamma_k. \quad (130)$$

*Proof.* By (128),  $\hat{B}\Gamma_k > 4M_0$ . Therefore, equation (125) gives

$$\mathcal{M}_k \leq \hat{B}\Gamma_k, \quad \frac{\alpha_k^3}{1 - \alpha_k} = \frac{\Gamma_k}{72\mathcal{M}_k} \geq \frac{1}{72\hat{B}}.$$

If  $\alpha_k \leq 1/2$ , then

$$\alpha_k^3 \geq \frac{1}{144\hat{B}}.$$

If  $\alpha_k > 1/2$ , the bound  $\alpha_k > 1/2$  holds directly. Hence,

$$\alpha_k \geq \min \left\{ \frac{1}{2}, \frac{1}{(144\hat{B})^{1/3}} \right\} \geq \frac{1}{2 + (144\hat{B})^{1/3}}.$$

This proves (129). Equation (130) follows from  $\Gamma_{k+1} = (1 - \alpha_k)\Gamma_k$  and  $1 - t \leq \exp(-t)$ .  $\square$

**Lemma D.21** (Near regime). *Suppose that  $f$  is convex, either (GH) or (LH) holds, and  $\Gamma_K \leq \Gamma_{\text{tr}}^{\text{ad}}$ . Then, for every  $j \geq K$ ,*

$$\mathcal{M}_j \leq 4M_0. \quad (131)$$

Moreover,

$$A_{K+n}^{1/3} \geq A_K^{1/3} + \frac{n}{3} \left( \frac{\Gamma_0}{288M_0} \right)^{1/3}, \quad n \geq 1, \quad (132)$$

and

$$\Gamma_{K+n} \leq \frac{7776M_0}{n^3}, \quad (133)$$

$$f(w_{K+n}) - f^* \leq \frac{5184M_0R^3}{n^3}, \quad n \geq 1. \quad (134)$$

*Proof.* The sequence  $\{\Gamma_k\}_{k \geq 0}$  is decreasing. Thus,  $\widehat{B}\Gamma_j \leq 4M_0$  for every  $j \geq K$ , and equation (125) proves (131). By (77),

$$a_{j+1}^3 = \frac{\Gamma_0 A_{j+1}^2}{72\mathcal{M}_j} \geq \frac{\Gamma_0 A_{j+1}^2}{288M_0}.$$

Therefore,

$$\begin{aligned} A_{j+1}^{1/3} - A_j^{1/3} &= \frac{a_{j+1}}{A_{j+1}^{2/3} + A_{j+1}^{1/3}A_j^{1/3} + A_j^{2/3}} \\ &\geq \frac{a_{j+1}}{3A_{j+1}^{2/3}} \geq \frac{1}{3} \left( \frac{\Gamma_0}{288M_0} \right)^{1/3}. \end{aligned}$$

Summing this inequality proves (132). Hence,

$$A_{K+n} \geq \frac{n^3\Gamma_0}{7776M_0}, \quad \Gamma_{K+n} = \frac{\Gamma_0}{A_{K+n}} \leq \frac{7776M_0}{n^3}.$$

This proves (133). Combining it with (127) proves (134).  $\square$

*Proof of Theorem D.16.* If  $\nabla f(x_0) = 0$ , then  $x_0$  is optimal by convexity. Otherwise, Lemma D.15 shows that the initialization backtracking terminates, gives (113), and proves the stated bound on the number of rejected initialization trials.

Lemma D.19 gives (127). It remains to consider  $0 < \varepsilon < f(w_0) - f^*$ ; otherwise the convergence claim and the rejected-trial bound are immediate. In this case, condition (113) implies  $\Gamma_0 > 3\varepsilon/R^3$ , so the first index below is positive. Let  $N$  be the first index satisfying

$$\Gamma_N \leq \frac{3\varepsilon}{2R^3}.$$

Then (127) gives  $f(w_N) - f^* \leq \varepsilon$ .

As long as  $\Gamma_k > \max\{\Gamma_{\text{tr}}^{\text{ad}}, 3\varepsilon/(2R^3)\}$ , Lemma D.20 gives geometric contraction. Therefore, after at most

$$N_{\text{far}} := \left\lceil \left( 2 + (144\widehat{B})^{1/3} \right) \log \max \left\{ 1, \frac{\Gamma_0}{\max\{\Gamma_{\text{tr}}^{\text{ad}}, 3\varepsilon/(2R^3)\}} \right\} \right\rceil$$

iterations, either  $\Gamma_k \leq 3\varepsilon/(2R^3)$ , or the near regime begins. In the second case, Lemma D.21 shows that

$$N_{\text{near}} := \left\lceil \left( \frac{5184M_0R^3}{\varepsilon} \right)^{1/3} \right\rceil$$

additional iterations suffice. Consequently,  $N \leq N_{\text{far}} + N_{\text{near}}$ . By (128),

$$\frac{\Gamma_0}{\max\{\Gamma_{\text{tr}}^{\text{ad}}, 3\varepsilon/(2R^3)\}} = \min\left\{\frac{\widehat{B}\Gamma_0}{4M_0}, \frac{2\Gamma_0 R^3}{3\varepsilon}\right\}.$$

Thus,

$$N \leq \left\lceil \left(2 + (144\widehat{B})^{1/3}\right) \log \max\left\{1, \min\left\{\frac{\widehat{B}\Gamma_0}{4M_0}, \frac{2\Gamma_0 R^3}{3\varepsilon}\right\}\right\} \right\rceil + \left\lceil \left(\frac{5184M_0 R^3}{\varepsilon}\right)^{1/3} \right\rceil. \quad (135)$$

Under (GH), definition (118) gives

$$\widehat{B}^{1/3} = \mathcal{O}\left(1 + \sqrt{M_1}R\right).$$

Under (LH), it gives

$$\widehat{B}^{1/3} = \mathcal{O}\left(1 + \sqrt{M_1}R + \frac{R}{\rho}\right).$$

Substituting these estimates into (135) proves (119) and (120).

The minimality of  $N$  gives

$$\Gamma_{N-1} > \frac{3\varepsilon}{2R^3}, \quad \frac{2M_0}{\Gamma_{N-1}} < \frac{4M_0 R^3}{3\varepsilon}.$$

Therefore, (126) proves that the main backtracking loops reject at most  $\left\lceil \log_2 \max\left\{1, B, \frac{4M_0 R^3}{3\varepsilon}\right\} \right\rceil$  trials.  $\square$

## E Suboptimal A-NPE Method

In this section, we propose suboptimal adaptive A-NPE method, which matches the convergence rate of Nesterov-type acceleration in exact Hessian case. For (H) and (IH) it achieves convergence rate  $\mathcal{O}\left(\frac{LR^3}{N^3} + \frac{\delta R^2}{N^2}\right)$ .

We repeat the two backtracking tests for convenience:

$$\text{PD}(x, \tau) : H(x) + \tau I \succ 0, \quad \text{MS}(x, \tau, s) : \|\nabla f(x + s) + \tau s\| \leq \frac{\tau}{2}\|s\|. \quad (136)$$

For  $w, v \in \mathbb{R}^d$  and  $\tau > 0$ , we first check  $\text{PD}(w, \tau)$ . If it holds, we compute the trial step

$$(H(w) + \tau I)s = -\nabla f(v), \quad y = v + s, \quad (137)$$

and accept it when  $\text{MS}(v, \tau, s)$  holds.

The initialization backtracking keeps  $x_0$ ,  $\nabla f(x_0)$ , and  $H(x_0)$  fixed. In the main loop,  $H(w_k)$  remains fixed during backtracking, whereas  $v_k$  and  $\nabla f(v_k)$  are recomputed after each rejection.

For the convergence bounds of Algorithm 6, we use the convention  $\rho = +\infty$  under (GH) and define

$$B := \max\left\{1, 8\sqrt{M_1}R, \frac{8R}{\rho}\right\}. \quad (140)$$

If  $R = 0$ , then  $x_0$  is already a minimizer. For  $R > 0$ , set

$$\varepsilon_{\text{tr}} := \max\left\{\frac{20M_0 R^3}{B^3}, \frac{4\delta R^2}{B^2}\right\}.$$

---

**Algorithm 6** Adaptive A-NPE with an inexact Hessian

---

1: **Input:**  $x_0 \in \mathbb{R}^d$  and  $\tau_0 > 0$ .

2:  $(w_1, \tau_1) := \text{MSBACKTRACK}(x_0, \tau_0)$ .

3: Set

$$A_1 := \frac{1}{\tau_1}, \quad z_1 := x_0 - A_1 \nabla f(w_1). \quad (138)$$

4: **for**  $k \geq 1$  **do**

5:   Query  $H(w_k)$  once.

6:   Set

$$\alpha_k := \frac{2}{1 + \sqrt{1 + 4A_k \tau_k}}, \quad A_{k+1} := \frac{A_k}{1 - \alpha_k}, \quad a_{k+1} := \alpha_k A_{k+1}. \quad (139)$$

7:   Set  $v_k := (1 - \alpha_k)w_k + \alpha_k z_k$ .

8:   **if**  $\text{PD}(w_k, \tau_k)$  **then**

9:     Compute  $s_k$  and  $w_{k+1}$  according to (137) with  $w = w_k$ ,  $v = v_k$ , and  $\tau = \tau_k$ .

10:   **while not**  $\text{PD}(w_k, \tau_k)$  **or not**  $\text{MS}(v_k, \tau_k, s_k)$  **do**

11:     Set  $\tau_k := 2\tau_k$ .

12:     Recompute  $\alpha_k$ ,  $A_{k+1}$ , and  $a_{k+1}$  according to (139).

13:     Recompute  $v_k := (1 - \alpha_k)w_k + \alpha_k z_k$ .

14:     **if**  $\text{PD}(w_k, \tau_k)$  **then**

15:       Recompute  $s_k$  and  $w_{k+1}$  according to (137).

16:   Set

$$z_{k+1} := z_k - a_{k+1} \nabla f(w_{k+1}), \quad \tau_{k+1} := (1 - \alpha_k) \tau_k.$$


---

**Theorem E.1.** Suppose that  $f$  is convex, (IH) holds, and either (GH) or (LH) holds. Assume that  $R > 0$ . For every  $\varepsilon > 0$ , the method generates an iterate  $w_N$  satisfying  $f(w_N) - f^* \leq \varepsilon$  within

$$N = \mathcal{O} \left( B \left( 1 + \log_+ \frac{R^2}{2A_1 \max\{\varepsilon, \varepsilon_{\text{tr}}\}} \right) + \left( \frac{M_0 R^3}{\varepsilon} \right)^{1/3} + \sqrt{\frac{\delta R^2}{\varepsilon}} \right)$$

iterations. The number of rejections is at most

$$\left\lceil \log_2 \max \left\{ 1, \frac{4\delta}{\tau_0}, \frac{2\sqrt{L(x_0)\|\nabla f(x_0)\|}}{\tau_0}, \frac{2\|\nabla f(x_0)\|}{\rho\tau_0} \right\} \right\rceil + \left\lceil \log_2 \max \left\{ B^2, \frac{4\delta R^2}{\varepsilon}, \left( \frac{20M_0 R^3}{\varepsilon} \right)^{2/3} \right\} \right\rceil.$$

Moreover, there exists an integer  $K \geq 1$  with  $K = \mathcal{O} \left( B \max \left\{ 1, \log \frac{R^2}{2A_1 \varepsilon_{\text{tr}}} \right\} \right)$  such that

$$\begin{aligned} f(w_k) - f^* &= \mathcal{O} \left( \frac{R^2}{A_1} \exp \left( -\frac{c(k-1)}{B} \right) \right), & 1 \leq k \leq K, \\ f(w_{K+n}) - f^* &= \mathcal{O} \left( \frac{M_0 R^3}{(B+n)^3} + \frac{\delta R^2}{(B+n)^2} \right), & n = 0, 1, \dots \end{aligned}$$

Here  $c > 0$  and the constants in  $\mathcal{O}(\cdot)$  are universal.

## E.1 Auxiliary Lemmas

**Lemma E.2.** For every main-loop trial of Algorithm 6, the parameters defined in (139) satisfy

$$A_{k+1} = A_k + a_{k+1}, \quad \frac{a_{k+1}^2}{A_{k+1}} = \frac{1}{\tau_k}, \quad v_k = \frac{A_k w_k + a_{k+1} z_k}{A_{k+1}}, \quad A_{k+1} \tau_k = \frac{1}{\alpha_k^2}.$$

*Proof.* By (139), we have

$$\alpha_k = \frac{2}{1 + \sqrt{1 + 4A_k\tau_k}} = \frac{\sqrt{1 + 4A_k\tau_k} - 1}{2A_k\tau_k}.$$

Thus  $\alpha_k$  is the positive root of the quadratic equation

$$A_k\tau_k\alpha_k^2 + \alpha_k = 1, \quad \frac{\alpha_k^2}{1 - \alpha_k} = \frac{1}{A_k\tau_k}. \quad (141)$$

Using (139) and (141) gives

$$A_{k+1} - a_{k+1} = (1 - \alpha_k)A_{k+1} = A_k, \quad \frac{a_{k+1}^2}{A_{k+1}} = \alpha_k^2 A_{k+1} = \frac{\alpha_k^2 A_k}{1 - \alpha_k} = \frac{1}{\tau_k}.$$

The definition of  $v_k$  and the identities above imply

$$v_k = (1 - \alpha_k)w_k + \alpha_k z_k = \frac{A_k}{A_{k+1}}w_k + \frac{a_{k+1}}{A_{k+1}}z_k = \frac{A_k w_k + a_{k+1} z_k}{A_{k+1}},$$

$$A_{k+1}\tau_k = \frac{A_{k+1}^2}{a_{k+1}^2} = \frac{1}{\alpha_k^2}.$$

□

**Lemma E.3.** Suppose that either (GH) or (LH) holds. Fix  $w, v, s \in \mathbb{R}^d$ , set  $y := v + s$ , and let  $H(w)$  satisfy (IH) at  $w$ . Under (LH), assume additionally that

$$\|v - w\| + \|s\| \leq \rho.$$

Then

$$\|\nabla f(y) - \nabla f(v) - H(w)s\| \leq \left( \delta + L(y) \left( \|v - w\| + \frac{3}{2}\|s\| \right) \right) \|s\|. \quad (142)$$

*Proof.* By Lemmas A.1 A.2 and assumption of the lemma

$$\|\nabla f(y) - \nabla f(v) - \nabla^2 f(y)s\| = \|\nabla f(y - s) - \nabla f(y) + \nabla^2 f(y)s\| \leq \frac{L(y)}{2}\|s\|^2.$$

Also,

$$\|y - w\| = \|v + s - w\| \leq \|v - w\| + \|s\|,$$

which is at most  $\rho$  under (LH). Applying (GH) or (LH) with base point  $y$ , together with (IH) at  $w$ , gives

$$\begin{aligned} \|\nabla^2 f(y) - H(w)\| &\leq \|\nabla^2 f(y) - \nabla^2 f(w)\| + \|\nabla^2 f(w) - H(w)\| \\ &\leq L(y)\|y - w\| + \delta \leq \delta + L(y)(\|v - w\| + \|s\|). \end{aligned}$$

Adding and subtracting  $\nabla^2 f(y)s$ , we obtain

$$\begin{aligned} \|\nabla f(y) - \nabla f(v) - H(w)s\| &\leq \|\nabla f(y) - \nabla f(v) - \nabla^2 f(y)s\| + \|\nabla^2 f(y) - H(w)\|\|s\| \\ &\leq \frac{L(y)}{2}\|s\|^2 + (\delta + L(y)(\|v - w\| + \|s\|))\|s\| = \left( \delta + L(y) \left( \|v - w\| + \frac{3}{2}\|s\| \right) \right) \|s\|. \end{aligned}$$

This proves (142). □

For a trial step (137), the linear system gives

$$\nabla f(y) + \tau s = \nabla f(y) - \nabla f(v) - H(w)s.$$

## E.2 Step Estimates

**Lemma E.4.** *Let  $f$  be convex and differentiable. Fix  $w, z \in \mathbb{R}^d$  and  $A \geq 0$ . Choose  $a > 0$  and  $\tau > 0$  such that*

$$A_+ := A + a, \quad \frac{a^2}{A_+} = \frac{1}{\tau}, \quad v := \frac{A}{A_+}w + \frac{a}{A_+}z.$$

*Let  $\text{MS}(v, \tau, s)$  in (136) hold, i.e.*

$$\|\nabla f(v + s) + \tau s\| \leq \frac{\tau}{2}\|s\|, \quad (143)$$

*and define*

$$y := v + s, \quad z_+ := z - a\nabla f(y).$$

*Then*

$$A_+(f(y) - f^*) + \frac{1}{2}\|z_+ - x^*\|^2 \leq A(f(w) - f^*) + \frac{1}{2}\|z - x^*\|^2 - \frac{3}{8}A_+\tau\|s\|^2. \quad (144)$$

*Proof.* Set  $g := \nabla f(y)$ . Squaring (143) gives

$$\|g\|^2 + 2\tau \langle g, s \rangle + \tau^2\|s\|^2 = \|g + \tau s\|^2 \leq \frac{\tau^2}{4}\|s\|^2.$$

Rearranging and dividing by  $2\tau > 0$ , we obtain

$$\langle g, s \rangle \leq -\frac{1}{2\tau}\|g\|^2 - \frac{3\tau}{8}\|s\|^2. \quad (145)$$

Convexity at  $y$  gives

$$f(w) \geq f(y) + \langle g, w - y \rangle, \quad f^* \geq f(y) + \langle g, x^* - y \rangle.$$

Multiply these inequalities by  $A$  and  $a$ , respectively, and add them:

$$Af(w) + af^* \geq A_+f(y) + \langle g, Aw + ax^* - A_+y \rangle.$$

The identities  $A_+v = Aw + az$  and  $y = v + s$  imply

$$Aw + ax^* - A_+y = Aw + ax^* - (Aw + az) - A_+s = a(x^* - z) - A_+s.$$

Substituting this identity and using  $A_+ = A + a$  yields

$$A_+(f(y) - f^*) \leq A(f(w) - f^*) + a \langle g, z - x^* \rangle + A_+ \langle g, s \rangle. \quad (146)$$

The update  $z_+ = z - ag$  gives

$$\frac{1}{2}\|z_+ - x^*\|^2 = \frac{1}{2}\|z - x^* - ag\|^2 = \frac{1}{2}\|z - x^*\|^2 - a \langle g, z - x^* \rangle + \frac{a^2}{2}\|g\|^2. \quad (147)$$

Adding (146) and (147) gives

$$A_+(f(y) - f^*) + \frac{1}{2}\|z_+ - x^*\|^2 \leq A(f(w) - f^*) + \frac{1}{2}\|z - x^*\|^2 + A_+ \langle g, s \rangle + \frac{a^2}{2}\|g\|^2.$$

Finally, (145) and  $a^2 = A_+/\tau$  give

$$A_+ \langle g, s \rangle + \frac{a^2}{2}\|g\|^2 \leq \left( \frac{a^2}{2} - \frac{A_+}{2\tau} \right) \|g\|^2 - \frac{3}{8}A_+\tau\|s\|^2 = -\frac{3}{8}A_+\tau\|s\|^2.$$

Substitution proves (144).  $\square$

Lemma E.2 shows that the weights used in the main loop of Algorithm 6 satisfy the relations assumed in Lemma E.4.

To bound the potential after initialization step in Algorithm 6, we apply Lemma E.4 with  $A = 0$  and  $w = z = x_0$ . For the accepted step, take  $a = A_+ = A_1 = 1/\tau_1$  and  $\tau = \tau_1$ . These choices satisfy the weight condition, and the updates in (138) give  $y = w_1$  and  $z_+ = z_1$ . Since  $A_1\tau_1 = 1$ , the lemma yields

$$A_1(f(w_1) - f^*) + \frac{1}{2}\|z_1 - x^*\|^2 + \frac{3}{8}\|w_1 - x_0\|^2 \leq \frac{R^2}{2}. \quad (148)$$

For the accepted iterates, write

$$\mathcal{E}_k := A_k(f(w_k) - f^*) + \frac{1}{2}\|z_k - x^*\|^2.$$

For any accepted iterate  $w_k$ , summing (144) over  $i = 1, \dots, k-1$  and using the initialization bound gives

$$\mathcal{E}_k + \frac{3}{8} \sum_{i=1}^{k-1} A_{i+1}\tau_i\|s_i\|^2 \leq \mathcal{E}_1 \leq \frac{R^2}{2}, \quad (149)$$

where  $s_i = w_{i+1} - v_i$  and  $\tau_i$  is accepted on iteration  $i$ . Since all terms on the left-hand side of (149) are nonnegative,

$$A_k(f(w_k) - f^*) \leq \mathcal{E}_k \leq \frac{R^2}{2}, \quad \sum_{i=1}^{k-1} A_{i+1}\tau_i\|s_i\|^2 \leq \frac{4}{3}R^2.$$

Consider any trial satisfying the conditions of Lemma E.4 with  $A = A_k$ ,  $w = w_k$ , and  $z = z_k$ , and write  $\alpha = a/A_+$ . Let  $y := v + s$  be the trial point. For an accepted main-loop step of Algorithm 6,  $y = v_k + s_k = w_{k+1}$ . The left-hand side of (144) is nonnegative. Hence, by (149),

$$\frac{3}{8}A_+\tau\|s\|^2 \leq \mathcal{E}_k \leq \frac{R^2}{2}.$$

The MS test (143) also gives

$$\|\nabla f(y)\| \leq \|\nabla f(y) + \tau s\| + \tau\|s\| \leq \frac{\tau}{2}\|s\| + \tau\|s\| = \frac{3\tau}{2}\|s\|.$$

Using  $A_+\tau = A_+^2/a^2 = 1/\alpha^2$ , we obtain

$$\|s\| \leq \frac{2}{\sqrt{3}}\alpha R, \quad \|\nabla f(y)\| \leq \frac{3}{2}\tau\|s\|. \quad (150)$$

**Lemma E.5.** *For every  $k \geq 1$ , the accepted iterates of Algorithm 6 satisfy*

$$\|z_k - x^*\| \leq R, \quad (151)$$

$$\|w_k - x^*\| \leq \left(1 + \frac{2}{\sqrt{3}}\right)R, \quad \|z_k - w_k\| \leq \left(2 + \frac{2}{\sqrt{3}}\right)R. \quad (152)$$

*Proof.* By (149),

$$\frac{1}{2}\|z_k - x^*\|^2 \leq \mathcal{E}_k \leq \frac{R^2}{2},$$

which gives (151). We prove the bound on  $\|w_k - x^*\|$  by induction. The initialization bound (148) gives

$$\begin{aligned}\|w_1 - x_0\| &\leq \frac{2}{\sqrt{3}}R, \\ \|w_1 - x^*\| &\leq \|w_1 - x_0\| + \|x_0 - x^*\| \leq \left(1 + \frac{2}{\sqrt{3}}\right)R.\end{aligned}$$

Suppose the bound holds for  $w_k$ . For the next accepted step,  $w_{k+1} = (1 - \alpha_k)w_k + \alpha_k z_k + s_k$ . Using (150) and (151), we obtain

$$\begin{aligned}\|w_{k+1} - x^*\| &\leq (1 - \alpha_k)\|w_k - x^*\| + \alpha_k\|z_k - x^*\| + \|s_k\| \\ &\leq (1 - \alpha_k)\left(1 + \frac{2}{\sqrt{3}}\right)R + \alpha_k R + \frac{2}{\sqrt{3}}\alpha_k R = \left(1 + \frac{2}{\sqrt{3}}\right)R.\end{aligned}$$

This completes the induction. The bound on  $\|z_k - w_k\|$  follows from

$$\|z_k - w_k\| \leq \|z_k - x^*\| + \|w_k - x^*\| \leq \left(2 + \frac{2}{\sqrt{3}}\right)R.$$

□

Fix already accepted  $w_k, z_k$ . These points remain fixed during backtracking. For  $v = (1 - \alpha)w_k + \alpha z_k$  with any  $\alpha \in (0, 1)$ , inequality (152) gives

$$\|v - w_k\| = \alpha\|z_k - w_k\| \leq \left(2 + \frac{2}{\sqrt{3}}\right)\alpha R.$$

This bound holds for every trial, whether or not the MS test passes.

### E.3 Explicit Choice of Regularization Parameter

With  $B$  defined in (140), set

$$Q(A) := \max\left\{B^2, 8\delta A, (40M_0RA)^{2/3}\right\}, \quad A > 0. \quad (153)$$

**Lemma E.6.** *Suppose that  $f$  is convex and either (GH) or (LH) holds. Fix an accepted state  $A_k, w_k, z_k$  of Algorithm 6 and a matrix  $H(w_k)$  satisfying (IH) at  $w_k$ . Every main-loop trial with*

$$\tau \geq \frac{Q(A_k)}{A_k} \quad (154)$$

*satisfies PD( $w_k, \tau$ ) and MS( $v, \tau, s$ ) in (136), where the trial weights are defined by (139) with  $\tau_k = \tau$ ,  $v = (1 - \alpha_k)w_k + \alpha_k z_k$ , and  $s$  is given by (137) with  $w = w_k$ .*

*Proof.* Keep  $A_k, w_k, z_k, H(w_k)$  fixed and vary  $q \geq Q(A_k)$  continuously. For each  $q$ , use the weights in (139) with  $\tau_k = \tau(q)$ , where

$$\tau(q) := \frac{q}{A_k}, \quad \alpha(q) := \frac{2}{1 + \sqrt{1 + 4q}}, \quad v(q) := (1 - \alpha(q))w_k + \alpha(q)z_k.$$

Convexity and (IH) give  $H(w_k) \succeq -\delta I$ . Since  $q \geq Q(A_k) \geq \max\{1, 8\delta A_k\}$ , we have

$$H(w_k) + \tau(q)I \succeq (\tau(q) - \delta)I \succeq \frac{7}{8}\tau(q)I \succ 0.$$

Thus,  $\text{PD}(w_k, \tau(q))$  holds throughout this range, and we can define

$$\begin{aligned} s(q) &:= -(H(w_k) + \tau(q)I)^{-1} \nabla f(v(q)), & y(q) &:= v(q) + s(q), \\ d(q) &:= \|v(q) - w_k\| = \alpha(q)\|z_k - w_k\|, & r(q) &:= \|s(q)\|. \end{aligned}$$

All these quantities are continuous because  $H(w_k)$  is fixed and the inverse exists. As  $q \rightarrow +\infty$ ,

$$\tau(q) \rightarrow +\infty, \quad \alpha(q) \rightarrow 0, \quad v(q) \rightarrow w_k, \quad d(q) \rightarrow 0, \quad r(q) \leq \frac{\|\nabla f(v(q))\|}{\tau(q) - \delta} \rightarrow 0, \quad y(q) \rightarrow w_k.$$

By Lemma E.3 and (137), the following conditions are sufficient for the test (136):

$$d(q) + r(q) \leq \rho \quad \text{only under (LH)}, \quad \delta + L(y(q)) \left( d(q) + \frac{3}{2}r(q) \right) \leq \frac{\tau(q)}{2}. \quad (155)$$

Indeed, if (155) holds, then Lemma E.3 and (137) give

$$\begin{aligned} \|\nabla f(y(q)) + \tau(q)s(q)\| &= \|\nabla f(y(q)) - \nabla f(v(q)) - H(w_k)s(q)\| \\ &\leq \left( \delta + L(y(q)) \left( d(q) + \frac{3}{2}r(q) \right) \right) r(q) \leq \frac{\tau(q)}{2} \|s(q)\|. \end{aligned}$$

Since  $L(y(q)) \rightarrow L(w_k)$ ,

$$\frac{\delta + L(y(q)) \left( d(q) + \frac{3}{2}r(q) \right)}{\tau(q)} \rightarrow 0.$$

Hence, all sufficiently large  $q$  satisfy (155) strictly.

Let  $\mathcal{U}_k$  be the set of  $q \geq Q(A_k)$  for which at least one condition in (155) fails. If  $\mathcal{U}_k = \emptyset$ , the statement follows. Otherwise, set  $q^\dagger := \sup \mathcal{U}_k$ . Since the conditions in (155) hold for all sufficiently large  $q$ , the set  $\mathcal{U}_k$  is bounded above. Hence,  $q^\dagger < +\infty$ . By continuity,  $q^\dagger$  lies on the boundary of the admissible set. Hence, the conditions in (155) hold at  $q^\dagger$ , and at least one is an equality. A dagger below denotes evaluation at  $q^\dagger$ .

The boundary trial therefore satisfies the test (155). Moreover, (149) gives  $\mathcal{E}_k \leq \frac{R^2}{2}$ . Consequently, the bounds (150) apply to this trial. Together with (152), they give

$$r^\dagger \leq \frac{2}{\sqrt{3}}\alpha^\dagger R, \quad \|\nabla f(y^\dagger)\| \leq \frac{3}{2}\tau^\dagger r^\dagger \leq \sqrt{3}\tau^\dagger \alpha^\dagger R, \quad d^\dagger = \alpha^\dagger \|z_k - w_k\| \leq \left(2 + \frac{2}{\sqrt{3}}\right) \alpha^\dagger R. \quad (156)$$

In particular,

$$d^\dagger + r^\dagger \leq d^\dagger + \frac{3}{2}r^\dagger \leq \left(2 + \frac{5}{\sqrt{3}}\right) \alpha^\dagger R \leq 5\alpha^\dagger R. \quad (157)$$

The definition of  $\alpha(q)$  gives

$$\alpha^\dagger = \frac{2}{1 + \sqrt{1 + 4q^\dagger}} \leq \frac{1}{\sqrt{q^\dagger}}.$$

By (153), we have  $q^\dagger \geq Q(A_k) \geq B^2$ . Under (LH), definition (140) also gives  $B \geq \frac{8R}{\rho}$ . Together with (157), this gives

$$d^\dagger + r^\dagger \leq \frac{5R}{\sqrt{q^\dagger}} \leq \frac{5R}{B} \leq \frac{5}{8}\rho < \rho.$$

Using  $L(y^\dagger) = M_0 + M_1 \|\nabla f(y^\dagger)\|$ , equations (156) and (157) give

$$\begin{aligned} \frac{\delta + L(y^\dagger) \left(d^\dagger + \frac{3}{2}r^\dagger\right)}{\tau^\dagger} &\leq \frac{\delta}{\tau^\dagger} + \frac{5M_0 R \alpha^\dagger}{\tau^\dagger} + 5\sqrt{3}M_1 R^2 (\alpha^\dagger)^2 \\ &\leq \frac{\delta A_k}{q^\dagger} + \frac{5M_0 R A_k}{(q^\dagger)^{3/2}} + \frac{5\sqrt{3}M_1 R^2}{q^\dagger} \leq \frac{1}{8} + \frac{1}{8} + \frac{5\sqrt{3}}{64} < \frac{1}{2}, \end{aligned}$$

where we used  $q^\dagger \geq Q(A_k)$ , (153), and (140). Thus, all conditions in (155) are strict at  $q^\dagger$ , contradicting its boundary property. Therefore,  $\mathcal{U}_k = \emptyset$ , and every trial satisfying (154) passes both tests.  $\square$

#### E.4 Main-Loop Backtracking

**Lemma E.7.** *Suppose that  $f$  is convex, (IH) holds, and either (GH) or (LH) holds. Every main backtracking loop of Algorithm 6 terminates. For every  $k \geq 1$ , let  $\tau_k$  denote the parameter accepted at iteration  $k$ . Then*

$$A_k \tau_k \leq 2Q(A_k). \quad (158)$$

For every  $K \geq 1$ , let  $J_K$  be the total number of rejected main trials during iterations  $1, \dots, K$ , excluding initialization. Then

$$J_K \leq \lceil \log_2 Q(A_K) \rceil. \quad (159)$$

*Proof.* We prove by induction on  $k$  that iteration  $k$  is reached after finitely many trials. The case  $k = 1$  follows from Lemma C.2 with  $(w_1, \tau_1) = \text{MSBACKTRACK}(x_0, \tau_0)$  and  $\bar{\tau}_0 := \bar{\tau}(x_0)$ . Suppose this holds for some  $k \geq 1$ . Then (149) and (152) hold. Since  $A_k$  is fixed, repeated doubling reaches  $\tau \geq \frac{Q(A_k)}{A_k}$ , and Lemma E.6 guarantees acceptance. Thus, iteration  $k + 1$  is reached after finitely many trials, completing the induction.

We next track the product of the current weight and trial parameter. By (138), before the first main trial this product is

$$A_1 \tau_1 = \frac{1}{\tau_1} \tau_1 = 1.$$

After accepting iteration  $k$ , the first trial at iteration  $k + 1$  uses  $\tau = (1 - \alpha_k) \tau_k$ . By (139), its product is

$$A_{k+1}(1 - \alpha_k) \tau_k = \frac{A_k}{1 - \alpha_k} (1 - \alpha_k) \tau_k = A_k \tau_k.$$

Every rejected trial doubles  $\tau_k$ , while  $A_k$  remains unchanged. Therefore,

$$A_k \tau_k = 2^{J_k}, \quad k \geq 1. \quad (160)$$

Fix  $K \geq 1$ . The weights increase by (139), and  $Q$  is nondecreasing by (153). Hence,  $Q(A_i) \leq Q(A_K)$  for every  $i \leq K$ . If  $J_K = 0$ , then (160) and  $Q(A_K) \geq B^2 \geq 1$  give both (158) and (159).

Suppose that  $J_K \geq 1$ , and let  $i \leq K$  be the iteration at which the last rejection occurs. For this trial, the product of  $A_i$  and the trial parameter is  $2^{J_K - 1}$ . By Lemma E.6, the trial would be accepted if this product were at least  $Q(A_i)$ . Therefore,

$$2^{J_K - 1} < Q(A_i) \leq Q(A_K).$$

Multiplying by 2 and using (160), we obtain

$$A_K \tau_K = 2^{J_K} < 2Q(A_K),$$

which proves (158). Taking logarithms in  $2^{J_K-1} < Q(A_K)$  gives  $J_K < 1 + \log_2 Q(A_K)$ . Since  $J_K$  is an integer,

$$J_K \leq \lceil \log_2 Q(A_K) \rceil,$$

which proves (159).  $\square$

### E.5 Proof of Theorem E.1

For the next lemma, recall the definitions in (140) and (153):

$$B = \begin{cases} \max \{1, 8\sqrt{M_1}R\}, & \text{under (GH),} \\ \max \left\{1, 8\sqrt{M_1}R, \frac{8R}{\rho}\right\}, & \text{under (LH),} \end{cases}$$

$$Q(A) = \max \left\{ B^2, 8\delta A, (40M_0RA)^{2/3} \right\}, \quad A > 0.$$

We also recall the error threshold

$$\varepsilon_{\text{tr}} = \max \left\{ \frac{20M_0R^3}{B^3}, \frac{4\delta R^2}{B^2} \right\}. \quad (161)$$

For  $R > 0$ , set

$$A_{\text{tr}} := \min \left\{ \frac{B^3}{40M_0R}, \frac{B^2}{8\delta} \right\} = \frac{R^2}{2\varepsilon_{\text{tr}}}. \quad (162)$$

Here  $\frac{B^2}{8\delta} = +\infty$  when  $\delta = 0$ . In particular,

$$\sqrt{Q(A)} = \max \left\{ B, (40M_0RA)^{1/3}, \sqrt{8\delta A} \right\}, \quad A > 0.$$

The threshold  $A_{\text{tr}}$  in (162) is the largest weight for which  $Q(A) = B^2$ .

**Lemma E.8.** *Under the assumptions of Lemma E.7, suppose that  $R > 0$ . For the accepted steps of Algorithm 6, the following bounds hold for every  $k \geq 1$ . If  $A_k \leq A_{\text{tr}}$ , then*

$$A_{k+1} \geq \exp \left( \frac{1}{3B} \right) A_k. \quad (163)$$

If  $A_k \geq A_{\text{tr}}$ , then

$$\sqrt{Q(A_{k+1})} - \sqrt{Q(A_k)} \geq \frac{1}{9}. \quad (164)$$

In particular, for any  $K \geq 1$  with  $A_K \geq A_{\text{tr}}$  and every integer  $n \geq 0$ ,

$$\sqrt{Q(A_{K+n})} \geq \sqrt{Q(A_K)} + \frac{n}{9} \geq B + \frac{n}{9}. \quad (165)$$

*Proof.* By (158), the accepted parameter satisfies  $A_k \tau_k \leq 2Q(A_k)$ . Since  $\sqrt{Q(A_k)} \geq B \geq 1$ , the definition of  $\alpha_k$  in (139) gives

$$\frac{1}{\alpha_k} = \frac{1 + \sqrt{1 + 4A_k \tau_k}}{2} \leq \frac{1 + \sqrt{1 + 8Q(A_k)}}{2} \leq 3\sqrt{Q(A_k)}.$$

Hence,

$$\alpha_k \geq \frac{1}{3\sqrt{Q(A_k)}}, \quad k \geq 1. \quad (166)$$

If  $A_k \leq A_{\text{tr}}$ ,  $\max \left\{ (40M_0RA_k)^{1/3}, \sqrt{8\delta A_k} \right\} \leq B$ . Thus, by definition of  $Q(A_k)$ , we have  $B = \sqrt{Q(A_k)}$ . By (166),  $\alpha_k \geq \frac{1}{3B}$ . Using (139) and  $1 - t \leq \exp(-t)$  for  $0 \leq t < 1$ , we obtain

$$A_{k+1} = \frac{A_k}{1 - \alpha_k} \geq \exp(\alpha_k) A_k \geq \exp\left(\frac{1}{3B}\right) A_k,$$

which proves (163).

Since  $A \geq A_{\text{tr}}$ ,  $B \leq \max \left\{ (40M_0RA)^{1/3}, \sqrt{8\delta A} \right\} = \sqrt{Q(A)}$ . Suppose that  $A_k \geq A_{\text{tr}}$ . Since  $A_{k+1} > A_k \geq A_{\text{tr}}$ , the identity holds for  $A = A_k$  and  $A = A_{k+1}$ . Since  $0 < 1 - \alpha_k < 1$ , substitution of (139) gives

$$\begin{aligned} \sqrt{Q(A_{k+1})} &= \max \left\{ \frac{(40M_0RA_k)^{1/3}}{(1 - \alpha_k)^{1/3}}, \frac{\sqrt{8\delta A_k}}{(1 - \alpha_k)^{1/2}} \right\} \\ &\geq \frac{\max \left\{ (40M_0RA_k)^{1/3}, \sqrt{8\delta A_k} \right\}}{(1 - \alpha_k)^{1/3}} = \frac{\sqrt{Q(A_k)}}{(1 - \alpha_k)^{1/3}}. \end{aligned}$$

The inequality  $(1 - t)^{-1/3} \geq 1 + \frac{t}{3}$  for  $0 \leq t < 1$  and (166) now imply

$$\sqrt{Q(A_{k+1})} - \sqrt{Q(A_k)} \geq \sqrt{Q(A_k)} \left( (1 - \alpha_k)^{-1/3} - 1 \right) \geq \frac{\alpha_k \sqrt{Q(A_k)}}{3} \geq \frac{1}{9}.$$

This proves (164). Finally, if  $A_K \geq A_{\text{tr}}$ , then  $A_j \geq A_{\text{tr}}$  for every  $j \geq K$ . Summing (164) gives

$$\sqrt{Q(A_{K+n})} = \sqrt{Q(A_K)} + \sum_{j=K}^{K+n-1} \left( \sqrt{Q(A_{j+1})} - \sqrt{Q(A_j)} \right) \geq \sqrt{Q(A_K)} + \frac{n}{9} \geq B + \frac{n}{9},$$

which proves (165). □

We restate Theorem E.1 for convenience.

**Theorem E.9.** *Suppose that  $f$  is convex, (IH) holds, and either (GH) or (LH) holds. Assume that  $R > 0$ . All backtracking loops in Algorithm 6 terminate. For every  $\varepsilon > 0$ , the method generates an iterate  $w_N$  satisfying  $f(w_N) - f^* \leq \varepsilon$  within*

$$N = \mathcal{O} \left( B \left( 1 + \log_+ \frac{R^2}{2A_1 \max \{\varepsilon, \varepsilon_{\text{tr}}\}} \right) + \left( \frac{M_0 R^3}{\varepsilon} \right)^{1/3} + \sqrt{\frac{\delta R^2}{\varepsilon}} \right) \quad (167)$$

*iterations. Under (GH), initialization rejects at most*

$$\left\lceil \log_2 \max \left\{ 1, \frac{4\delta}{\tau_0}, \frac{2\sqrt{L(x_0)} \|\nabla f(x_0)\|}{\tau_0} \right\} \right\rceil \quad (168)$$

*trials. Under (LH), the maximum additionally includes  $\frac{2\|\nabla f(x_0)\|}{\rho\tau_0}$ . The total number of rejected main-loop trials before producing  $w_N$  is at most*

$$\left\lceil \log_2 \max \left\{ B^2, \frac{4\delta R^2}{\varepsilon}, \left( \frac{20M_0 R^3}{\varepsilon} \right)^{2/3} \right\} \right\rceil. \quad (169)$$

Moreover, there exists an integer  $K \geq 1$  with

$$K \leq 1 + \left\lceil 3B \log_+ \frac{R^2}{2A_1 \varepsilon_{\text{tr}}} \right\rceil. \quad (170)$$

For this  $K$ ,

$$f(w_k) - f^* \leq \frac{R^2}{2A_1} \exp\left(-\frac{k-1}{3B}\right), \quad 1 \leq k \leq K, \quad (171)$$

$$f(w_{K+n}) - f^* \leq \max \left\{ \frac{20M_0 R^3}{(B+n/9)^3}, \frac{4\delta R^2}{(B+n/9)^2} \right\}, \quad n = 0, 1, \dots \quad (172)$$

*Proof.* Lemmas C.2 and E.7 show that all backtracking loops terminate. By (149),

$$f(w_k) - f^* \leq \frac{\mathcal{E}_k}{A_k} \leq \frac{R^2}{2A_k}, \quad k \geq 1. \quad (173)$$

Set  $K := \min \{k \geq 1 : A_k \geq A_{\text{tr}}\}$ , with  $K = +\infty$  if this set is empty. If  $A_1 \geq A_{\text{tr}}$ , then  $K = 1$ , and (170) holds. Otherwise, set

$$m := \left\lceil 3B \log \frac{A_{\text{tr}}}{A_1} \right\rceil.$$

If  $A_j < A_{\text{tr}}$  for  $j = 1, \dots, m$ , then (163) yields

$$A_{m+1} \geq A_1 \exp\left(\frac{m}{3B}\right) \geq A_1 \exp\left(\log \frac{A_{\text{tr}}}{A_1}\right) = A_{\text{tr}}.$$

The threshold is therefore reached by index  $m+1$ . Since  $A_{\text{tr}} = \frac{R^2}{2\varepsilon_{\text{tr}}}$ , this proves (170).

Fix  $1 \leq k \leq K$ . By the definition of  $K$ , we have  $A_j < A_{\text{tr}}$  for  $1 \leq j < k$ . Applying (163) successively and using (173) gives

$$\begin{aligned} A_k &\geq A_1 \exp\left(\frac{k-1}{3B}\right), \\ f(w_k) - f^* &\leq \frac{R^2}{2A_k} \leq \frac{R^2}{2A_1} \exp\left(-\frac{k-1}{3B}\right). \end{aligned}$$

This proves (171).

For every  $A \geq A_{\text{tr}}$ ,

$$\sqrt{Q(A)} = \max \left\{ (40M_0 R A)^{1/3}, \sqrt{8\delta A} \right\}.$$

Hence,  $Q(A)^{3/2} \geq 40M_0 R A$  and  $Q(A) \geq 8\delta A$ , so

$$\frac{20M_0 R^3}{Q(A)^{3/2}} \leq \frac{R^2}{2A}, \quad \frac{4\delta R^2}{Q(A)} \leq \frac{R^2}{2A}.$$

At least one of these inequalities is an equality, because  $\sqrt{Q(A)}$  equals at least one of the two terms defining it. Therefore,

$$\frac{R^2}{2A} = \max \left\{ \frac{20M_0 R^3}{Q(A)^{3/2}}, \frac{4\delta R^2}{Q(A)} \right\}, \quad A \geq A_{\text{tr}}. \quad (174)$$

For  $A = A_{K+n}$ , combine (173), (174), and (165) to obtain

$$\begin{aligned} f(w_{K+n}) - f^* &\leq \frac{R^2}{2A_{K+n}} = \max \left\{ \frac{20M_0R^3}{Q(A_{K+n})^{3/2}}, \frac{4\delta R^2}{Q(A_{K+n})} \right\} \\ &\leq \max \left\{ \frac{20M_0R^3}{(B+n/9)^3}, \frac{4\delta R^2}{(B+n/9)^2} \right\}. \end{aligned}$$

This proves (172).

It remains to bound the iteration and backtracking counts. Fix  $\varepsilon > 0$ . Set

$$A_\varepsilon := \frac{R^2}{2\varepsilon}, \quad N := \min \{k \geq 1 : A_k \geq A_\varepsilon\}.$$

By (165) and (174),  $\frac{R^2}{2A_{K+n}} \rightarrow 0$  as  $n \rightarrow +\infty$ . Thus,  $N$  is finite, and (173) gives  $f(w_N) - f^* \leq \varepsilon$ .

First, suppose that  $\varepsilon \geq \varepsilon_{\text{tr}}$ . Then  $A_\varepsilon \leq A_{\text{tr}}$ , so every iteration with  $A_k < A_\varepsilon$  satisfies (163). The same argument used to prove (170), with  $A_\varepsilon$  in place of  $A_{\text{tr}}$ , gives

$$N \leq 1 + \left\lceil 3B \log_+ \frac{A_\varepsilon}{A_1} \right\rceil = 1 + \left\lceil 3B \log_+ \frac{R^2}{2A_1\varepsilon} \right\rceil.$$

Now suppose that  $0 < \varepsilon < \varepsilon_{\text{tr}}$ . Set

$$n := \left\lceil 9 \max \left\{ \left( \frac{20M_0R^3}{\varepsilon} \right)^{1/3}, 2\sqrt{\frac{\delta R^2}{\varepsilon}} \right\} \right\rceil.$$

Equations (174) and (165) give

$$\frac{R^2}{2A_{K+n}} \leq \max \left\{ \frac{20M_0R^3}{(B+n/9)^3}, \frac{4\delta R^2}{(B+n/9)^2} \right\} \leq \varepsilon.$$

Hence  $A_{K+n} \geq A_\varepsilon$ , and  $N \leq K + n$ . Using (170), we obtain

$$N \leq 1 + \left\lceil 3B \log_+ \frac{A_{\text{tr}}}{A_1} \right\rceil + \left\lceil 9 \max \left\{ \left( \frac{20M_0R^3}{\varepsilon} \right)^{1/3}, 2\sqrt{\frac{\delta R^2}{\varepsilon}} \right\} \right\rceil.$$

Combining the two cases and using (161) and  $B \geq 1$  proves (167).

We now bound the number of rejected trials before producing  $w_N$ . If  $N = 1$ , no main trial is needed. If  $N \geq 2$ , the main iterations needed to compute  $w_N$  are  $1, \dots, N-1$ . The minimality of  $N$  gives  $A_{N-1} < A_\varepsilon$ . Since  $Q$  is nondecreasing, Lemma E.7 yields

$$J_{N-1} \leq \lceil \log_2 Q(A_{N-1}) \rceil \leq \lceil \log_2 Q(A_\varepsilon) \rceil.$$

Substituting  $A_\varepsilon = \frac{R^2}{2\varepsilon}$  into (153) gives

$$Q(A_\varepsilon) = \max \left\{ B^2, \frac{4\delta R^2}{\varepsilon}, \left( \frac{20M_0R^3}{\varepsilon} \right)^{2/3} \right\}.$$

This proves the rejection bound (169). By Lemma C.2, initialization rejects at most

$$\left\lceil \log_2 \max \left\{ 1, \frac{\bar{\tau}_0}{\tau_0} \right\} \right\rceil$$

trials, where  $\bar{\tau}_0 = \bar{\tau}(x_0)$  is defined in (17). Substituting (17) gives the initialization bound (168) and its counterpart under (LH).

□

## E.6 Oracle Complexity

**Corollary E.10.** *Under the assumptions of Theorem E.9, fix  $\varepsilon > 0$  and let  $N$  be the first index with  $A_N \geq R^2/(2\varepsilon)$ . Let  $\bar{\tau}_0 := \bar{\tau}(x_0)$ , where  $\bar{\tau}(x)$  is defined in (17). Then Algorithm 6 computes  $w_N$  using at most  $N$  Hessian evaluations and at most*

$$2N + 3 + \log_{2,+} \frac{\bar{\tau}_0}{\tau_0} + 2 \log_2 \max \left\{ B^2, \frac{4\delta R^2}{\varepsilon}, \left( \frac{20M_0 R^3}{\varepsilon} \right)^{2/3} \right\}$$

*gradient evaluations.*

*Proof.* Initialization uses one Hessian, the gradient at  $x_0$ , and at most one gradient per trial. By Lemma C.2, it has at most  $2 + \log_{2,+} \frac{\bar{\tau}_0}{\tau_0}$  trials. Each of the  $N - 1$  main iterations uses one Hessian  $H(w_k)$  and at most two gradients per trial,  $\nabla f(v_k)$  and  $\nabla f(v_k + s_k)$ , since  $v_k$  changes after each rejection. These iterations have  $N - 1$  accepted trials, and (169) bounds the number of rejected ones. The updates reuse the gradient at the accepted point. Hence, the number of Hessian evaluations is at most  $1 + (N - 1) = N$ , and the number of gradient evaluations is at most

$$\begin{aligned} & 1 + \left( 2 + \log_{2,+} \frac{\bar{\tau}_0}{\tau_0} \right) + 2 \left( N - 1 + \left\lceil \log_2 \max \left\{ B^2, \frac{4\delta R^2}{\varepsilon}, \left( \frac{20M_0 R^3}{\varepsilon} \right)^{2/3} \right\} \right\rceil \right) \\ & \leq 2N + 3 + \log_{2,+} \frac{\bar{\tau}_0}{\tau_0} + 2 \log_2 \max \left\{ B^2, \frac{4\delta R^2}{\varepsilon}, \left( \frac{20M_0 R^3}{\varepsilon} \right)^{2/3} \right\}. \end{aligned}$$

□

## F Proof of Algorithm 3

We restate Algorithm 3.

### F.1 Initialization and Backtracking

Lemma C.2 guarantees termination of all calls to MSBACKTRACK in Algorithm 7.

**Lemma F.1.** *Suppose that  $f$  is convex, (IH) holds, and either (GH) or (LH) holds. Initialization in Algorithm 7 (line 2) uses at most*

$$1 + \left\lceil \log_2 \max \left\{ 1, \frac{\bar{\tau}_0}{\eta_0} \right\} \right\rceil$$

*trials, where  $\bar{\tau}_0 = \bar{\tau}(x_0)$  from (17), and accepted parameter satisfies*

$$\eta_0 \leq \eta_1 \leq \max \{ \eta_0, 2\bar{\tau}_0 \}.$$

*The algorithm either returns a minimizer or produces an initial state  $A_1, w_1, z_1, \eta_1$  satisfying  $A_1 \eta_1 = 1$  and*

$$A_1(f(w_1) - f^*) + \frac{1}{2} \|z_1 - x^*\|^2 + \frac{3}{8} \|w_1 - x_0\|^2 \leq \frac{R^2}{2}. \quad (179)$$

*Proof.* If  $\nabla f(x_0) = 0$ , the algorithm returns  $x_0$  without any trials. Otherwise, Lemma C.2 with  $x = x_0$ ,  $\eta = \eta_0$ , and  $\bar{\tau}(x_0) = \bar{\tau}_0$  proves the bound. For the initialization (175), Lemma E.4 applies with  $A = 0$ ,  $w = z = x_0$ ,  $a = A_+ = A_1 = 1/\eta_1$ , and  $\tau = \eta_1$ . The same updates give  $y = w_1$  and  $z_+ = z_1$ . Since  $A_1 \tau = 1$ , the lemma yields (179). □

---

**Algorithm 7** Adaptive damped A-NPE with an inexact Hessian

---

1: **Input:**  $x_0 \in \mathbb{R}^d$  and  $\eta_0 > 0$ .

2:  $(w_1, \eta_1) := \text{MSBACKTRACK}(x_0, \eta_0)$ .

3: Set

$$A_1 := \frac{1}{\eta_1}, \quad z_1 := x_0 - A_1 \nabla f(w_1). \quad (175)$$

4: **for**  $k \geq 1$  **do**

5: Set

$$a'_k := \frac{1 + \sqrt{1 + 4\eta_k A_k}}{2\eta_k}, \quad A'_k := A_k + a'_k, \quad v_k := \frac{A_k w_k + a'_k z_k}{A'_k}. \quad (176)$$

6:  $(y_k, \tau_k) := \text{MSBACKTRACK}(v_k, \eta_k)$ .

7: Set

$$\gamma_k := \frac{\eta_k}{\tau_k}, \quad a_{k+1} := \gamma_k a'_k, \quad A_{k+1} := A_k + a_{k+1}. \quad (177)$$

8: Set

$$w_{k+1} := \frac{(1 - \gamma_k)A_k w_k + \gamma_k A'_k y_k}{A_{k+1}}, \quad z_{k+1} := z_k - a_{k+1} \nabla f(y_k). \quad (178)$$

9: **if**  $\tau_k = \eta_k$  **then**

10: Set  $\eta_{k+1} := \frac{\eta_k}{2}$ .

11: **else**

12: Set  $\eta_{k+1} := 2\eta_k$ .

---

## F.2 Step Estimates

**Lemma F.2.** *Let  $f$  be convex. Every completed main iteration of Algorithm 7 satisfies*

$$\mathcal{E}_{k+1} \leq \mathcal{E}_k - \frac{3}{8} A'_k \eta_k \|s_k\|^2, \quad (180)$$

where  $\mathcal{E}_k := A_k(f(w_k) - f^*) + \frac{1}{2}\|z_k - x^*\|^2$ .

*Proof.* By (176) and (177),  $(a'_k)^2 = A'_k/\eta_k$  and  $\gamma_k \tau_k = \eta_k$ , so

$$\frac{a_{k+1}^2}{\gamma_k A'_k} = \frac{1}{\tau_k}, \quad v_k = \frac{\gamma_k A_k w_k + a_{k+1} z_k}{\gamma_k A'_k}.$$

Apply Lemma E.4 with  $A = \gamma_k A_k$ ,  $a = a_{k+1}$ ,  $A_+ = \gamma_k A'_k$ ,  $w = w_k$ ,  $z = z_k$ , and  $\tau = \tau_k$ . By (178),  $z_+ = z_{k+1}$ , so it gives

$$\gamma_k A'_k (f(y_k) - f^*) + \frac{1}{2}\|z_{k+1} - x^*\|^2 \leq \gamma_k A_k (f(w_k) - f^*) + \frac{1}{2}\|z_k - x^*\|^2 - \frac{3}{8} A'_k \eta_k \|s_k\|^2. \quad (181)$$

Since  $0 < \gamma_k \leq 1$  and

$$\frac{(1 - \gamma_k)A_k}{A_{k+1}} + \frac{\gamma_k A'_k}{A_{k+1}} = \frac{A_k + \gamma_k a'_k}{A_{k+1}} = 1,$$

convexity of  $f$  and (178) give

$$A_{k+1}(f(w_{k+1}) - f^*) \leq (1 - \gamma_k)A_k(f(w_k) - f^*) + \gamma_k A'_k(f(y_k) - f^*). \quad (182)$$

Using (182) and then (181), we obtain

$$\begin{aligned}
\mathcal{E}_{k+1} &= A_{k+1}(f(w_{k+1}) - f^*) + \frac{1}{2}\|z_{k+1} - x^*\|^2 \\
&\leq (1 - \gamma_k)A_k(f(w_k) - f^*) + \gamma_k A'_k(f(y_k) - f^*) + \frac{1}{2}\|z_{k+1} - x^*\|^2 \\
&\leq (1 - \gamma_k)A_k(f(w_k) - f^*) + \gamma_k A_k(f(w_k) - f^*) + \frac{1}{2}\|z_k - x^*\|^2 - \frac{3}{8}A'_k\eta_k\|s_k\|^2 \\
&= \mathcal{E}_k - \frac{3}{8}A'_k\eta_k\|s_k\|^2.
\end{aligned}$$

□

Summing (180) over  $i = 1, \dots, k-1$  and using (179), we obtain

$$\mathcal{E}_k + \frac{3}{8} \sum_{i=1}^{k-1} A'_i \eta_i \|s_i\|^2 \leq \mathcal{E}_1 \leq \frac{R^2}{2}. \quad (183)$$

Since all terms on the left-hand side of (183) are nonnegative,

$$f(w_k) - f^* \leq \frac{R^2}{2A_k}, \quad \sum_{i=1}^{k-1} A'_i \eta_i \|s_i\|^2 \leq \frac{4}{3}R^2.$$

For every completed main iteration, (180) and (183) give

$$\frac{3}{8}A'_k\eta_k\|s_k\|^2 \leq \mathcal{E}_k - \mathcal{E}_{k+1} \leq \mathcal{E}_k \leq \frac{R^2}{2}.$$

The MS test (143) also gives

$$\|\nabla f(y_k)\| \leq \|\nabla f(y_k) + \tau_k s_k\| + \tau_k \|s_k\| \leq \frac{3}{2}\tau_k \|s_k\|.$$

Using  $(a'_k)^2 = A'_k/\eta_k$ , we obtain

$$\|s_k\| \leq \frac{2R}{\sqrt{3A'_k\eta_k}} = \frac{2a'_k}{\sqrt{3}A'_k}R, \quad \|\nabla f(y_k)\| \leq \frac{3}{2}\tau_k \|s_k\|. \quad (184)$$

**Lemma F.3.** *For every  $k \geq 1$ , the points  $w_k, z_k, v_k$  generated by Algorithm 7 satisfy*

$$\|z_k - x^*\| \leq R, \quad (185)$$

$$\|w_k - x^*\| \leq \left(1 + \frac{2}{\sqrt{3}}\right)R, \quad \|v_k - x^*\| \leq \left(1 + \frac{2}{\sqrt{3}}\right)R. \quad (186)$$

*Proof.* By (183),

$$\frac{1}{2}\|z_k - x^*\|^2 \leq \mathcal{E}_k \leq \frac{R^2}{2},$$

which gives (185). We prove the bound on  $\|w_k - x^*\|$  by induction. The initialization bound (179) gives

$$\|w_1 - x^*\| \leq \|w_1 - x_0\| + \|x_0 - x^*\| \leq \left(1 + \frac{2}{\sqrt{3}}\right)R.$$

Suppose the bound holds for  $w_k$ . Substituting  $y_k = v_k + s_k$  and  $A'_k v_k = A_k w_k + a'_k z_k$  into (178) yields

$$w_{k+1} = \frac{(1 - \gamma_k)A_k w_k + \gamma_k A'_k (v_k + s_k)}{A_{k+1}} = \frac{A_k}{A_{k+1}} w_k + \frac{a_{k+1}}{A_{k+1}} z_k + \frac{\gamma_k A'_k}{A_{k+1}} s_k.$$

Using (184), (185), and  $\gamma_k a'_k = a_{k+1}$ , we obtain

$$\begin{aligned} \|w_{k+1} - x^*\| &\leq \frac{A_k}{A_{k+1}} \|w_k - x^*\| + \frac{a_{k+1}}{A_{k+1}} \|z_k - x^*\| + \frac{\gamma_k A'_k}{A_{k+1}} \|s_k\| \\ &\leq \frac{A_k}{A_{k+1}} \left(1 + \frac{2}{\sqrt{3}}\right) R + \frac{a_{k+1}}{A_{k+1}} R + \frac{2a_{k+1}}{\sqrt{3}A_{k+1}} R = \left(1 + \frac{2}{\sqrt{3}}\right) R. \end{aligned}$$

This completes the induction. Finally,

$$\|v_k - x^*\| \leq \frac{A_k}{A'_k} \|w_k - x^*\| + \frac{a'_k}{A'_k} \|z_k - x^*\| \leq \frac{A_k}{A'_k} \left(1 + \frac{2}{\sqrt{3}}\right) R + \frac{a'_k}{A'_k} R \leq \left(1 + \frac{2}{\sqrt{3}}\right) R,$$

which proves (186).  $\square$

For iterations of Algorithm 7 with  $\gamma_k < 1$ , apply Lemma C.3 with  $x = v_k$ ,  $y = y_k$ ,  $\eta = \eta_k$ , and  $\tau = \tau_k$ . If  $\tau_k \geq 16\delta$ , (20) bounds the step length from below. As in Carmon et al. (2022, Appendix A), we combine this bound with (183) to control the number of these iterations.

### F.3 Proof of Theorem 4.1

Define

$$\begin{aligned} B &:= \begin{cases} \max \left\{ 1, (6\sqrt{M_1}R)^{2/3} \right\}, & \text{under (GH),} \\ \max \left\{ 1, (6\sqrt{M_1}R)^{2/3}, \left(\frac{2R}{\rho}\right)^{2/3} \right\}, & \text{under (LH),} \end{cases} \\ \varepsilon_{\text{tr}} &:= \max \left\{ \frac{48M_0R^3}{B^{7/2}}, \frac{16\delta R^2}{B^2} \right\}. \end{aligned} \tag{187}$$

Equivalently, with  $r_0$  defined in (18),  $B = \max \left\{ 1, \left(\frac{R}{r_0}\right)^{2/3} \right\}$ , where  $R/(+\infty) = 0$ . For  $A > 0$ , define

$$\Lambda(A) := \max \left\{ 32\delta, (96M_0R)^{4/7} A^{-3/7}, \frac{B^2}{A} \right\}. \tag{188}$$

The function  $\Lambda$  is nonincreasing, and  $A\Lambda(A)$  is nondecreasing with  $A\Lambda(A) \geq B^2 \geq 1$ .

**Lemma F.4.** *Suppose that  $f$  is convex, (IH) holds, and either (GH) or (LH) holds. Assume that  $R > 0$  and set*

$$A_{\text{tr}} := \min \left\{ \frac{B^{7/2}}{96M_0R}, \frac{B^2}{32\delta} \right\}, \tag{189}$$

where  $B^2/(32\delta) = +\infty$  when  $\delta = 0$ . The weights  $A_k$ ,  $k \geq 1$ , generated by Algorithm 7 satisfy

$$k - 1 \leq \frac{16}{3} B + 2 \log_2 B + \left(6B + \frac{3}{2}\right) \log_+ \frac{\min \{A_k, A_{\text{tr}}\}}{A_1} + 35 \left( \sqrt{A_k \Lambda(A_k)} - B \right). \tag{190}$$

*Proof.* Fix  $k \geq 1$  and define

$$\begin{aligned}\mathcal{S}_{k-1}^- &:= \{i \in \{1, \dots, k-1\} : \tau_i = \eta_i, \eta_i \leq 2\Lambda(A_i)\}, \\ \mathcal{S}_{k-1}^+ &:= \{i \in \{1, \dots, k-1\} : \tau_i > \eta_i, \eta_i \geq \Lambda(A_i)/2\}.\end{aligned}$$

We first bound  $|\mathcal{S}_{k-1}^-|$  in terms of  $A_k$ . On these iterations, the bound  $\eta_i \leq 2\Lambda(A_i)$  gives an increase of  $A_i$ . For  $i \in \mathcal{S}_{k-1}^-$ , we have  $\gamma_i = 1$ ,  $A_{i+1} = A'_i$ , and  $(a'_i)^2 = A'_i/\eta_i$ . Hence,

$$\frac{a'_i}{A_i} = \sqrt{\frac{A'_i}{\eta_i A_i^2}} \geq \frac{1}{\sqrt{\eta_i A_i}} \geq \frac{1}{\sqrt{2A_i \Lambda(A_i)}}.$$

Using  $\log(1+t) \geq t/(1+t)$  for  $t \geq 0$  gives

$$\log \frac{A_{i+1}}{A_i} = \log \left(1 + \frac{a'_i}{A_i}\right) \geq \log \left(1 + \frac{1}{\sqrt{2A_i \Lambda(A_i)}}\right) \geq \frac{1}{\sqrt{2A_i \Lambda(A_i)} + 1} \geq \frac{1}{3\sqrt{A_i \Lambda(A_i)}}.$$

Since  $\sqrt{t\Lambda(t)}$  is nondecreasing, each  $i \in \mathcal{S}_{k-1}^-$  therefore satisfies

$$\int_{A_i}^{A_{i+1}} \sqrt{\frac{\Lambda(t)}{t}} dt = \int_{A_i}^{A_{i+1}} \frac{\sqrt{t\Lambda(t)}}{t} dt \geq \sqrt{A_i \Lambda(A_i)} \log \frac{A_{i+1}}{A_i} \geq \frac{1}{3}.$$

Summing over  $\mathcal{S}_{k-1}^-$  and using the nonnegativity of the integrand, we obtain

$$|\mathcal{S}_{k-1}^-| = \sum_{i \in \mathcal{S}_{k-1}^-} 1 \leq 3 \sum_{i \in \mathcal{S}_{k-1}^-} \int_{A_i}^{A_{i+1}} \sqrt{\frac{\Lambda(t)}{t}} dt \leq 3 \sum_{i=1}^{k-1} \int_{A_i}^{A_{i+1}} \sqrt{\frac{\Lambda(t)}{t}} dt = 3 \int_{A_1}^{A_k} \sqrt{\frac{\Lambda(t)}{t}} dt. \quad (191)$$

We next bound  $|\mathcal{S}_{k-1}^+|$  using (183). We first bound the term  $\frac{3}{8}A'_i\eta_i\|s_i\|^2$  from below. For  $i \in \mathcal{S}_{k-1}^+$ , we have  $\tau_i > \eta_i \geq \frac{\Lambda(A_i)}{2} \geq 16\delta$ . Thus, by Lemma C.3, (20) gives

$$\|s_i\| \geq \min \left\{ r_0, \frac{\tau_i}{48M_0} \right\} \geq \min \left\{ r_0, \frac{\Lambda(A_i)}{96M_0} \right\}.$$

Since  $A'_i \geq A_i$  and  $\eta_i \geq \Lambda(A_i)/2$ , it follows that

$$\begin{aligned}\frac{3}{8}A'_i\eta_i\|s_i\|^2 &\geq \frac{3}{16}A_i\Lambda(A_i) \min \left\{ r_0^2, \frac{\Lambda(A_i)^2}{(96M_0)^2} \right\} \\ &= \frac{3}{16\sqrt{A_i\Lambda(A_i)}} \min \left\{ (A_i\Lambda(A_i))^{3/2} r_0^2, \frac{A_i^{3/2}\Lambda(A_i)^{7/2}}{(96M_0)^2} \right\}.\end{aligned} \quad (192)$$

We now use the definition of  $\Lambda$  (188) to bound both terms in the last minimum. For  $r_0 < +\infty$ , the definition of  $B$  and (188) give

$$(A_i\Lambda(A_i))^{3/2} r_0^2 \geq B^3 r_0^2 \geq \left(\frac{R}{r_0}\right)^2 r_0^2 = R^2.$$

Also, (188) gives

$$\frac{A_i^{3/2}\Lambda(A_i)^{7/2}}{(96M_0)^2} \geq \frac{A_i^{3/2} \left((96M_0 R)^{4/7} A_i^{-3/7}\right)^{7/2}}{(96M_0)^2} = R^2.$$

Substituting these bounds in (192) and using that  $A_i \leq A_k$  and  $A\Lambda(A)$  is nondecreasing, we obtain

$$\frac{3}{8} A'_i \eta_i \|s_i\|^2 \geq \frac{3R^2}{16\sqrt{A_i\Lambda(A_i)}} \geq \frac{3R^2}{16\sqrt{A_k\Lambda(A_k)}}.$$

Summing over  $i \in \mathcal{S}_{k-1}^>$  and using (183), we obtain

$$\frac{3R^2}{16\sqrt{A_k\Lambda(A_k)}} |\mathcal{S}_{k-1}^>| = \sum_{i \in \mathcal{S}_{k-1}^>} \frac{3R^2}{16\sqrt{A_k\Lambda(A_k)}} \leq \frac{3}{8} \sum_{i \in \mathcal{S}_{k-1}^>} A'_i \eta_i \|s_i\|^2 \leq \frac{3}{8} \sum_{i=1}^{k-1} A'_i \eta_i \|s_i\|^2 \leq \frac{R^2}{2}.$$

Therefore,

$$|\mathcal{S}_{k-1}^>| \leq \frac{8}{3} \sqrt{A_k\Lambda(A_k)}. \quad (193)$$

It remains to bound the number  $(k-1) - |\mathcal{S}_{k-1}^=| - |\mathcal{S}_{k-1}^>|$  of iterations outside  $\mathcal{S}_{k-1}^= \cup \mathcal{S}_{k-1}^>$ . For these iterations, there are two cases:

- First, if  $\tau_i = \eta_i$ , then  $\eta_i > 2\Lambda(A_i)$ ,  $\eta_{i+1} = \frac{\eta_i}{2}$ .
- Second, if  $\tau_i > \eta_i$ , then  $\eta_i < \frac{\Lambda(A_i)}{2}$ ,  $\eta_{i+1} = 2\eta_i$ .

To measure the change relative to  $\Lambda(A_i)$ , set  $V_i := \left| \log \frac{\eta_i}{\Lambda(A_i)} \right|$ . In the first case,  $\eta_{i+1} > \Lambda(A_i)$ , so

$$\left| \log \frac{\eta_{i+1}}{\Lambda(A_i)} \right| = \log \frac{\eta_i}{2\Lambda(A_i)} = \log \frac{\eta_i}{\Lambda(A_i)} - \log 2 = V_i - \log 2.$$

In the second case,  $\eta_{i+1} < \Lambda(A_i)$ , so

$$\left| \log \frac{\eta_{i+1}}{\Lambda(A_i)} \right| = -\log \frac{2\eta_i}{\Lambda(A_i)} = -\log \frac{\eta_i}{\Lambda(A_i)} - \log 2 = V_i - \log 2.$$

For  $i \in \mathcal{S}_{k-1}^= \cup \mathcal{S}_{k-1}^>$ , the update of  $\eta_i$  gives  $\left| \log \frac{\eta_{i+1}}{\Lambda(A_i)} \right| \leq V_i + \log 2$ .

Since  $\Lambda(A_{i+1}) \leq \Lambda(A_i)$ ,

$$V_{i+1} = \left| \log \frac{\eta_{i+1}}{\Lambda(A_{i+1})} + \log \frac{\Lambda(A_i)}{\Lambda(A_{i+1})} \right| \leq \left| \log \frac{\eta_{i+1}}{\Lambda(A_i)} \right| + \log \frac{\Lambda(A_i)}{\Lambda(A_{i+1})}.$$

Summing over  $i = 1, \dots, k-1$  and telescoping the logarithms gives

$$\begin{aligned} V_k - V_1 &\leq \left( |\mathcal{S}_{k-1}^=| + |\mathcal{S}_{k-1}^>| \right) \log 2 - \left( (k-1) - |\mathcal{S}_{k-1}^=| - |\mathcal{S}_{k-1}^>| \right) \log 2 + \sum_{i=1}^{k-1} \log \frac{\Lambda(A_i)}{\Lambda(A_{i+1})} \\ &= \left( 2 \left( |\mathcal{S}_{k-1}^=| + |\mathcal{S}_{k-1}^>| \right) - (k-1) \right) \log 2 + \log \frac{\Lambda(A_1)}{\Lambda(A_k)}. \end{aligned}$$

Initialization gives  $\eta_1 = 1/A_1$ . Since  $A_1\Lambda(A_1) \geq 1$ ,

$$V_1 = \left| \log \frac{1}{A_1\Lambda(A_1)} \right| = \log (A_1\Lambda(A_1)).$$

Also, monotonicity of  $A\Lambda(A)$  implies

$$\log \frac{\Lambda(A_1)}{\Lambda(A_k)} = \log \frac{A_k}{A_1} + \log \frac{A_1\Lambda(A_1)}{A_k\Lambda(A_k)} \leq \log \frac{A_k}{A_1}.$$

Using  $V_k \geq 0$ , combining the logarithms, and dividing by  $\log 2$ , we obtain

$$k - 1 \leq 2 \left( |\mathcal{S}_{k-1}^-| + |\mathcal{S}_{k-1}^+| \right) + \log_2 (A_k \Lambda(A_1)). \quad (194)$$

Substituting (191) and (193) into (194) gives

$$k - 1 \leq 6 \int_{A_1}^{A_k} \sqrt{\frac{\Lambda(t)}{t}} dt + \frac{16}{3} \sqrt{A_k \Lambda(A_k)} + \log_2 (A_k \Lambda(A_1)). \quad (195)$$

We now estimate the integral and logarithmic term in (195). By (188) and (189),

$$\sqrt{A\Lambda(A)} = \begin{cases} B, & 0 < A \leq A_{\text{tr}}, \\ \max \left\{ (96M_0RA)^{2/7}, \sqrt{32\delta A} \right\}, & A > A_{\text{tr}}. \end{cases}$$

Consider an interval  $[u, v] \subset [A_{\text{tr}}, +\infty)$  on which the same term attains the maximum. If  $\sqrt{t\Lambda(t)} = (96M_0Rt)^{2/7}$  on this interval, then

$$\int_u^v \sqrt{\frac{\Lambda(t)}{t}} dt = (96M_0R)^{2/7} \int_u^v t^{-5/7} dt = \frac{7}{2} \left( \sqrt{v\Lambda(v)} - \sqrt{u\Lambda(u)} \right), \quad \log \frac{v}{u} = \frac{7}{2} \log \frac{\sqrt{v\Lambda(v)}}{\sqrt{u\Lambda(u)}}.$$

If  $\sqrt{t\Lambda(t)} = \sqrt{32\delta t}$  on this interval, then

$$\int_u^v \sqrt{\frac{\Lambda(t)}{t}} dt = \sqrt{32\delta} \int_u^v t^{-1/2} dt = 2 \left( \sqrt{v\Lambda(v)} - \sqrt{u\Lambda(u)} \right), \quad \log \frac{v}{u} = 2 \log \frac{\sqrt{v\Lambda(v)}}{\sqrt{u\Lambda(u)}}.$$

Both coefficients are at most  $7/2$ . Summing over the intervals above  $A_{\text{tr}}$ , the intermediate endpoint terms cancel in both bounds. Below  $A_{\text{tr}}$ , we have  $\sqrt{t\Lambda(t)} = B$  and integrate  $B/t$ . Thus, for every  $A \geq A_1$ ,

$$\int_{A_1}^A \sqrt{\frac{\Lambda(t)}{t}} dt \leq B \log_+ \frac{\min \{A, A_{\text{tr}}\}}{A_1} + \frac{7}{2} \left( \sqrt{A\Lambda(A)} - B \right), \quad (196)$$

$$\log \frac{A}{A_1} \leq \log_+ \frac{\min \{A, A_{\text{tr}}\}}{A_1} + \frac{7}{2} \log \frac{\sqrt{A\Lambda(A)}}{B}. \quad (197)$$

These bounds also hold when  $A_1 > A_{\text{tr}}$ , since  $\sqrt{A_1\Lambda(A_1)} \geq B$ . Using  $\log_2 t \leq \frac{3}{2} \log t$  for  $t \geq 1$ , (197), and the monotonicity of  $A\Lambda(A)$ , we obtain

$$\begin{aligned} \log_2 (A_k \Lambda(A_1)) &= \log_2 \frac{A_k}{A_1} + \log_2 (A_1 \Lambda(A_1)) \leq \frac{3}{2} \log \frac{A_k}{A_1} + 2 \log_2 B + 3 \log \frac{\sqrt{A_k \Lambda(A_k)}}{B} \\ &\leq 2 \log_2 B + \frac{3}{2} \log_+ \frac{\min \{A_k, A_{\text{tr}}\}}{A_1} + \frac{33}{4} \log \frac{\sqrt{A_k \Lambda(A_k)}}{B}. \end{aligned} \quad (198)$$

Substituting (196) and (198) into (195), we obtain

$$\begin{aligned} k - 1 &\leq \frac{16}{3} B + 2 \log_2 B + \left( 6B + \frac{3}{2} \right) \log_+ \frac{\min \{A_k, A_{\text{tr}}\}}{A_1} \\ &\quad + \frac{79}{3} \left( \sqrt{A_k \Lambda(A_k)} - B \right) + \frac{33}{4} \log \frac{\sqrt{A_k \Lambda(A_k)}}{B}. \end{aligned}$$

Finally, since  $\sqrt{A_k \Lambda(A_k)} \geq B \geq 1$ ,

$$\log \frac{\sqrt{A_k \Lambda(A_k)}}{B} \leq \frac{\sqrt{A_k \Lambda(A_k)} - B}{B}, \quad \frac{79}{3} + \frac{33}{4B} \leq \frac{415}{12} < 35.$$

Substituting these inequalities proves (190).  $\square$

We restate Theorem 4.1 with explicit constants.

**Theorem F.5.** Suppose that  $f$  is convex, (IH) holds, and either (GH) or (LH) holds. Let  $\bar{\tau}_0 := \bar{\tau}(x_0)$ , where  $\bar{\tau}(x)$  is defined in (17). For every  $\varepsilon > 0$ , the method generates an iterate  $w_N$  satisfying  $f(w_N) - f^* \leq \varepsilon$  within

$$N \leq 2 + \frac{16}{3}B + 2\log_2 B + \left(6B + \frac{3}{2}\right) \log_+ \frac{\max\{\eta_0, \bar{\tau}_0\} R^2}{\max\{\varepsilon, \varepsilon_{\text{tr}}\}} + 35 \left( \max \left\{ B, \left( \frac{48M_0 R^3}{\varepsilon} \right)^{2/7}, 4\sqrt{\frac{\delta R^2}{\varepsilon}} \right\} - B \right) \quad (199)$$

outer iterations, including initialization, each using at most one Hessian evaluation. Moreover, set

$$K := 2 + \left\lfloor \frac{16}{3}B + 2\log_2 B + \left(6B + \frac{3}{2}\right) \log_+ \frac{\max\{\eta_0, \bar{\tau}_0\} R^2}{\varepsilon_{\text{tr}}} \right\rfloor. \quad (200)$$

For this  $K$ ,

$$f(w_k) - f^* \leq \max\{\eta_0, \bar{\tau}_0\} R^2 \exp \left( -\frac{\max\left\{0, k-1 - \frac{16}{3}B - 2\log_2 B\right\}}{6B + \frac{3}{2}} \right), \quad 1 \leq k < K, \quad (201)$$

$$f(w_{K+n}) - f^* \leq \max \left\{ \frac{48M_0 R^3}{(B+n/35)^{7/2}}, \frac{16\delta R^2}{(B+n/35)^2} \right\}, \quad n = 0, 1, \dots \quad (202)$$

*Proof.* Lemmas C.2 and F.1 show that all backtracking loops terminate. If initialization returns a minimizer, all claims are immediate. By (183),

$$f(w_k) - f^* \leq \frac{\mathcal{E}_k}{A_k} \leq \frac{R^2}{2A_k}. \quad (203)$$

Since  $a_{k+1} > 0$ , the  $A_k$  strictly increase.

Fix  $\varepsilon > 0$  and set  $A_\varepsilon := R^2/(2\varepsilon)$ . If  $A_1 \geq A_\varepsilon$ , initialization suffices by (203). Otherwise, by (190), unless a minimizer is returned earlier, there exists a first index  $N$  with  $A_N \geq A_\varepsilon$ . Then  $N \geq 2$ ,  $A_{N-1} < A_\varepsilon \leq A_N$ , and (203) gives  $f(w_N) - f^* \leq \varepsilon$ . Applying (190) at  $N-1$  and using monotonicity of its right-hand side, we obtain

$$N \leq 2 + \frac{16}{3}B + 2\log_2 B + \left(6B + \frac{3}{2}\right) \log_+ \frac{\min\{A_\varepsilon, A_{\text{tr}}\}}{A_1} + 35 \left( \sqrt{A_\varepsilon \Lambda(A_\varepsilon)} - B \right).$$

By (187) and (189),  $A_{\text{tr}} = R^2/(2\varepsilon_{\text{tr}})$ . Hence,

$$\log_+ \frac{\min\{A_\varepsilon, A_{\text{tr}}\}}{A_1} = \log_+ \frac{R^2}{2A_1 \max\{\varepsilon, \varepsilon_{\text{tr}}\}},$$

$$\sqrt{A_\varepsilon \Lambda(A_\varepsilon)} = \max \left\{ B, \left( \frac{48M_0 R^3}{\varepsilon} \right)^{2/7}, 4\sqrt{\frac{\delta R^2}{\varepsilon}} \right\}.$$

Next, by Lemma F.1 and (175) we have

$$\frac{1}{A_1} = \eta_1 \leq \max\{\eta_0, 2\bar{\tau}_0\} \leq 2 \max\{\eta_0, \bar{\tau}_0\}. \quad (204)$$

Hence,

$$\frac{R^2}{2A_1} \leq \max\{\eta_0, \bar{\tau}_0\} R^2, \quad 2 \max\{\eta_0, \bar{\tau}_0\} A_{\text{tr}} = \frac{\max\{\eta_0, \bar{\tau}_0\} R^2}{\varepsilon_{\text{tr}}}, \quad (205)$$

and

$$\log_+ \frac{R^2}{2A_1 \max\{\varepsilon, \varepsilon_{\text{tr}}\}} \leq \log_+ \frac{\max\{\eta_0, \bar{\tau}_0\} R^2}{\max\{\varepsilon, \varepsilon_{\text{tr}}\}}.$$

Substituting these estimates proves (199). If a zero gradient is queried before reaching  $A_\varepsilon$ , let  $N$  count the outer iterations up to that return. The last completed weight still satisfies  $A_{N-1} < A_\varepsilon$ , so the same estimate applies, and the returned point is a minimizer. There is one Hessian evaluation during initialization and at most one on each main iteration, since all inner trials reuse that Hessian.

We now prove (201) and (202) for iterates produced before a zero-gradient return. The right-hand side of (190) is nonincreasing in  $A_1$ , and  $1/A_1 \leq 2 \max\{\eta_0, \bar{\tau}_0\}$  by (204). Hence, (190) remains valid with  $1/A_1$  replaced by  $2 \max\{\eta_0, \bar{\tau}_0\}$ . First, let  $1 \leq k < K$ . If  $A_k \leq A_{\text{tr}}$ , then  $\sqrt{A_k \Lambda(A_k)} = B$ , and (190) gives

$$k - 1 \leq \frac{16}{3}B + 2 \log_2 B + \left(6B + \frac{3}{2}\right) \log(2 \max\{\eta_0, \bar{\tau}_0\} A_k).$$

If  $A_k > A_{\text{tr}}$ , then (200), (205), and  $2 \max\{\eta_0, \bar{\tau}_0\} A_k \geq 1$  give

$$\begin{aligned} k - 1 &\leq K - 2 \leq \frac{16}{3}B + 2 \log_2 B + \left(6B + \frac{3}{2}\right) \log_+(2 \max\{\eta_0, \bar{\tau}_0\} A_{\text{tr}}) \\ &\leq \frac{16}{3}B + 2 \log_2 B + \left(6B + \frac{3}{2}\right) \log(2 \max\{\eta_0, \bar{\tau}_0\} A_k). \end{aligned}$$

Thus, in both cases, since  $\log(2 \max\{\eta_0, \bar{\tau}_0\} A_k) \geq 0$ ,

$$\log(2 \max\{\eta_0, \bar{\tau}_0\} A_k) \geq \frac{\max\left\{0, k - 1 - \frac{16}{3}B - 2 \log_2 B\right\}}{6B + \frac{3}{2}}.$$

Combining this bound with (203) proves (201).

Next, let  $k = K + n$  with  $n \geq 0$ . By (200) and (205),

$$\frac{16}{3}B + 2 \log_2 B + \left(6B + \frac{3}{2}\right) \log_+(2 \max\{\eta_0, \bar{\tau}_0\} A_{\text{tr}}) + n < k - 1.$$

Since  $\min\{A_k, A_{\text{tr}}\} \leq A_{\text{tr}}$ , (190) also gives

$$k - 1 \leq \frac{16}{3}B + 2 \log_2 B + \left(6B + \frac{3}{2}\right) \log_+(2 \max\{\eta_0, \bar{\tau}_0\} A_{\text{tr}}) + 35 \left(\sqrt{A_k \Lambda(A_k)} - B\right).$$

Combining these inequalities yields

$$\sqrt{A_k \Lambda(A_k)} > B + \frac{n}{35} \geq B.$$

Therefore, by (188),

$$\max\left\{(96M_0 R A_k)^{2/7}, \sqrt{32\delta A_k}\right\} > B + \frac{n}{35}, \quad \frac{1}{A_k} \leq \max\left\{\frac{96M_0 R}{(B + n/35)^{7/2}}, \frac{32\delta}{(B + n/35)^2}\right\}.$$

Multiplying the last inequality by  $R^2/2$  and using (203) proves (202). After a zero-gradient return, the output sequence consists of minimizers, so both rate bounds continue to hold.  $\square$

#### F.4 Oracle Complexity

In this section, we bound the numbers of Hessian and gradient evaluations of Algorithm 7. Let

$$\bar{g} := \max_{\|x-x^*\| \leq (1+2/\sqrt{3})R} \|\nabla f(x)\|. \quad (206)$$

By (186),  $\|\nabla f(v_i)\| \leq \bar{g}$  for any  $v_i$ . Recall that  $r_0$  is defined in (18).

**Corollary F.6.** *Under the assumptions of Theorem F.5, fix  $\varepsilon > 0$ . Let  $N$  be the number of outer iterations, including initialization, until  $A_N \geq R^2/(2\varepsilon)$  or an earlier zero-gradient return. Then Algorithm 7 uses at most  $N$  Hessian evaluations and at most*

$$\begin{aligned} & 7N + \log_{2,+} \frac{\bar{\tau}_0}{\eta_0} + 30\sqrt{\frac{2\delta R^2}{\varepsilon}} + 120 \left( \frac{2M_0 R^3}{\varepsilon} \right)^{2/7} \\ & + 8B \log_{2,+} \frac{\delta + \bar{g}/r_0}{\delta + M_0 r_0} \left( 1 + \log_+ \frac{4 \max\{\eta_0, \bar{\tau}_0\} R^2}{\varepsilon} \right)^{2/3} \end{aligned} \quad (207)$$

gradient evaluations.

We prove Corollary F.6 after three auxiliary lemmas. Lemma F.9 bounds the total number of rejected trials. Its proof uses the bounds on the sums of  $\eta_i^{-1/2}$  and  $(\|s_i\|/\eta_i)^{2/7}$  from Lemma F.7 and the bound on the number of rejected iterations with  $\|s_i\| > r_0$  from Lemma F.8. By (183) and  $A_i \leq A'_i$ ,

$$\sum_{i=1}^m A_i \eta_i \|s_i\|^2 \leq \frac{4}{3} R^2, \quad m \geq 1. \quad (208)$$

**Lemma F.7.** *Under the assumptions of Theorem F.5, every completed main iteration of Algorithm 7 satisfies*

$$\eta_i A_i \geq \frac{1}{2}, \quad A_{i+1} \leq (2 + \sqrt{3}) A_i < 4A_i. \quad (209)$$

For every  $m \geq 1$  and  $A = A_{m+1}$ ,

$$\sum_{i=1}^m \eta_i^{-1/2} \leq 5\sqrt{A}, \quad \sum_{i=1}^m \left( \frac{\|s_i\|}{\eta_i} \right)^{2/7} \leq 20(RA)^{2/7}, \quad (210)$$

$$\sum_{i=1}^m \frac{1}{\sqrt{\eta_i A_i}} \leq 8 \log 6 + 4 \log \frac{A}{A_1}. \quad (211)$$

*Proof.* Initialization gives  $\eta_1 A_1 = 1$ . If  $\tau_i > \eta_i$ , then  $\eta_{i+1} = 2\eta_i$  and  $A_{i+1} > A_i$ . If  $\tau_i = \eta_i$ , then  $A_{i+1} = A'_i$ ,  $a'_i \geq 1/\eta_i$ , and

$$\eta_{i+1} A_{i+1} = \frac{1}{2} \eta_i A'_i = \frac{1}{2} (\eta_i a'_i)^2 \geq \frac{1}{2}.$$

Induction proves the first bound in (209). Since  $\eta_i A_i \geq 1/2$  and  $0 < \gamma_i \leq 1$ ,

$$\frac{A_{i+1}}{A_i} \leq \frac{A'_i}{A_i} = 1 + \frac{1 + \sqrt{1 + 4\eta_i A_i}}{2\eta_i A_i} \leq 2 + \sqrt{3} < 4.$$

On every iteration with  $\tau_i = \eta_i$ , we have  $a'_i = \sqrt{A_{i+1}/\eta_i}$  and

$$\sqrt{A_{i+1}} - \sqrt{A_i} = \frac{a'_i}{\sqrt{A_{i+1}} + \sqrt{A_i}} = \frac{\sqrt{A_{i+1}}}{\sqrt{\eta_i} (\sqrt{A_{i+1}} + \sqrt{A_i})} \geq \frac{1}{2\sqrt{\eta_i}}.$$

All weight increments are nonnegative, so

$$\sum_{\substack{1 \leq i \leq m \\ \tau_i = \eta_i}} \eta_i^{-1/2} \leq 2(\sqrt{A} - \sqrt{A_1}).$$

The update of  $\eta_i$  gives  $\eta_{i+1}^{-1/2} = \sqrt{2}\eta_i^{-1/2}$  when  $\tau_i = \eta_i$ , and  $\eta_{i+1}^{-1/2} = \eta_i^{-1/2}/\sqrt{2}$  otherwise. Telescoping and  $\eta_1^{-1/2} = \sqrt{A_1}$  yield

$$\begin{aligned} (1 - 1/\sqrt{2}) \sum_{\substack{1 \leq i \leq m \\ \tau_i > \eta_i}} \eta_i^{-1/2} &= (\sqrt{2} - 1) \sum_{\substack{1 \leq i \leq m \\ \tau_i = \eta_i}} \eta_i^{-1/2} + \eta_1^{-1/2} - \eta_{m+1}^{-1/2}, \\ \sum_{i=1}^m \eta_i^{-1/2} &\leq (1 + \sqrt{2}) \sum_{\substack{1 \leq i \leq m \\ \tau_i = \eta_i}} \eta_i^{-1/2} + (2 + \sqrt{2})\sqrt{A_1} \leq 2(1 + \sqrt{2})\sqrt{A} - \sqrt{2}\sqrt{A_1} \leq 5\sqrt{A}. \end{aligned}$$

Set  $S_0 = \sqrt{A_1}$  and  $S_i = S_0 + \sum_{j=1}^i \eta_j^{-1/2}$ . Applying the preceding bound to each shorter prefix gives

$$S_m \leq 6\sqrt{A}, \quad S_{i-1} \leq 6\sqrt{A_i}, \quad S_i \leq (6 + \sqrt{2})\sqrt{A_i} \leq 8\sqrt{A_i}. \quad (212)$$

Here  $\eta_i^{-1/2} \leq \sqrt{2A_i}$  follows from (209). Since  $\eta_i^{-1/2} = S_i - S_{i-1}$ , integration of the decreasing function  $1/t$  yields

$$\sum_{i=1}^m \frac{1}{\sqrt{\eta_i A_i}} \leq 8 \sum_{i=1}^m \frac{S_i - S_{i-1}}{S_i} \leq 8 \int_{S_0}^{S_m} \frac{dt}{t} \leq 8 \log 6 + 4 \log \frac{A}{A_1}.$$

This proves (211).

For the second bound in (210), Hölder's inequality gives

$$\begin{aligned} \sum_{i=1}^m \left( \frac{\|s_i\|}{\eta_i} \right)^{2/7} &= \sum_{i=1}^m \left( A_i \eta_i \|s_i\|^2 \right)^{1/7} \left( \frac{1}{\eta_i^{1/2} A_i^{1/6}} \right)^{6/7} \leq \left( \sum_{i=1}^m A_i \eta_i \|s_i\|^2 \right)^{1/7} \left( \sum_{i=1}^m \frac{1}{\eta_i^{1/2} A_i^{1/6}} \right)^{6/7}, \\ \sum_{i=1}^m \frac{1}{\eta_i^{1/2} A_i^{1/6}} &\leq 2 \sum_{i=1}^m \frac{S_i - S_{i-1}}{S_i^{1/3}} \leq 2 \int_{S_0}^{S_m} t^{-1/3} dt \leq 3 S_m^{2/3} \leq 3 \cdot 6^{2/3} A^{1/3}. \end{aligned}$$

The last display uses (212). Substituting these bounds and (208) gives

$$\sum_{i=1}^m \left( \frac{\|s_i\|}{\eta_i} \right)^{2/7} \leq (4R^2/3)^{1/7} (3 \cdot 6^{2/3} A^{1/3})^{6/7} \leq 20(RA)^{2/7}.$$

□

**Lemma F.8.** *Under the assumptions of Theorem F.5, for  $m \geq 1$  and  $A = A_{m+1}$ , define*

$$\mathcal{I}_m := \{i \in \{1, \dots, m\} : \tau_i > \eta_i, \ \|s_i\| > r_0\}.$$

Then

$$|\mathcal{I}_m| \leq 8B \left( 1 + \log \frac{A}{A_1} \right)^{2/3}. \quad (213)$$

If  $r_0 = +\infty$ , then  $\mathcal{I}_m$  is empty.

*Proof.* Let  $r_0 < +\infty$ . By (187),  $(R/r_0)^{2/3} \leq B$ . Hölder's inequality, (208), and (211) give

$$\begin{aligned} |\mathcal{I}_m| &\leq r_0^{-2/3} \sum_{i \in \mathcal{I}_m} \|s_i\|^{2/3} = r_0^{-2/3} \sum_{i \in \mathcal{I}_m} \left( A_i \eta_i \|s_i\|^2 \right)^{1/3} \left( \frac{1}{\sqrt{\eta_i A_i}} \right)^{2/3} \\ &\leq r_0^{-2/3} \left( \sum_{i=1}^m A_i \eta_i \|s_i\|^2 \right)^{1/3} \left( \sum_{i=1}^m \frac{1}{\sqrt{\eta_i A_i}} \right)^{2/3} \leq (4/3)^{1/3} B \left( 8 \log 6 + 4 \log \frac{A}{A_1} \right)^{2/3}. \end{aligned}$$

Since  $(4/3)^{1/3}(8 \log 6)^{2/3} < 8$  and  $8 \log 6 > 4$ , this proves (213).  $\square$

**Lemma F.9.** *Under the assumptions of Theorem F.5, let  $b_i := \log_2(\tau_i/\eta_i)$  be the number of rejected trials in main iteration  $i$ , including failed PD tests. For every  $m \geq 1$  and  $A = A_{m+1}$ ,*

$$\sum_{i=1}^m b_i \leq 5m + 30\sqrt{\delta A} + 120(M_0 R A)^{2/7} + 8B \log_{2,+} \frac{\delta + \bar{g}/r_0}{\delta + M_0 r_0} \left( 1 + \log \frac{A}{A_1} \right)^{2/3}, \quad (214)$$

where the last term is absent if  $r_0 = +\infty$ .

*Proof.* If  $\tau_i = \eta_i$ , then  $b_i = 0$ . Let  $\tau_i > \eta_i$ . If  $\|s_i\| \leq r_0$ , then Lemma C.3 gives  $\tau_i \leq 8\delta + 24M_0\|s_i\|$ . If  $\|s_i\| > r_0$ , then (IH), convexity, and (206) give

$$\begin{aligned} (\tau_i - \delta)\|s_i\|^2 &\leq \langle (H(v_i) + \tau_i I)s_i, s_i \rangle = -\langle \nabla f(v_i), s_i \rangle \leq \bar{g}\|s_i\|, \\ \tau_i &\leq \delta + \frac{\bar{g}}{r_0} \leq \max \left\{ 1, \frac{\delta + \bar{g}/r_0}{\delta + M_0 r_0} \right\} (\delta + M_0\|s_i\|). \end{aligned}$$

Since  $8\delta + 24M_0\|s_i\| \leq 32\eta_i \left( 1 + \frac{\delta}{\eta_i} \right) \left( 1 + \frac{M_0\|s_i\|}{\eta_i} \right)$ , in both cases

$$b_i \leq 5 + \log_2 \left( 1 + \frac{\delta}{\eta_i} \right) + \log_2 \left( 1 + \frac{M_0\|s_i\|}{\eta_i} \right) + \mathbf{1}_{\{i \in \mathcal{I}_m\}} \log_{2,+} \frac{\delta + \bar{g}/r_0}{\delta + M_0 r_0}.$$

Since  $\log(1+x) = \int_0^x \frac{dt}{1+t} \leq \frac{x^p}{p}$  for  $x \geq 0$  and  $p \in (0, 1]$ , we have  $\log_2(1+x) \leq 6x^p$  for  $p \in \left\{ \frac{1}{2}, \frac{2}{7} \right\}$ . Hence,

$$\sum_{i=1}^m b_i \leq 5m + 6\sqrt{\delta} \sum_{i=1}^m \eta_i^{-1/2} + 6M_0^{2/7} \sum_{i=1}^m \left( \frac{\|s_i\|}{\eta_i} \right)^{2/7} + |\mathcal{I}_m| \log_{2,+} \frac{\delta + \bar{g}/r_0}{\delta + M_0 r_0}.$$

Substituting (210) and (213) proves (214).  $\square$

*Proof of Corollary F.6.* By Lemma F.1, initialization rejects at most  $1 + \log_{2,+} \frac{\bar{\tau}_0}{\eta_0}$  trials. If  $A_1 \geq R^2/(2\varepsilon)$ , then  $N = 1$  and there are no main iterations. Otherwise, the minimality of  $N$  and (209) give  $A_N < 4A_{N-1} < 2R^2/\varepsilon$ . Together with (204), this gives

$$\sqrt{\delta A_N} \leq \sqrt{\frac{2\delta R^2}{\varepsilon}}, \quad (M_0 R A_N)^{2/7} \leq \left( \frac{2M_0 R^3}{\varepsilon} \right)^{2/7}, \quad \frac{A_N}{A_1} \leq \frac{4 \max \{\eta_0, \bar{\tau}_0\} R^2}{\varepsilon}.$$

Substituting these bounds into (214) with  $m = N - 1$  and adding the initialization rejections, we obtain that the total number of rejected trials is at most the right-hand side of (207) with  $7N$  replaced by  $5N$ . Each outer iteration makes one call  $\text{MSBACKTRACK}(x, \eta)$ , which uses  $H(x)$ ,  $\nabla f(x)$ , and one gradient for each trial with a positive definite matrix. The updates reuse the gradient at the accepted point. Hence, the method uses at most  $N$  Hessian evaluations, and the number of gradient evaluations is at most  $2N$  plus the number of rejected trials. This proves (207).  $\square$

## G Extra Experiments

### G.1 Experiment Details

Table 2 gives the initial parameters for experiments on Figure 1. ACRN and A-ACRN use  $\delta_0 = 0$  and adapt  $\mathcal{M}$  by doubling at each rejected trial (Kamzolov et al., 2025b).

Table 2: Initial parameters of the runs in Figure 1, from  $x_0 = -1$ .

|                                             | gisette | german_numer | ijcnn1  | abalone            |
|---------------------------------------------|---------|--------------|---------|--------------------|
| $L(x_0)$                                    | 0.1     | 0.1          | 0.1     | 789.5              |
| $\ \nabla f(x_0)\ $                         | 0.4233  | 0.3268       | 0.07210 | 19.48              |
| $\ x_0\ $                                   | 70.71   | 4.899        | 4.690   | 3                  |
| $\eta_0$ (Algorithms 2, 3, 6)               | 0.2057  | 0.1808       | 0.08491 | 124.0              |
| $H_0$ (AdaN)                                | 0.05    | 0.05         | 0.05    | 394.8              |
| $\gamma$ (Extra-Newton)                     | 141.4   | 9.798        | 9.381   | 6                  |
| $\beta_0$ (Extra-Newton)                    | 846.6   | 3.138        | 0.6345  | $5.537 \cdot 10^5$ |
| $\mathcal{M}_0$ (ACRN, A-ACRN, Algorithm 5) | 0.1     | 0.1          | 0.1     | 789.5              |

### G.2 Sensitivity to Initial Parameters

We start from  $x_0 = -1$ , as in Figure 1, and run Algorithm 2, AdaN and ACRN with exact Hessians. Their initial parameters  $\eta_0$ ,  $H_0$  and  $M_0$  take the same value  $c$  from the grid  $\{10^{-10}, 10^{-9}, \dots, 10^{10}\}$ . Figure 3 shows the number of iterations to reach  $f(x_k) - f^* \leq 10^{-10}$  for every  $c$ , and Figure 4 shows the convergence for five values of  $c$ .

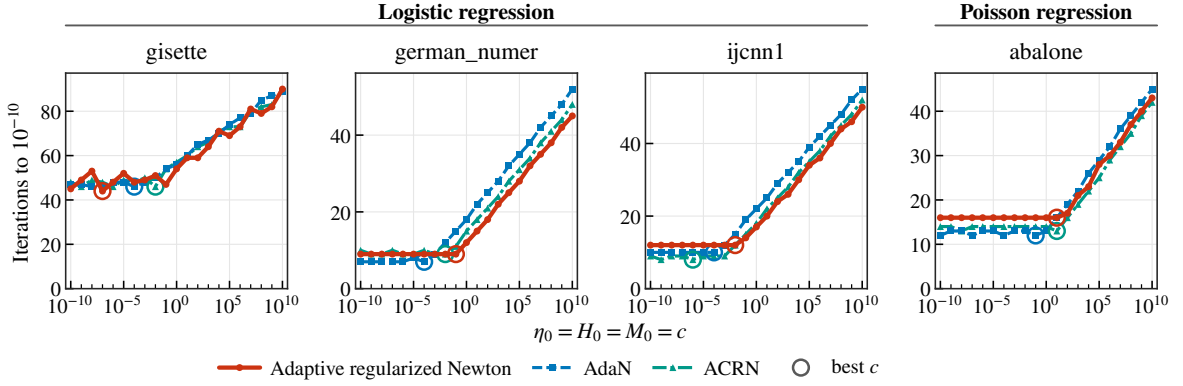


Figure 3: Iterations to reach  $f(x_k) - f^* \leq 10^{-10}$  for  $\eta_0 = H_0 = M_0 = c$ . Circles mark the best  $c$  of each method.

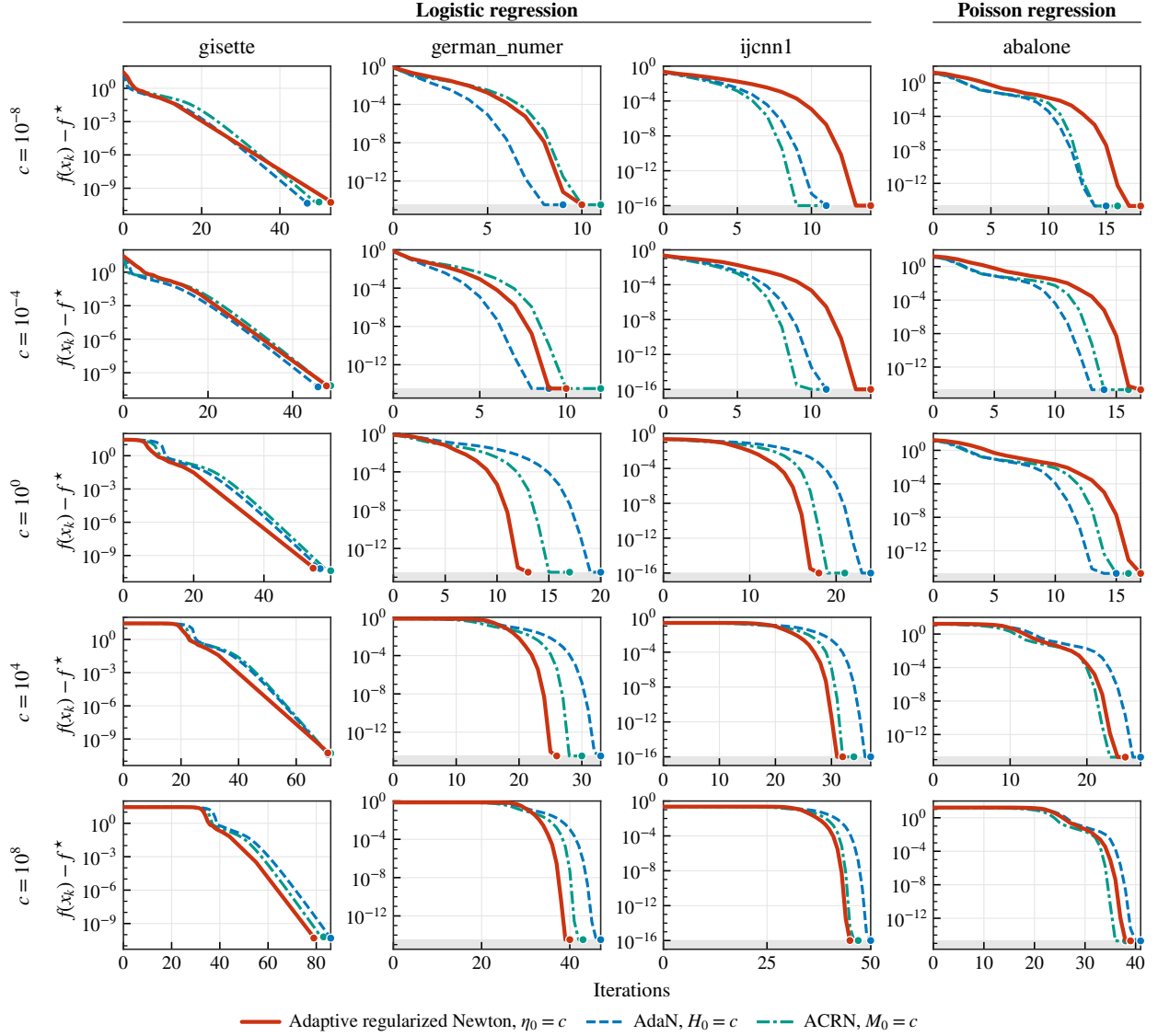


Figure 4: Convergence of Algorithm 2, AdaN and ACRN for  $\eta_0 = H_0 = M_0 = c$ , one row per value of  $c$ .